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(54) **AIR CONDITIONER AND SELF-CLEANING
CONTROL METHOD THEREFOR**

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(57) **ABSTRACT**

The air conditioner includes an indoor unit, an outdoor unit, an indoor temperature measurement apparatus, and a controller. The indoor unit includes an indoor heat exchanger. The outdoor unit includes an outdoor heat exchanger. The indoor temperature measurement apparatus is configured to measure the indoor ambient temperature. The controller is configured to: control the air conditioner to enter a self-cleaning mode in response to a received self-cleaning instruction, causing the heat exchanger to be cleaned to operate as an evaporator and cause ice to form on a surface of the heat exchanger to be cleaned; adjust an operating parameter of the air conditioner according to the indoor ambient temperature; and control the air conditioner to enter the defrosting stage, where the heat exchanger to be cleaned is the outdoor heat exchanger or the indoor heat exchanger.

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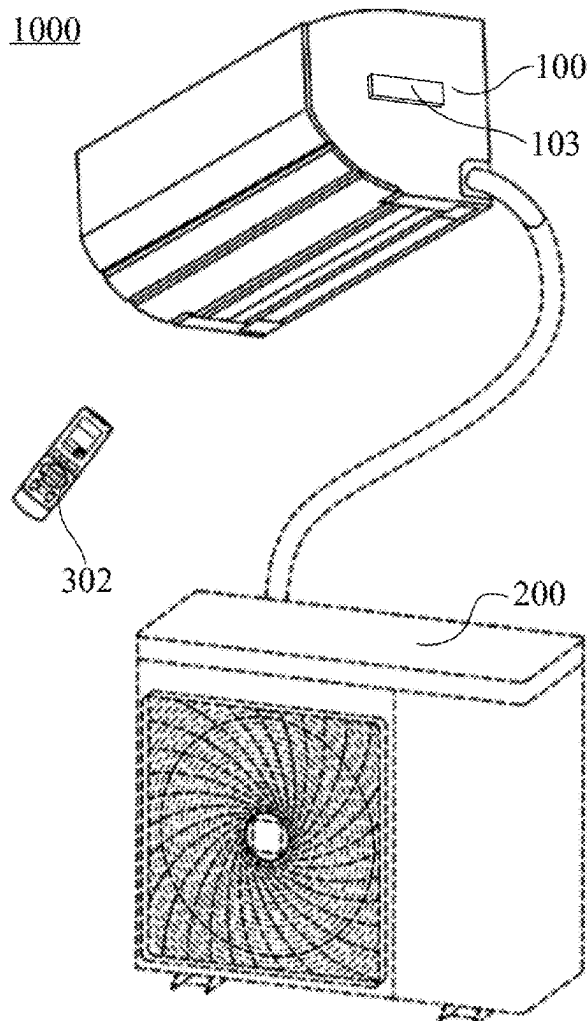
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135195, filed on Nov. 29, 2023.

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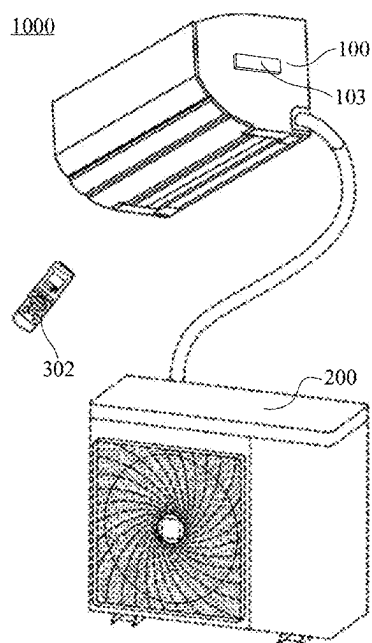


FIG. 1

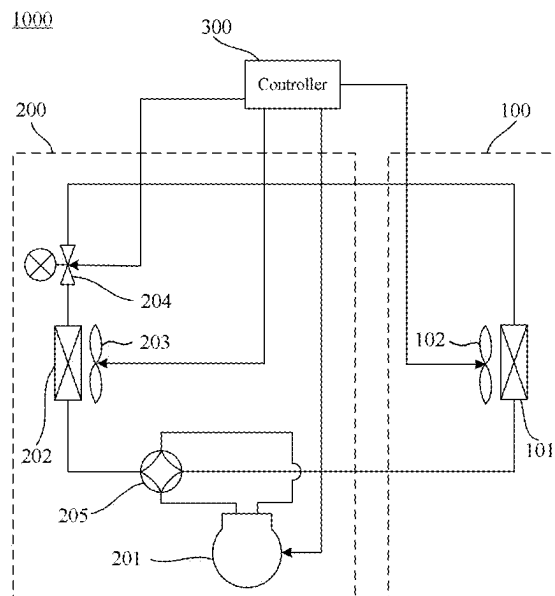


FIG. 2

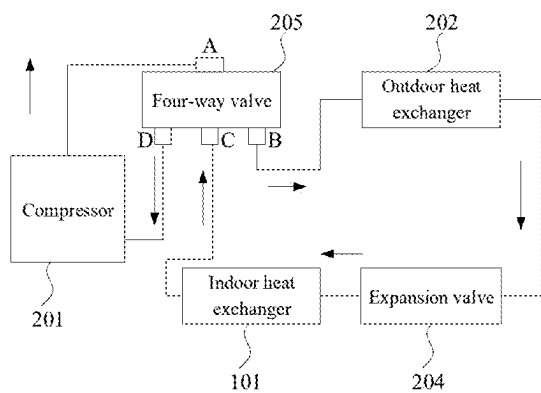


FIG. 3

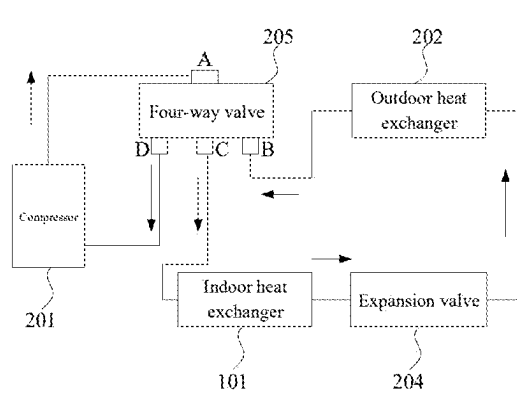


FIG. 4

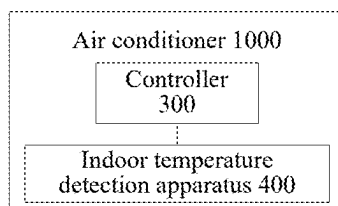


FIG. 5

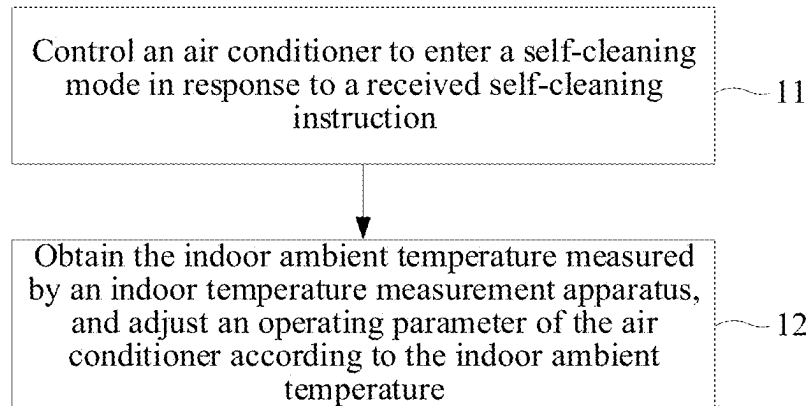


FIG. 6

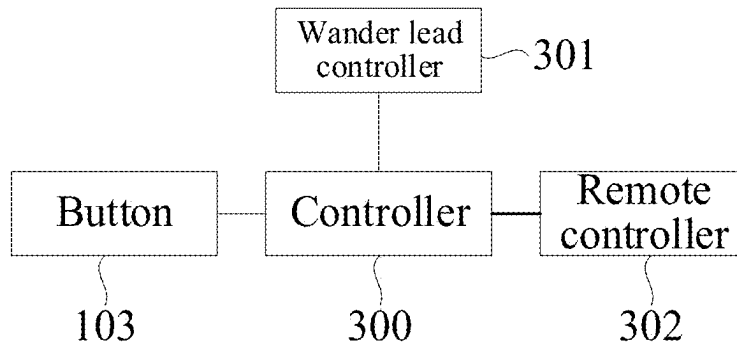


FIG. 7

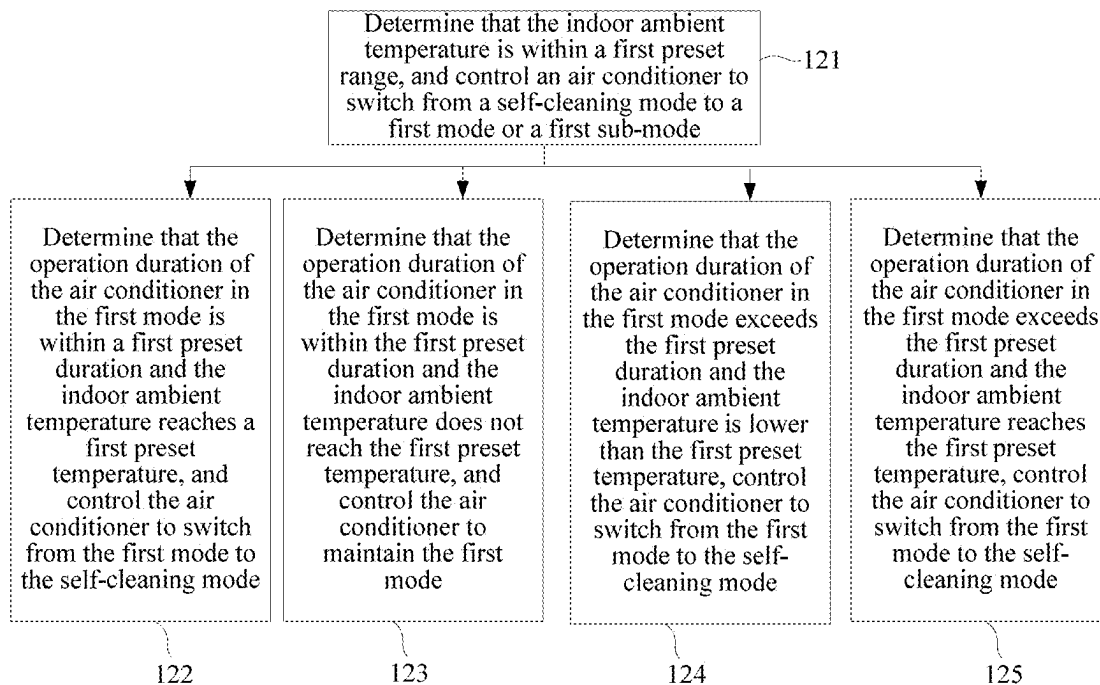


FIG. 8

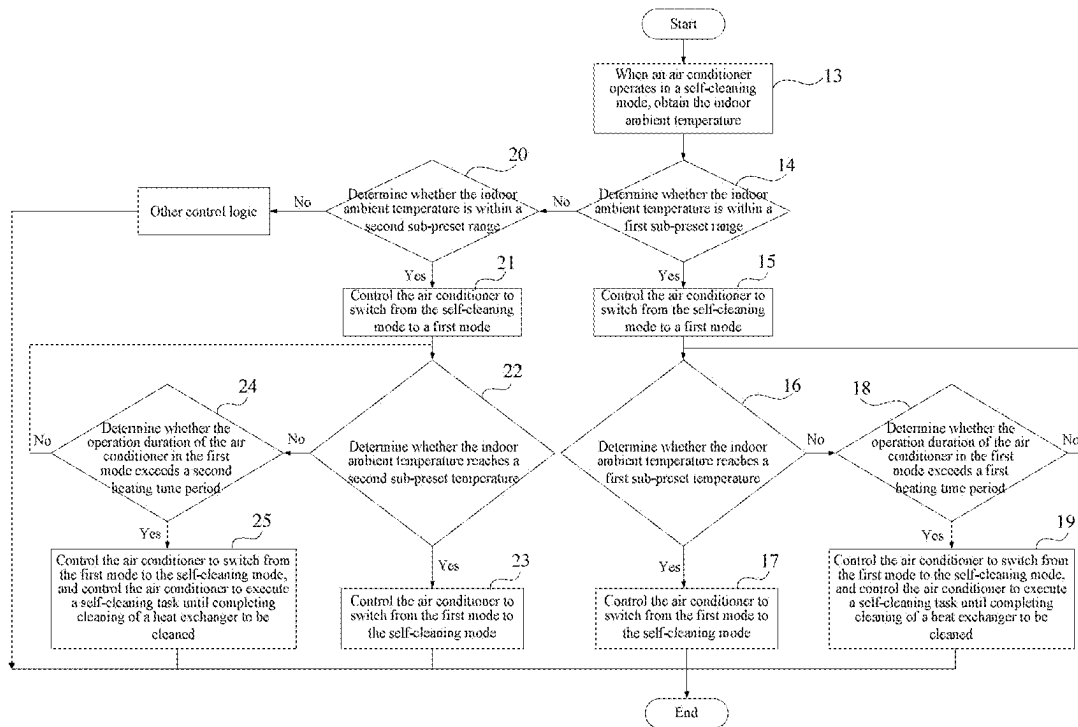


FIG. 9

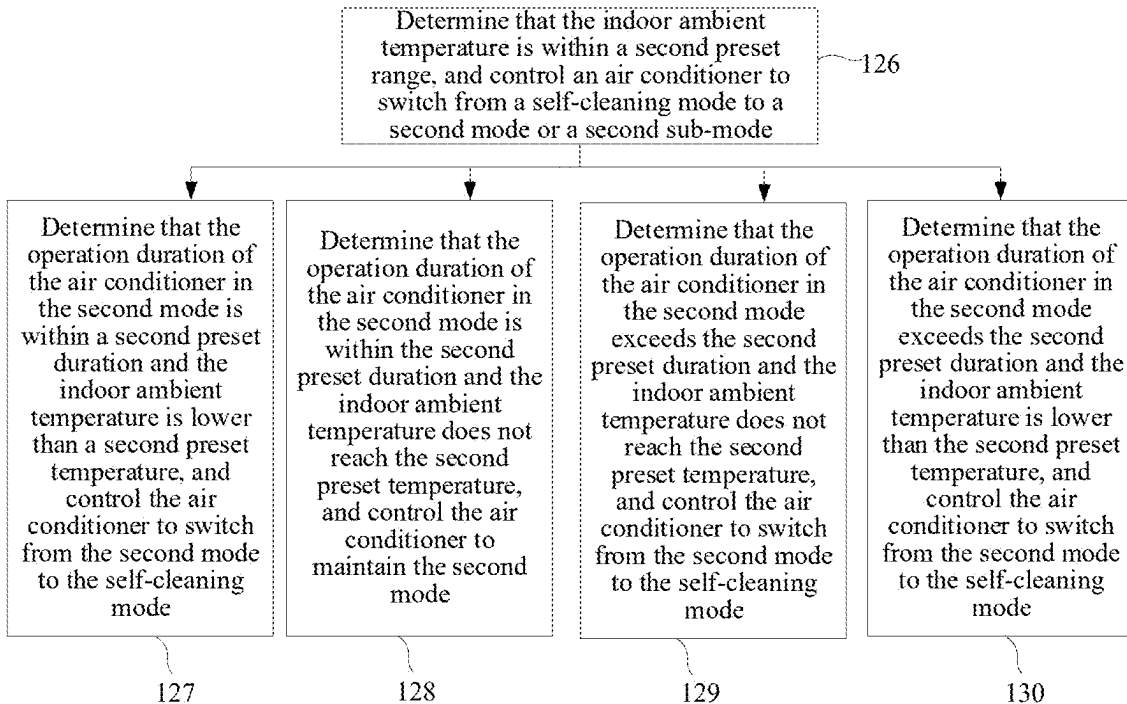


FIG. 10

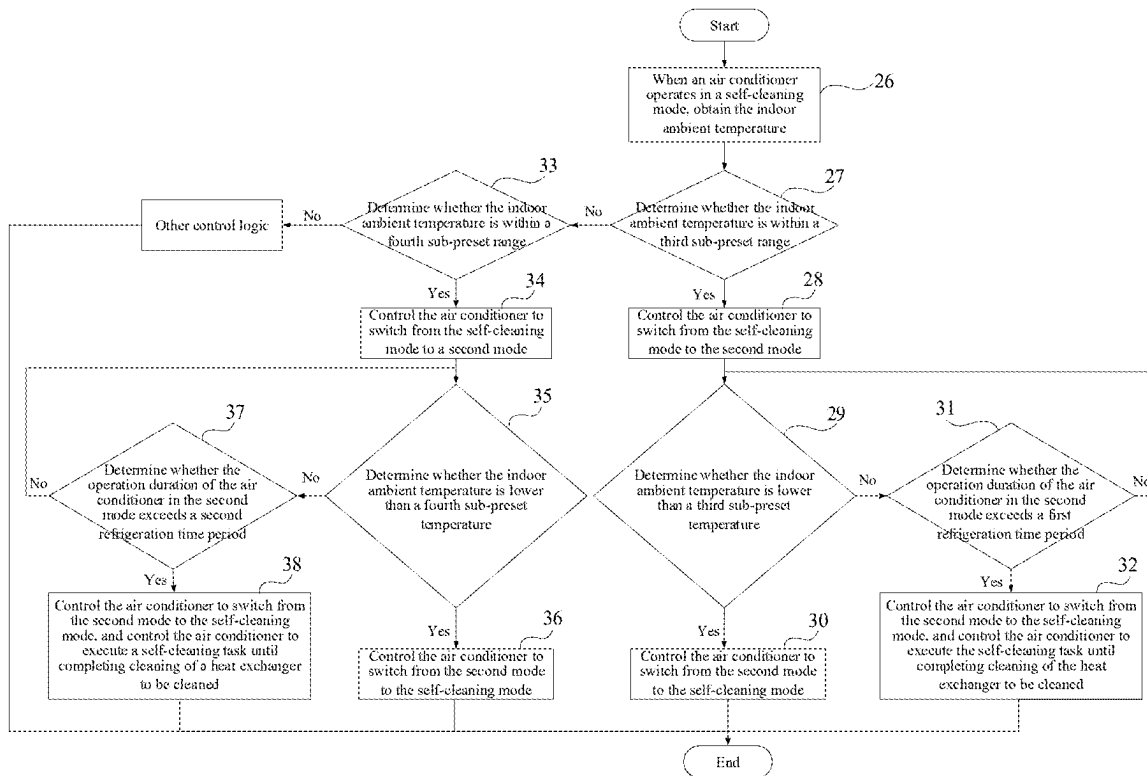


FIG. 11

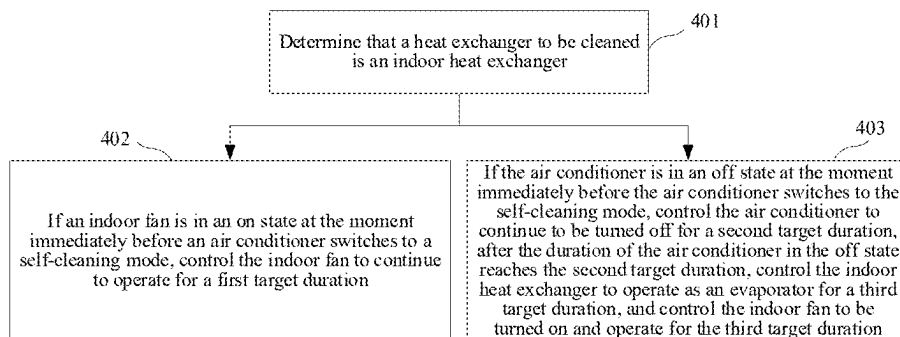


FIG. 12

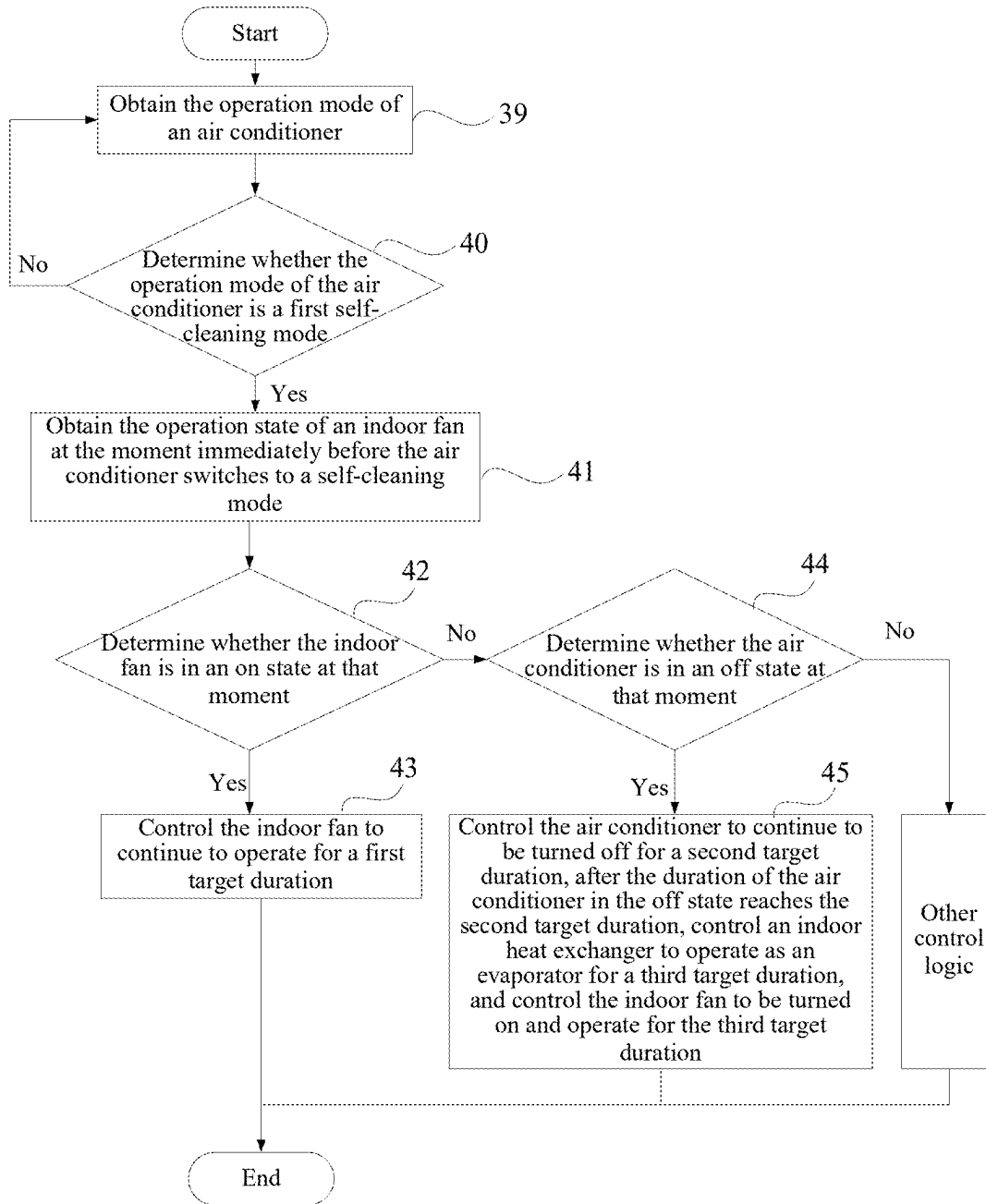


FIG. 13

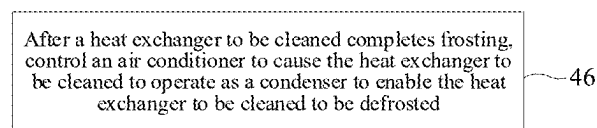


FIG. 14

Before an air conditioner switches between a first mode and a first self-cleaning mode, or before the air conditioner switches between a second mode and a second self-cleaning mode, control the air conditioner to be shut down for a fourth target duration

FIG. 15

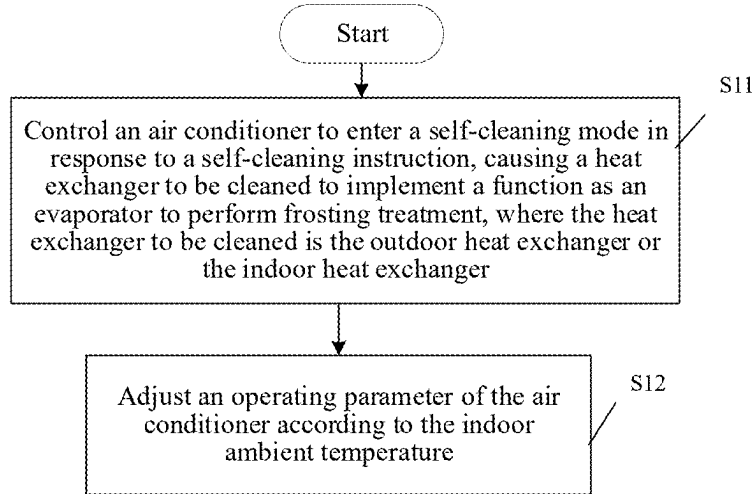


FIG. 16

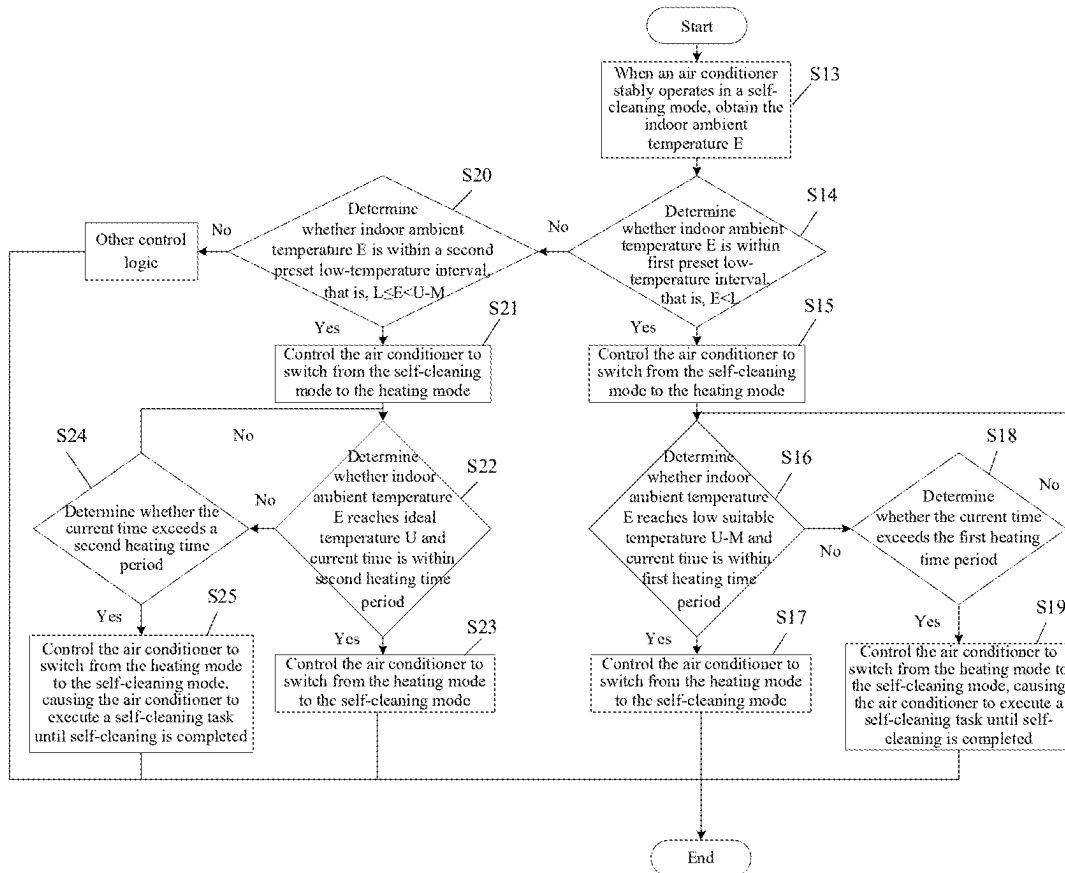


FIG. 17

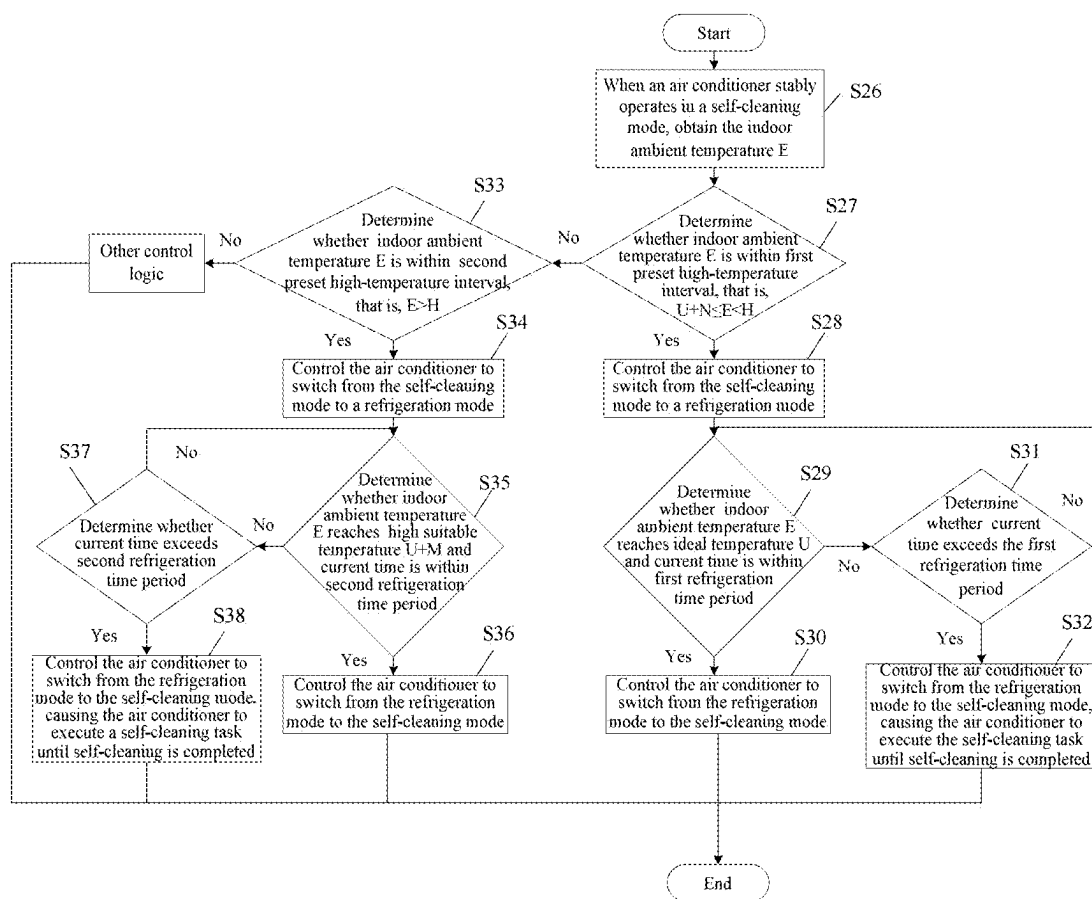


FIG. 18

AIR CONDITIONER AND SELF-CLEANING CONTROL METHOD THEREFOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of International Application No. PCT/CN2023/135195, filed Nov. 29, 2023, which claims priority to International Patent Application No. PCT/CN2023/084828, filed on Mar. 29, 2023, and Chinese Patent Application No. 202211520667.5, filed on Nov. 30, 2022. The entire disclosures of the above-identified applications are hereby incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of air conditioning, in particular to an air conditioner and a self-cleaning control method therefor.

BACKGROUND

[0003] The air conditioner is one of the commonly-used electric appliances in the family life. The refrigeration cycle or heating cycle of the air conditioner is usually executed by means of a compressor, a condenser, an expansion valve, and an evaporator, making heat be transferred from a fluid with the relatively low temperature to a fluid with the relatively high temperature, thereby achieving refrigeration or heating of the air conditioner.

SUMMARY

[0004] On one hand, an air conditioner is provided. The air conditioner includes an indoor unit, an outdoor unit, an indoor temperature detection apparatus, and a controller. The indoor unit includes an indoor heat exchanger. The outdoor unit includes a compressor, an outdoor heat exchanger, and an expansion valve. The indoor temperature detection apparatus is configured to detect the indoor ambient temperature. The controller is configured to: control the air conditioner to enter a self-cleaning mode in response to a received self-cleaning instruction, causing the heat exchanger to be cleaned to operate as an evaporator to cause the heat exchanger to be cleaned to perform frosting treatment; adjust an operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, and control the air conditioner to enter the defrosting stage of the heat exchanger to be cleaned after the frosting treatment ends. The heat exchanger to be cleaned is the outdoor heat exchanger or the indoor heat exchanger. The self-cleaning instruction includes a first instruction and a second instruction. The first instruction is configured to instruct cleaning of the indoor heat exchanger, and the second instruction is configured to instruct cleaning of the outdoor heat exchanger.

[0005] On the other hand, a self-cleaning control method for an air conditioner is provided. The method is applied to a controller of the air conditioner. The air conditioner includes an indoor unit, an outdoor unit, and an indoor temperature detection apparatus. The indoor unit includes an indoor heat exchanger. The outdoor unit includes a compressor, an outdoor heat exchanger, and an expansion valve. The indoor temperature detection apparatus is configured to detect the indoor ambient temperature. The method includes: controlling the air conditioner to enter a self-cleaning mode in response to a received self-cleaning

instruction, causing the heat exchanger to be cleaned to operate as an evaporator to cause the heat exchanger to be cleaned to perform frosting treatment; adjusting an operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, and controlling the air conditioner to enter the defrosting stage of the heat exchanger to be cleaned after the frosting treatment ends. The heat exchanger to be cleaned is the outdoor heat exchanger or the indoor heat exchanger. The self-cleaning instruction includes a first instruction and a second instruction. The first instruction is configured to instruct cleaning of the indoor heat exchanger, and the second instruction is configured to instruct cleaning of the outdoor heat exchanger.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a structure diagram of an air conditioner according to some embodiments.

[0007] FIG. 2 is another structure diagram of an air conditioner according to some embodiments.

[0008] FIG. 3 is a schematic diagram of a refrigerant flow direction of an air conditioner in a refrigeration mode according to some embodiments.

[0009] FIG. 4 is a schematic diagram of a refrigerant flow direction of an air conditioner in a heating mode according to some embodiments.

[0010] FIG. 5 is a block diagram of an air conditioner according to some embodiments.

[0011] FIG. 6 is a flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0012] FIG. 7 is a block diagram of a communication system of an air conditioner according to some embodiments.

[0013] FIG. 8 is another flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0014] FIG. 9 is yet another flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0015] FIG. 10 is yet another flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0016] FIG. 11 is yet another flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0017] FIG. 12 is yet another flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0018] FIG. 13 is yet another flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0019] FIG. 14 is yet another flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0020] FIG. 15 is yet another flow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0021] FIG. 16 is a workflow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0022] FIG. 17 is another workflow chart of execution steps of a controller of an air conditioner according to some embodiments.

[0023] FIG. 18 is another workflow chart of execution steps of a controller of an air conditioner according to some embodiments.

DESCRIPTION OF THE EMBODIMENTS

[0024] Some embodiments of the present disclosure will be described clearly and completely below in conjunction with the accompanying drawings. However, the described embodiments are only part of embodiments of the present disclosure, not all of them. Based on the embodiments provided by the present disclosure, all other embodiments obtained by those ordinarily skilled in the art fall within the scope of protection of the present disclosure.

[0025] Unless the context requires otherwise, throughout the description and claims, the term “comprise” and other forms thereof, such as the third-person singular form “comprises” and the present participle form “comprising” are construed in an open, inclusive meaning, that is, “comprising, but not limited to”. In the description, the terms “one embodiment”, “some embodiments”, “exemplary embodiments”, “example”, “specific example”, or “some examples”, etc. are intended to indicate that a particular feature, structure, material, or characteristic related to the embodiment or example is included in at least one embodiment or example of the present disclosure. The schematic illustration of the above terms does not necessarily refer to the same embodiment or example. Furthermore, the specific features, structures, materials, or characteristics described may be included in any one or more embodiments or examples in any suitable manner.

[0026] Hereinafter, the terms “first” and “second” are for descriptive purposes only, and are not to be understood as indicating or implying relative importance or as implicitly indicating the number of technical features indicated. Thus, features limited by “first” and “second” may expressly or implicitly include one or more features. In the description of embodiments of the present disclosure, unless otherwise specified, “a plurality” means two or more.

[0027] In describing some embodiments, the expressions “coupled” and “connected” and extensions thereof may be used. The term “connected” is to be understood in a broad sense, for example, “connected” may refer to a fixed connection, may also refer to a detachable connection, or an integral connection; and it may refer to a direct connection or an indirect connected through an intermediate medium. When describing some embodiments, the term “coupled” may be used to indicate that two or more components are in direct physical contact or electric contact. However, the term “coupled” or “communicatively coupled” may also refer to that two or more components are not in direct contact, but still cooperate or interact with each other. The embodiments disclosed herein are not necessarily limited to the contents herein.

[0028] “A and/or B” includes three combinations of only A, only B, and a combination of A and B.

[0029] The use of “suitable for” or “configured to” herein means open and inclusive language that does not exclude devices suitable for or configured to perform additional tasks or steps.

[0030] As used herein, according to the context, the term “if” is optionally construed as “when”, “upon”, “in response to determining”, or “in response to detecting”. Similarly, according to the context, the phrase “if it is determined . . .” or “if it is detected that [the stated conditions or events]”

is optionally construed as “when it is determined . . .”, “in response to determining . . .”, “when it is detected that [the stated conditions or events]”, or “in response to determining [the stated conditions or events]”.

[0031] The air contains dust, so that during the operation process of an air conditioner, the dust is easily accumulated on a heat exchanger (such as an indoor heat exchanger or an outdoor heat exchanger) of the air conditioner, which affects the heat exchange efficiency of the heat exchanger. Besides, if the dust is accumulated in the air conditioner for a long time, bacteria are bred easily, and the quality of the air is affected. For the air conditioner, the problem of dust accumulation of the heat exchanger is often solved by means of frosting and defrosting the heat exchanger to be cleaned. For example, the air conditioner runs a refrigeration or heating mode to cause ice to form on a surface of the heat exchanger to be cleaned, then the air conditioner is defrosted such that the ice on the surface of the heat exchanger to be cleaned thaws and melts into water, and the water flows downwards to take away the dust on the heat exchanger to be cleaned, thereby cleaning the heat exchanger to be cleaned. However, in the self-cleaning process, the indoor heat exchanger serving as a condenser or an evaporator may operate as an evaporator or a condenser before self-cleaning. Therefore, the indoor ambient temperature changes easily, which affects the heating or refrigeration effect of the air conditioner.

[0032] Some embodiments of the present disclosure provide an air conditioner 1000. FIG. 1 is a structure diagram of the air conditioner according to some embodiments. As shown in FIG. 1, the air conditioner (hereinafter also referred to as air-conditioning) 1000 includes an indoor unit 100 and an outdoor unit 200. The indoor unit 100 and the outdoor unit 200 are connected by means of a pipeline to transfer a refrigerant. The indoor unit 100 is configured to adjust the temperature and humidity of the indoor air. The outdoor unit 200 is mounted outdoors, and the indoor unit 100 is mounted indoors. Taking FIG. 1 as an example, the air conditioner 1000 is a split air conditioner (such as a wall-mounted air conditioner), and the indoor unit 100 is hung on an indoor wall. Of course, the air conditioner 1000 in some embodiments of the present disclosure may also be an integrated air conditioner (such as a window type air conditioner), which is not limited in the present disclosure. Besides, the indoor unit 100 in FIG. 1 is located indoors, and the outdoor unit 200 is located outdoors, so that the outdoor unit 200 is represented by dashed lines in FIG. 1.

[0033] FIG. 2 is another structure diagram of the air conditioner according to some embodiments. In some embodiments, as shown in FIG. 2, the indoor unit 100 includes an indoor heat exchanger 101 and an indoor fan 102. The outdoor unit 200 includes a compressor 201, an outdoor heat exchanger 202, an outdoor fan 203, an expansion valve 204, and a four-way valve 205. The compressor 201, the outdoor heat exchanger 202, the expansion valve 204, and the indoor heat exchanger 101 which are sequentially connected form a refrigerant loop. The refrigerant flows circularly in the refrigerant loop and exchanges heat with the ambient air by means of the outdoor heat exchanger 202 and the indoor heat exchanger 101 respectively, so as to implement a refrigeration mode or a heating mode of the air conditioner 1000.

[0034] The indoor heat exchanger 101 is configured to perform heat exchange between the indoor air and the

refrigerant transferred in the indoor heat exchanger **101**. For example, the indoor heat exchanger **101** works as the evaporator in the refrigeration mode of the air conditioner **1000**, so that the refrigerant subjected to heat dissipation by means of the outdoor heat exchanger **202** absorbs heat of the indoor air by means of the indoor heat exchanger **101** to be evaporated. The indoor heat exchanger **101** works as the condenser in the heating mode of the air conditioner **1000**, so that the refrigerant subjected to absorbing heat by means of the outdoor heat exchanger **202** dissipates the heat to the indoor air by means of the indoor heat exchanger **101** to be condensed.

[0035] The indoor fan **102** is configured to suck the indoor air into the indoor unit **100** through an indoor air inlet of the indoor unit **100**, and send out the indoor air subjected to exchanging heat with the indoor heat exchanger **101** through an indoor air outlet of the indoor unit **100**. The indoor fan **102** provides power for the flow of the indoor air.

[0036] The compressor **201** is configured to compress the refrigerant such that a low-pressure refrigerant is compressed to form a high-pressure refrigerant.

[0037] The outdoor heat exchanger **202** is configured to perform heat exchange between the outdoor air and the refrigerant transferred in the outdoor heat exchanger **202**. For example, the outdoor heat exchanger **202** works as the condenser in the refrigeration mode of the air conditioner **1000**, so that the refrigerant compressed by means of the compressor **201** dissipates heat to the outdoor air by means of the outdoor heat exchanger **202** to be condensed. The outdoor heat exchanger **202** works as the evaporator in the heating mode of the air conditioner **1000**, so that the decompressed refrigerant absorbs heat of the outdoor air by means of the outdoor heat exchanger **202** to be evaporated.

[0038] The outdoor fan **203** is configured to suck the outdoor air into the outdoor unit **200** through an outdoor air inlet of the outdoor unit **200**, and send out the outdoor air exchanging heat with the outdoor heat exchanger **202** through an outdoor air outlet of the outdoor unit **200**. The outdoor fan **203** provides power for the flow of the outdoor air, so that the outdoor air flows through the outdoor heat exchanger **202** to exchange heat with the refrigerant in the outdoor heat exchanger **202**.

[0039] The expansion valve **204** is connected between the outdoor heat exchanger **202** and the indoor heat exchanger **101**, and the pressure of the refrigerant flowing through the outdoor heat exchanger **202** and the indoor heat exchanger **101** is adjusted by means of the opening degree of the expansion valve **204**, thereby adjusting the flow rate of the refrigerant circulating between the outdoor heat exchanger **202** and the indoor heat exchanger **101**. The flow rate and pressure of the refrigerant circulating between the outdoor heat exchanger **202** and the indoor heat exchanger **101** will affect the heat exchange performance of the outdoor heat exchanger **202** and the indoor heat exchanger **101**. The opening degree of the expansion valve **204** is adjustable, thereby controlling the flow rate and pressure of the refrigerant flowing through the expansion valve **204**. For example, the expansion valve **204** enables a liquid refrigerant condensed in the condenser to be expanded into a low-pressure liquid refrigerant. It should be noted that in some embodiments of the present disclosure, the expansion valve **204** being arranged in the outdoor unit **200** is taken as an

example for illustration. Of course, in some embodiments, the expansion valve **204** may also be arranged in the indoor unit **100**.

[0040] The four-way valve **205** is connected in the refrigerant loop, and is configured to switch the flow direction of the refrigerant in the refrigerant loop to enable the air conditioner **1000** to execute the refrigeration mode or the heating mode.

[0041] FIG. 3 and FIG. 4 are taken as examples to explain the refrigeration mode and the heating mode of the air conditioner **1000** in detail below. FIG. 3 is a schematic diagram of a refrigerant flow direction of the air conditioner in the refrigeration mode according to some embodiments. FIG. 4 is a schematic diagram of a refrigerant flow direction of the air conditioner in the heating mode according to some embodiments.

[0042] As shown in FIG. 3 and FIG. 4, an exhaust port of the compressor **201** is connected to a first end A of the four-way valve **205**, a second end B of the four-way valve **205** is connected to a first end of the outdoor heat exchanger **202**, a second end of the outdoor heat exchanger **202** is connected to a first end of the expansion valve **204**, a second end of the expansion valve **204** is connected to a first end of the indoor heat exchanger **101**, a second end of the indoor heat exchanger **101** is connected to a third end C of the four-way valve **205**, and a fourth end D of the four-way valve **205** is connected to an air suction port of the compressor **201**. Thus, the flow direction of the refrigerant in the refrigerant loop can be controlled by controlling the state of the four-way valve **205**, thereby controlling the operation mode of the air conditioner **1000**.

[0043] As shown in FIG. 3, under the condition that the air conditioner **1000** operates in the refrigeration mode, the first end A and the second end B of the four-way valve **205** are in communication with each other, and the third end C and the fourth end D of the four-way valve **205** are in communication with each other. The compressor **201** compresses a gaseous refrigerant into a high-temperature and high-pressure refrigerant and discharge the refrigerant by means of the exhaust port. The discharged refrigerant flows to the first end A of the four-way valve **205**, and flows to the first end of the outdoor heat exchanger **202** from the second end B of the four-way valve **205**. The outdoor heat exchanger **202** serves as the condenser to condense the compressed refrigerant into the liquid refrigerant. During the condensing process, the refrigerant releases heat to the ambient environment (outdoors). The condensed refrigerant flows into the first end of the expansion valve **204** through the second end of the outdoor heat exchanger **202**, and the expansion valve **204** expands the refrigerant into the low-pressure liquid refrigerant. The expanded refrigerant flows to the indoor heat exchanger **101** through the second end of the expansion valve **204**. The indoor heat exchanger **101** serves as the evaporator to evaporate the expanded refrigerant. During the evaporation process, the refrigerant absorbs the heat of the ambient environment (indoors), thereby refrigerating the indoor environment. The evaporated refrigerant returns to the compressor **201** through the third end C and the fourth end D of the four-way valve **205**.

[0044] As shown in FIG. 4, under the condition that the air conditioner **1000** operates in the heating mode, the first end A and the third end C of the four-way valve **205** are in communication with each other, and the second end B and the fourth end D of the four-way valve **205** are in commu-

nication with each other. The compressor **201** compresses the gaseous refrigerant into the high-temperature and high-pressure refrigerant and discharge the refrigerant by means of the exhaust port. The discharged refrigerant flows to the first end A of the four-way valve **205**, and flows to the second end of the indoor heat exchanger **101** from the third end C of the four-way valve **205**. The indoor heat exchanger **101** serves as the condenser to condense the compressed refrigerant into the liquid refrigerant. During the condensing process, the refrigerant releases the heat to the ambient environment (indoors), thereby heating the indoor environment. The condensed refrigerant flows into the second end of the expansion valve **204** through the first end of the indoor heat exchanger **101**, and the expansion valve **204** expands the refrigerant into the low-pressure liquid refrigerant. The expanded refrigerant flows to the outdoor heat exchanger **202** through the first end of the expansion valve **204**. The outdoor heat exchanger **202** serves as the evaporator to evaporate the expanded refrigerant. During the evaporation process, the refrigerant absorbs the heat of the ambient environment (outdoors). The evaporated refrigerant returns to the compressor **201** through the second end B and the fourth end D of the four-way valve **205**.

[0045] FIG. 5 is a block diagram of the air conditioner according to some embodiments. In some embodiments, as shown in FIG. 5, the air conditioner **1000** further includes an indoor temperature detection apparatus **400**. The indoor temperature detection apparatus **400** may be arranged indoors, and is configured to detect the indoor ambient temperature. The specific indoor mounting position of the indoor temperature detection apparatus **400** may be set according to the actual requirements.

[0046] For example, the indoor temperature detection apparatus **400** is a temperature sensor. The temperature sensor refers to a sensor that can sense the temperature and convert temperature information into a usable output signal. Temperature sensors can be divided into contact type temperature sensors and non-contact type temperature sensors according to detection methods, or can be divided into thermistor temperature sensors and thermocouple temperature sensors according to sensor materials and characteristics of electronic components therein. It should be noted that the temperature sensor in some embodiments of the present disclosure can be selected according to the actual application requirements.

[0047] In some embodiments, as shown in FIG. 2, the air conditioner **1000** further includes a controller **300**. The controller **300** is configured to control various components of the air conditioner **1000** to work, thereby implementing various preset functions of the air conditioner **1000**. For example, the controller **300** controls the operating frequency of the compressor **201**, the opening degree of the expansion valve **204**, and the rotational speed of the indoor fan **102**. The controller **300** is connected to the compressor **201**, the expansion valve **204**, the outdoor fan **203** and the indoor fan **102** by means of data lines so as to transmit communication information. Besides, the controller **300** is coupled to the indoor temperature detection apparatus **400**, thereby obtaining the indoor ambient temperature detected by the indoor temperature detection apparatus **400**.

[0048] The controller **300** includes a central processing unit (CPU), a microprocessor, and an application specific integrated circuit (ASIC), and can be configured to execute corresponding operations described in the controller **300**

when the processor executes programs stored in a non-transitory computer-readable medium coupled to the controller **300**.

[0049] In some embodiments, the air conditioner **1000** further includes a display screen, and the display screen is configured to display state information of the air conditioner, for example, operating parameters (including the temperature, the humidity, the wind speed, etc.), operation modes (including the refrigeration mode, the heating mode, a self-cleaning mode, etc.), and fault conditions (network connection failure prompting, etc.). The display screen may be located on a housing of the air conditioner **1000**, and may be located on a control device (for example, a wander lead controller **301**, a remote controller **302**, etc.).

[0050] The execution steps of the controller **300** in some embodiments of the present disclosure are described in detail below.

[0051] FIG. 6 is a flow chart of the execution steps of the controller of the air conditioner according to some embodiments.

[0052] In some embodiments, the controller **300** is configured to execute the following step:

[0053] step **11**, controlling the air conditioner **1000** to enter the self-cleaning mode in response to a received self-cleaning instruction.

[0054] The operation process of the air conditioner **1000** in the self-cleaning mode includes: a frosting stage (also referred to as a refrigeration stage and a freezing stage) and a defrosting stage. When the air conditioner **1000** enters the self-cleaning mode, the air conditioner **1000** first enters the frosting stage and then enters the defrosting stage after meeting conditions. The frosting stage refers to that the heat exchanger to be cleaned operates as the evaporator, causing the heat exchanger to be cleaned to operate as the evaporator to cause ice to form on a surface of the heat exchanger to be cleaned.

[0055] In some embodiments, when an ice layer on the surface of the heat exchanger to be cleaned can meet the cleaning requirement, the air conditioner **1000** enters the defrosting stage. For example, if the duration of the heat exchanger to be cleaned operating as the evaporator reaches a preset time, or the temperature of a coil pipe in the heat exchanger to be cleaned reaches a preset temperature, or the temperature of the coil pipe in the heat exchanger to be cleaned reaches the preset temperature and lasts for a preset time, or the indoor ambient temperature reaches a preset temperature and lasts for a preset time, it indicates that the ice on the surface of the heat exchanger to be cleaned is enough to meet the cleaning requirements, the controller **300** determines that the heat exchanger to be cleaned completes frosting, and the air conditioner **1000** enters the defrosting stage.

[0056] The defrosting stage refers to that the heat exchanger to be cleaned operates as the condenser, causing the ice on the surface of the heat exchanger to be cleaned to melt, thereby cleaning the heat exchanger to be cleaned.

[0057] In some embodiments, when the ice layer on the surface of the heat exchanger to be cleaned completely melts, the air conditioner **1000** exits the defrosting stage. For example, if the heat exchanger to be cleaned is defrosted for a preset time, or the temperature of the coil pipe in the heat exchanger to be cleaned reaches a preset temperature, or the temperature of the coil pipe in the heat exchanger to be cleaned reaches the preset temperature and lasts for a preset

time, the controller **300** determines that the heat exchanger to be cleaned completes defrosting, the air conditioner **1000** exits the defrosting stage, and thus the air conditioner **1000** completes cleaning of the heat exchanger to be cleaned.

[0058] In some embodiments, after receiving a self-cleaning instruction, the controller **300** caches the current operation state of the air conditioner **1000**, for example, the current temperature and humidity, etc.

[0059] In some embodiments, the controller **300** sends its operation state to the control device (such as the wander lead controller **301**, a mobile phone APP, the remote controller **302**, etc.).

[0060] In some embodiments, the display screen or the control device of the air conditioner **1000** displays operation mode information and operation state information of the air conditioner **1000**, where the operation mode information includes a “self-cleaning marking”, which is configured to remind a user that the air conditioner **1000** enters the self-cleaning mode at present.

[0061] In some embodiments, in the frosting stage, a blower (that is, the fan) stops working and lasts for a preset time. The frosting stage aims to cause ice to form on the surface of the heat exchanger to be cleaned, wind is generated if the fan works, and the wind affects the frosting efficiency, so that the blower does not work in the frosting stage. It should be noted that if the heat exchanger to be cleaned in the current stage is the indoor heat exchanger, the blower corresponds to the indoor blower, and if the heat exchanger to be cleaned in the current stage is the outdoor heat exchanger, the blower corresponds to the outdoor blower.

[0062] In some embodiments, in the frosting stage, the display screen or the control device of the air conditioner **1000** displays the “self-cleaning mode”, the “frosting stage”, etc.

[0063] In some embodiments, in the defrosting stage, the compressor stops working for a preset time.

[0064] In some embodiments, in the defrosting stage, the blower works at a low rotational speed, for example, the refrigeration wind low rotational speed and the refrigeration silent rotational speed.

[0065] In some embodiments, in the defrosting stage, the opening degree of the expansion valve is reduced.

[0066] In some embodiments, the heat exchanger to be cleaned is the outdoor heat exchanger **202** or the indoor heat exchanger **101**. The self-cleaning instruction includes a first instruction and a second instruction. The self-cleaning mode includes a first self-cleaning mode and a second self-cleaning mode.

[0067] FIG. 7 is a block diagram of a communication system of the air conditioner according to some embodiments. As shown in FIG. 1 and FIG. 7, a button **103** is arranged on the air conditioner **1000** (such as the indoor unit **100**), and the user can input the self-cleaning instruction by means of the button **103**. Or the user can also input the self-cleaning instruction by means of the control device such as the wander lead controller **301**, the mobile phone APP, the remote controller **302**, etc. Or the user can also preset the self-cleaning instruction that can be triggered in a timed manner in the air conditioner **1000**, thereby implementing the timed self-cleaning function of the air conditioner **1000**.

[0068] When the controller **300** receives the first instruction, the controller **300** controls the air conditioner **1000** to enter the first self-cleaning mode. Under such condition, the

indoor heat exchanger **101** is the heat exchanger to be cleaned, and the controller **300** can control the flow direction of the refrigerant by means of the four-way valve **205** to enable the flow direction of the refrigerant to be the same as the flow direction of the refrigerant in the refrigeration mode, causing the indoor heat exchanger **101** to operate as the evaporator to cause ice to form on the surface of the indoor heat exchanger **101**. When the ice meets the cleaning requirements, the controller **300** changes the flow direction of the refrigerant again by means of the four-way valve **205**, causing the indoor heat exchanger **101** to operate as the condenser, thereby defrosting the indoor heat exchanger **101**. It should be noted that the controller **300** receiving the first instruction to enter the first self-cleaning mode is taken as an example below to explain the specific implementation process in detail. Moreover, the controller **300** receiving the second instruction to enter the second self-cleaning mode is no longer elaborated.

[0069] When the controller **300** receives the second instruction, the controller **300** controls the air conditioner **1000** to enter the second self-cleaning mode. Under such condition, the outdoor heat exchanger **202** is the heat exchanger to be cleaned, and the controller **300** can control the flow direction of the refrigerant by means of the four-way valve **205** to enable the flow direction of the refrigerant to be the same as the flow direction of the refrigerant in the heating mode, causing the outdoor heat exchanger **202** to operate as the evaporator to cause ice to form on the surface of the outdoor heat exchanger **202**. When the ice meets the cleaning requirements, the controller **300** changes the flow direction of the refrigerant again by means of the four-way valve **205**, causing the outdoor heat exchanger **202** to operate as the condenser, thereby defrosting the outdoor heat exchanger **202**.

[0070] Step **12**, obtaining the indoor ambient temperature detected by the indoor temperature detection apparatus, and adjusting an operating parameter of the air conditioner **1000** according to the indoor ambient temperature.

[0071] In some embodiments, step **12** includes controlling working parameters of the compressor and the expansion valve according to the indoor ambient temperature, including: in the frosting stage, the controller **300** being capable of periodically obtaining the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, controlling the operating parameters of the compressor and/or the expansion valve to be subjected to first adjustment; and when it is determined that the indoor ambient temperature is lower than the critical temperature, controlling the operating parameters of the compressor and/or the expansion valve to be subjected to second adjustment.

[0072] In some embodiments, in the frosting stage, the controller **300** can periodically obtain the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, the frequency of the compressor and the opening degree of the expansion valve are controlled to be increased; and when it is determined that the indoor ambient temperature is lower than the critical temperature, the frequency of the compressor and the opening degree of the expansion valve are controlled to be reduced.

[0073] In some embodiments, in the frosting stage, the controller **300** can periodically obtain the indoor ambient temperature, and when it is determined that the indoor

[0086] In some embodiments, in the frosting stage, the controller 300 can periodically obtain the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, the frequency of the compressor and the opening degree of the expansion valve are controlled to be reduced; and when it is determined that the indoor ambient temperature is lower than the critical temperature, the frequency of the compressor or the opening degree of the expansion valve is controlled to be increased.

[0087] In some embodiments, in the frosting stage, the controller 300 can periodically obtain the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, the frequency of the compressor or the opening degree of the expansion valve is controlled to be reduced; and when it is determined that the indoor ambient temperature is lower than the critical temperature, the frequency of the compressor or the opening degree of the expansion valve is controlled to be increased.

[0088] In some embodiments, in the frosting stage, the controller 300 can periodically obtain the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, the frequency of the compressor and the opening degree of the expansion valve are controlled to be reduced; and when it is determined that the indoor ambient temperature is lower than the critical temperature, the frequency of the compressor and the opening degree of the expansion valve are controlled unchanged.

[0089] In some embodiments, in the frosting stage, the controller 300 can periodically obtain the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, the frequency of the compressor or the opening degree of the expansion valve is controlled to be reduced; and when it is determined that the indoor ambient temperature is lower than the critical temperature, the frequency of the compressor and the opening degree of the expansion valve are controlled unchanged.

[0090] In some embodiments, in the frosting stage, the controller 300 can periodically obtain the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, the frequency of the compressor and the opening degree of the expansion valve are controlled to be reduced; and when it is determined that the indoor ambient temperature is lower than the critical temperature, the frequency of the compressor or the opening degree of the expansion valve is controlled unchanged.

[0091] In some embodiments, in the frosting stage, the controller 300 can periodically obtain the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, the frequency of the compressor or the opening degree of the expansion valve is controlled to be reduced; and when it is determined that the indoor ambient temperature is lower than the critical temperature, the frequency of the compressor or the opening degree of the expansion valve is controlled unchanged.

[0092] In some embodiments, the critical temperature is preset according to requirements, for example, 0 DEG C.

[0093] In some embodiments, the adjustment range of the frequency of the compressor and the adjustment range of the opening degree of the expansion valve can be set according to actual conditions.

[0094] It should be noted that the increase or reduction amplitude of the frequency of the compressor and the opening degree of the expansion valve can be the same or different. In the same embodiment, before and after the indoor ambient temperature meets the critical temperature, if the frequency of the compressor or the opening degree of the expansion valve is adjusted in the same direction (for example, increased), the increase or reduction amplitude can be the same or different. Secondly, in the above embodiments, the frequency of the compressor and the opening degree of the expansion valve are taken as examples for illustration, it is not limited to control the two components only, and it is not limited to control the two parameters, namely, the frequency of the compressor or the opening degree of the expansion valve, only.

[0095] In some embodiments, step 12 includes controlling the air conditioner 1000 to switch the mode according to the indoor ambient temperature, specifically including the following content.

[0096] In some embodiments, in the frosting stage, the controller 300 can periodically obtain the indoor ambient temperature, and the controller 300 can adjust the operation mode of the air conditioner 1000 according to the indoor ambient temperature, thereby changing the operating parameters of the air conditioner 1000, so as to avoid the influence of the excessively large change of the indoor ambient temperature on the refrigeration or heating effect of the air conditioner 1000. For the different heat exchangers to be cleaned, the flow directions of the refrigerant in the frosting stage are different, causing the indoor ambient temperature to drop or rise in the frosting stage, and then the indoor ambient temperature is prone to being too low or too high. Thus, for the indoor ambient temperature, a first preset range, a second preset range and a third preset range are preset. The first preset range corresponds to the condition that the indoor ambient temperature is too low, the third preset range corresponds to the condition that the indoor ambient temperature is relatively suitable, and the second preset range corresponds to the condition that the indoor ambient temperature is too high.

[0097] When the indoor ambient temperature is in the first preset range, the indoor ambient temperature is very low, and it is urgent to increase the indoor ambient temperature. If the heat exchanger to be cleaned continues to operate as the evaporator to cause ice to form on the surface of the heat exchanger to be cleaned, the indoor ambient temperature gets lower and lower, which further affects the user experience. Thus, the air conditioner 1000 needs to be controlled to switch from the self-cleaning mode to a first mode (for example, the heating mode) or a first sub-mode (for example, a heating sub-mode).

[0098] When the indoor ambient temperature is in the second preset range, the indoor ambient temperature is very high, and it is urgent to reduce the indoor ambient temperature. Although in the self-cleaning mode, the heat exchanger to be cleaned operates as the evaporator, and refrigeration can also be realized, the blower stops working in the self-cleaning mode, and cool air generated by the air conditioner 1000 cannot be quickly blown into the room. Thus, the air conditioner 1000 needs to be controlled to switch

from the self-cleaning mode to a second mode (for example, the refrigeration mode) or a second sub-mode (for example, a refrigeration sub-mode).

[0099] When the indoor ambient temperature is in the third preset range, the air conditioner 1000 is controlled to switch from the self-cleaning mode to a third mode (for example, the heating mode) or a third sub-mode (for example, a heating sub-mode).

[0100] The first sub-mode, the second sub-mode and the third sub-mode are the heating sub-mode, the refrigeration sub-mode, a state maintaining sub-mode and an air supply sub-mode under the self-cleaning mode, or other sub-modes in which the air conditioner can operation under the self-cleaning mode (including, but are not limited to a “refrigeration freezing mode 1”, a “refrigeration freezing mode 2”, and a fresh air sub-mode).

[0101] The first mode, the second mode and the third mode may be the heating mode, the refrigeration mode, the state maintaining mode, the air supply mode, or other modes in which the air conditioner can operation (including, but are not limited to a fresh air mode).

[0102] FIG. 8 is another flow chart of execution steps of the controller of the air conditioner according to some embodiments. In some embodiments, as shown in FIG. 8, step 12 executed by the controller 300 includes step 121 to step 124.

[0103] In step 121, it is determined that the indoor ambient temperature is in the first preset range, and the air conditioner 1000 is controlled to switch from the self-cleaning mode to the first mode or the first sub-mode. Switching to the first mode is taken as an example below for explanation.

[0104] Under the condition that the indoor ambient temperature is in the first preset range, that is, the indoor ambient temperature is relatively low, the controller 300 needs to control the air conditioner 1000 to switch from the self-cleaning mode to the first mode, so as to increase the indoor ambient temperature and enable the indoor ambient temperature to rise again to a suitable temperature range. In the first mode, at least one of the operating frequency of the compressor 201 or the opening degree of the expansion valve 204 in the air conditioner 1000 is changed, thereby adjusting (for example, increasing) the indoor ambient temperature.

[0105] In some embodiments, the indoor blower operates at the automatic wind speed.

[0106] In step 122, it is determined that the operation duration of the air conditioner 1000 in the first mode is within a first preset duration and the indoor ambient temperature reaches a first preset temperature, and the air conditioner 1000 is controlled to switch from the first mode to the self-cleaning mode. The first preset temperature here is higher than the upper limit value of the first preset range.

[0107] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the indoor ambient temperature rises to the first preset temperature within the first preset duration, the heating effect of the air conditioner 1000 is relatively good, and the indoor ambient temperature can meet user's requirements. Under such condition, the controller 300 can control the air conditioner 1000 to switch from the first mode to the self-cleaning mode.

[0108] In some embodiments, after the air conditioner 1000 switches from the first mode to the self-cleaning mode, the controller 300 still monitors the indoor ambient tem-

perature by means of the corresponding indoor temperature detection apparatus, thereby controlling the air conditioner 1000 to switch to the first mode when the indoor ambient temperature is relatively low.

[0109] In step 123, it is determined that the operation duration of the air conditioner 1000 in the first mode is within the first preset duration and the indoor ambient temperature does not reach the first preset temperature, and the controller 300 controls the air conditioner 1000 to maintain the first mode.

[0110] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the indoor ambient temperature does not rise to the first preset temperature within the first preset duration, the indoor ambient temperature does not meet the user's requirements, and then the air conditioner 1000 needs to continue the first mode.

[0111] In step 124, it is determined that the operation duration of the air conditioner 1000 in the first mode exceeds the first preset duration and the indoor ambient temperature is lower than the first preset temperature, and the air conditioner 1000 is controlled to switch from the first mode to the self-cleaning mode, thereby continuing to clean the heat exchanger to be cleaned until the air conditioner 1000 completes cleaning of the heat exchanger to be cleaned.

[0112] In step 125, it is determined that the operation duration of the air conditioner 1000 in the first mode exceeds the first preset duration and the indoor ambient temperature reaches the first preset temperature, and the air conditioner 1000 is controlled to switch from the first mode to the self-cleaning mode, thereby continuing to clean the heat exchanger to be cleaned until the air conditioner 1000 completes cleaning of the heat exchanger to be cleaned.

[0113] In order to improve the self-cleaning efficiency while meeting the user's requirements, the first preset duration is preset. Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the operation duration of the air conditioner 1000 in the first mode exceeds the first preset duration, the heating effect of the air conditioner 1000 is not obvious. Under such condition, if the air conditioner 1000 continues the first mode, the indoor ambient temperature rises slowly, resulting in prolonging the self-cleaning time. Thus, under such condition, in order to improve the self-cleaning efficiency, after the operation duration of the air conditioner 1000 in the first mode exceeds the first preset duration, the controller 300 can control the air conditioner 1000 to switch from the first mode to the self-cleaning mode.

[0114] In some embodiments, after the air conditioner 1000 switches from the first mode to the self-cleaning mode, the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes a self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0115] Thus, the controller 300 can control the air conditioner 1000 to switch between the first mode and the self-cleaning mode according to the indoor ambient temperature, that is, when the operation duration of the air conditioner 1000 in the first mode reaches the first preset duration or the indoor ambient temperature reaches the first preset temperature, the air conditioner 1000 is controlled to switch from the first mode to the self-cleaning mode, so as to implement step 12.

[0116] In order to further improve the self-cleaning efficiency, the first preset range, the first preset temperature and the first preset duration are further defined.

[0117] In some embodiments, the first preset range includes a first sub-preset range and a second sub-preset range, and the upper limit value of the first sub-preset range is less than the lower limit value of the second sub-preset range. The first preset temperature includes a first sub-preset temperature and a second sub-preset temperature. The first preset duration includes a first heating time period and a second heating time period. The first sub-preset temperature and the first heating time period correspond to the first sub-preset range, and the second sub-preset temperature and the second heating time period correspond to the second sub-preset range. Thus, by dividing the first preset range into the first sub-preset range and the second sub-preset range, the self-cleaning efficiency can be further improved while the user's requirements are met.

[0118] In some embodiments, the first sub-preset temperature is equal to the second sub-preset temperature.

[0119] In some embodiments, the first sub-preset temperature is different from the second sub-preset temperature.

[0120] In some embodiments, the first heating time period can be equal to the second heating time period.

[0121] In some embodiments, the first heating time period can be different from the second heating time period.

[0122] In some embodiments, during the process that the air conditioner 1000 operates in the self-cleaning mode, if the indoor ambient temperature is within the first sub-preset range, the current indoor ambient temperature is relatively low, and the controller 300 controls the air conditioner 1000 to enter the first mode.

[0123] Under such condition, in order to shorten the self-cleaning time as much as possible and improve the self-cleaning efficiency while meeting the user's requirements, the first sub-preset temperature and the first heating time period are set, so as to control the operation duration and heating effect of the air conditioner 1000 in the first mode.

[0124] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the indoor ambient temperature rises to (be greater than or equal to) the first sub-preset temperature within the first heating time period, the heating effect of the air conditioner 1000 is relatively good, and the indoor ambient temperature can meet the user's requirements. Under such condition, the controller 300 can control the air conditioner 1000 to switch from the first mode to the self-cleaning mode.

[0125] Further, after the air conditioner 1000 switches from the first mode to the self-cleaning mode, the controller 300 still monitors the indoor ambient temperature by means of the indoor temperature detection apparatus 400, thereby timely controlling the air conditioner 1000 to switch to the first mode when the indoor ambient temperature is relatively low.

[0126] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the indoor ambient temperature does not reach the first sub-preset temperature within the first heating time period, it is indicated that the indoor ambient temperature does not meet the user's requirements, and the controller 300 controls the air conditioner 1000 to continue the first mode.

[0127] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the

operation duration of the air conditioner 1000 in the first mode exceeds the first heating time period and the indoor ambient temperature is still lower than the first sub-preset temperature, the heating effect of the air conditioner 1000 is not obvious. Under such condition, if the air conditioner 1000 continues the first mode, the indoor ambient temperature rises slowly, resulting in prolonging the self-cleaning time. Thus, under such condition, in order to improve the self-cleaning efficiency, after the operation duration of the air conditioner 1000 in the first mode exceeds the first heating time period, the controller 300 can control the air conditioner 1000 to switch from the first mode to the self-cleaning mode, and the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes a self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0128] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the operation duration of the air conditioner 1000 in the first mode exceeds the first heating time period and the indoor ambient temperature reaches the first sub-preset temperature, the air conditioner 1000 is controlled to switch from the first mode to the self-cleaning mode, thereby continuing to clean the heat exchanger to be cleaned until the air conditioner 1000 completes cleaning of the heat exchanger to be cleaned.

[0129] Similar to the control process of the indoor ambient temperature in the first sub-preset range, during the process that the air conditioner 1000 operates in the self-cleaning mode, if the indoor ambient temperature is within the second sub-preset range, the current indoor ambient temperature is relatively low, and the controller 300 controls the air conditioner 1000 to enter the first mode.

[0130] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the indoor ambient temperature rises to (be greater than or equal to) the second sub-preset temperature within the second heating time period, the controller 300 can control the air conditioner 1000 to switch from the first mode to the self-cleaning mode. After the air conditioner 1000 switches from the first mode to the self-cleaning mode, the controller 300 still monitors the indoor ambient temperature by means of the indoor temperature detection apparatus 400, thereby timely controlling the air conditioner 1000 to switch to the first mode when the indoor ambient temperature is relatively low.

[0131] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the indoor ambient temperature does not reach the second sub-preset temperature within the second heating time period, the controller 300 controls the air conditioner 1000 to maintain the first mode.

[0132] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the operation duration of the air conditioner 1000 in the first mode exceeds the second heating time period and the indoor ambient temperature is still lower than the second sub-preset temperature, the controller 300 can control the air conditioner 1000 to switch from the first mode to the self-cleaning mode, and the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0133] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the first mode, if the operation duration of the air conditioner 1000 in the first mode exceeds the second heating time period and the indoor ambient temperature reaches the second sub-preset temperature, the controller 300 can control the air conditioner 1000 to switch from the first mode to the self-cleaning mode, and the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0134] It should be noted that the temperature within the second sub-preset range is higher than the temperature within the first sub-preset range, if the first sub-preset temperature is equal to the second sub-preset temperature, the temperature within the second sub-preset range is controlled to rise to the second sub-preset temperature, compared with the action of controlling the temperature within the first sub-preset range to rise to the first sub-preset temperature, the control difficulty is relatively low, the needed control time is relatively short, and thus under such condition, the second heating time period is shorter than the first heating time period. Besides, for the detailed description of the control process within the second sub-preset range, reference can be made to the aforementioned control process within the first sub-preset range, and it is not elaborated here.

[0135] The heating time needed to recover the different indoor ambient temperatures to the temperature needed by the user is different, so that in some embodiments of the present disclosure, by dividing the first preset range into the first sub-preset range and the second sub-preset range, and setting the different sub-preset temperatures and heating time periods in correspondence to the different sub-preset ranges, the operation time of the air conditioner 1000 in the first mode can be shortened, and the self-cleaning efficiency can be further improved.

[0136] FIG. 9 is yet another flow chart of execution steps of the controller of the air conditioner according to some embodiments. The steps are illustrated below in conjunction with FIG. 9. As shown in FIG. 9, the controller 300 is configured to:

[0137] in step 13, when the air conditioner 1000 operates in the self-cleaning mode, obtain the indoor ambient temperature E;

[0138] in step 14, determine whether the indoor ambient temperature E is within the first sub-preset range; if so, execute step 15, otherwise, execute step 20;

[0139] in step 15, control the air conditioner 1000 to switch from the self-cleaning mode to the first mode;

[0140] in step 16, determine whether the indoor ambient temperature E reaches the first sub-preset temperature; if so, execute step 17, otherwise, execute step 18;

[0141] in step 17, control the air conditioner 1000 to switch from the first mode to the self-cleaning mode;

[0142] in step 18, determine whether the operation duration of the air conditioner 1000 in the first mode exceeds the first heating time period; if so, execute step 19, otherwise, return to step 16;

[0143] in step 19, control the air conditioner 1000 to switch from the first mode to the self-cleaning mode, and control the air conditioner 1000 to execute the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned;

[0144] in step 20, determine whether the indoor ambient temperature E is within the second sub-preset range; if so, execute step 21, otherwise, execute other control logic, for example, subsequent step 26 to step 38;

[0145] in step 21, control the air conditioner 1000 to switch from the self-cleaning mode to the first mode;

[0146] in step 22, determine whether the indoor ambient temperature E reaches the second sub-preset temperature; if so, execute step 23, otherwise, execute step 24;

[0147] in step 23, control the air conditioner 1000 to switch from the first mode to the self-cleaning mode;

[0148] in step 24, determine whether the operation duration of the air conditioner 1000 in the first mode exceeds the second heating time period; if so, execute step 25, otherwise, return to step 22; and

[0149] in step 25, control the air conditioner 1000 to switch from the first mode to the self-cleaning mode, and control the air conditioner 1000 to execute the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0150] It should be noted that different users have different requirements for the suitable indoor ambient temperature, so that the second sub-preset temperature can be selected according to the requirement of most users, and is not limited in the present disclosure. In addition, the first sub-preset temperature can be preset according to the actual requirements. The first heating time period may be the same as or different from the second heating time period, and the two heating time periods can be preset according to the actual requirements and are not limited in the present disclosure.

[0151] The first preset range corresponding to the excessively low indoor ambient temperature is taken as an example above for explanation. The second preset range corresponding to the excessively high indoor ambient temperature is taken as an example below for explanation.

[0152] FIG. 10 is yet another flow chart of execution steps of the controller of the air conditioner according to some embodiments. In some embodiments, as shown in FIG. 10, the controller 300 executes the following content.

[0153] In step 126, it is determined that the indoor ambient temperature is in the second preset range, and the air conditioner 1000 is controlled to switch from the self-cleaning mode to the second mode or the second sub-mode. Switching to the second mode is taken as an example below for detailed introduction.

[0154] Specifically, under the condition that the indoor ambient temperature is in the second preset range, the indoor ambient temperature is relatively high, the controller 300 needs to control the air conditioner 1000 to switch from the self-cleaning mode to the second mode, so as to reduce the indoor ambient temperature and enable the indoor ambient temperature to drop to a suitable temperature range. Besides, in order to improve the self-cleaning efficiency while meeting the user's requirements, the second preset duration is preset. In the second mode, at least one of the operating frequency of the compressor 201 or the opening degree of the expansion valve 204 in the air conditioner 1000 is changed, thereby adjusting (for example, reducing) the indoor ambient temperature.

[0155] In some embodiments, the indoor blower operates at the automatic wind speed.

[0156] In step 127, it is determined that the operation duration of the air conditioner 1000 in the second mode is

within the second preset duration and the indoor ambient temperature is lower than a second preset temperature, and the air conditioner 1000 is controlled to switch from the second mode to the self-cleaning mode. The second preset temperature here is lower than the lower limit value of the second preset range.

[0157] Specifically, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the indoor ambient temperature drops to (for example, be lower than) the second preset temperature within the second preset duration, the refrigeration effect of the air conditioner 1000 is relatively good, and the indoor ambient temperature can meet user's requirements. Under such condition, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode. After the air conditioner 1000 switches from the second mode to the self-cleaning mode, the controller 300 still monitors the indoor ambient temperature by means of the indoor temperature detection apparatus 400, thereby timely controlling the air conditioner 1000 to switch to the second mode when the indoor ambient temperature is relatively high.

[0158] In step 128, it is determined that the operation duration of the air conditioner 1000 in the second mode is within the second preset duration and the indoor ambient temperature does not reach the second preset temperature, and the controller 300 controls the air conditioner 1000 to maintain the second mode.

[0159] In step 129, it is determined that the operation duration of the air conditioner 1000 in the second mode exceeds the second preset duration and the indoor ambient temperature does not reach the second preset temperature, and the air conditioner 1000 is controlled to switch from the second mode to the self-cleaning mode, thereby continuing to clean the heat exchanger to be cleaned until the air conditioner 1000 completes cleaning of the heat exchanger to be cleaned.

[0160] In step 130, it is determined that the operation duration of the air conditioner 1000 in the second mode exceeds the second preset duration and the indoor ambient temperature is lower than the second preset temperature, and the air conditioner 1000 is controlled to switch from the second mode to the self-cleaning mode, thereby continuing to clean the heat exchanger to be cleaned until the air conditioner 1000 completes cleaning of the heat exchanger to be cleaned.

[0161] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the operation duration of the air conditioner 1000 in the second mode exceeds the second preset duration and the indoor ambient temperature is still higher than the second preset temperature, the refrigeration effect of the air conditioner 1000 is not obvious. Under such condition, if the air conditioner 1000 continues the second mode, the indoor ambient temperature drops slowly, resulting in prolonging the self-cleaning time. Thus, under such condition, in order to improve the self-cleaning efficiency, after the operation duration of the air conditioner 1000 in the second mode exceeds the second preset duration, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode. Besides, after the air conditioner 1000 switches from the second mode to the self-cleaning mode, the controller 300 no longer switches the operation modes according to the indoor ambient tem-

perature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0162] Thus, the controller 300 can control the air conditioner 1000 to switch between the second mode and the self-cleaning mode according to the indoor ambient temperature, that is, when the operation duration of the air conditioner 1000 in the second mode exceeds the second preset duration or the indoor ambient temperature is lower than the second preset temperature, the air conditioner 1000 is controlled to switch from the second mode to the self-cleaning mode, so as to implement step 12.

[0163] In order to further improve the self-cleaning efficiency, the second preset range, the second preset temperature and the second preset duration are further defined.

[0164] In some embodiments, the second preset range includes a third sub-preset range and a fourth sub-preset range, and the upper limit value of the third sub-preset range is less than the lower limit value of the fourth sub-preset range. The second preset temperature includes a third sub-preset temperature and a fourth sub-preset temperature. The second preset duration includes a first refrigeration time period and a second refrigeration time period. The third sub-preset temperature and the first refrigeration time period correspond to the third sub-preset range, and the fourth sub-preset temperature and the second refrigeration time period correspond to the fourth sub-preset range. Thus, by dividing the second preset range into the third sub-preset range and the fourth sub-preset range, the self-cleaning efficiency can be further improved while the user's requirements are met.

[0165] In some embodiments, the third sub-preset temperature may be equal to the fourth sub-preset temperature.

[0166] In some embodiments, the first sub-preset temperature, the second sub-preset temperature, the third sub-preset temperature and the fourth sub-preset temperature may be equal to one another.

[0167] During the process that the air conditioner 1000 operates in the self-cleaning mode, if the indoor ambient temperature is within the third sub-preset range, the current indoor ambient temperature is relatively high, and the controller 300 controls the air conditioner 1000 to enter the second mode. In some embodiments, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the indoor ambient temperature drops to (be lower than) the third sub-preset temperature within the first refrigeration time period, the refrigeration effect of the air conditioner 1000 is relatively good, and the indoor ambient temperature can meet user's requirements. Under such condition, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode.

[0168] Further, after the air conditioner 1000 switches from the second mode to the self-cleaning mode, the controller 300 still monitors the indoor ambient temperature by means of the indoor temperature detection apparatus 400, thereby timely controlling the air conditioner 1000 to switch to the second mode when the indoor ambient temperature is relatively high.

[0169] In some embodiments, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the indoor ambient temperature does not reach the third sub-preset temperature within the first refrigeration time period, the controller 300 controls the air conditioner 1000 to maintain the second mode.

[0170] In some embodiments, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the operation duration of the air conditioner 1000 in the second mode exceeds the first refrigeration time period and the indoor ambient temperature does not reach the third sub-preset temperature, the refrigeration effect of the air conditioner 1000 is not obvious. Under such condition, if the air conditioner 1000 continues the second mode, the indoor ambient temperature drops slowly, resulting in prolonging the self-cleaning time. Thus, under such condition, in order to improve the self-cleaning efficiency, after the operation duration of the air conditioner 1000 in the second mode exceeds the first refrigeration time period, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode, and the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0171] In some embodiments, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the operation duration of the air conditioner 1000 in the second mode exceeds the first refrigeration time period and the indoor ambient temperature is lower than the third sub-preset temperature, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode, and the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0172] Similar to the control process of the indoor ambient temperature in the third sub-preset range, during the process that the air conditioner 1000 operates in the self-cleaning mode, if the indoor ambient temperature is within the fourth sub-preset range, the current indoor ambient temperature is relatively high, and the controller 300 controls the air conditioner 1000 to enter the second mode.

[0173] Under such condition, in order to shorten the self-cleaning time as much as possible and improve the self-cleaning efficiency while meeting the user's requirements, the fourth sub-preset temperature and the second refrigeration time period are set, so as to control the operation duration and refrigeration effect of the air conditioner 1000 in the second mode.

[0174] In some embodiments, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the indoor ambient temperature drops to (be lower than) the fourth sub-preset temperature within the second refrigeration time period, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode.

[0175] In some embodiments, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the indoor ambient temperature does not reach the fourth sub-preset temperature within the second refrigeration time period, the controller 300 controls the air conditioner 1000 to maintain the second mode.

[0176] In some embodiments, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the operation duration of the air conditioner 1000 in the second mode exceeds the second refrigeration time period and the indoor ambient temperature does not reach the fourth sub-preset temperature, the con-

troller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode, and the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0177] In some embodiments, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the operation duration of the air conditioner 1000 in the second mode exceeds the second refrigeration time period and the indoor ambient temperature is lower than the fourth sub-preset temperature, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode, and the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0178] It should be noted that the temperature within the fourth sub-preset range is higher than the temperature within the third sub-preset range, if the third sub-preset temperature is equal to the fourth sub-preset temperature, the temperature within the third sub-preset range is controlled to drop to the third sub-preset temperature, compared with the action of controlling the temperature within the fourth sub-preset range to drop to the fourth sub-preset temperature, the control difficulty is relatively low, the needed control time is relatively short, and thus under such condition, the third heating time period is shorter than the fourth heating time period. Besides, for the detailed description of the control process within the fourth sub-preset range, reference can be made to the aforementioned control process within the third sub-preset range, and it is not elaborated here.

[0179] The refrigeration time needed to recover the different indoor ambient temperatures to the temperature needed by the user is different, so that in some embodiments of the present disclosure, by dividing the second preset range into the third sub-preset range and the fourth sub-preset range, and setting the different sub-preset temperatures and refrigeration time periods in correspondence to the different sub-preset ranges, the operation time of the air conditioner 1000 in the second mode can be shortened, and the self-cleaning efficiency can be further improved.

[0180] FIG. 11 is yet another flow chart of execution steps of the controller of the air conditioner according to some embodiments. The steps are illustrated below in conjunction with FIG. 11. As shown in FIG. 11, the controller 300 is further configured to:

[0181] in step 26, when the air conditioner 1000 operates in the self-cleaning mode, obtain the indoor ambient temperature E;

[0182] in step 27, determine whether the indoor ambient temperature E is within the third sub-preset range; if so, execute step 28, otherwise, execute step 33;

[0183] in step 28, control the air conditioner 1000 to switch from the self-cleaning mode to the second mode;

[0184] in step 29, determine whether the indoor ambient temperature E is lower than the third sub-preset temperature; if so, execute step 30, otherwise, execute step 31;

[0185] in step 30, control the air conditioner 1000 to switch from the second mode to the self-cleaning mode;

[0186] in step 31, determine whether the operation duration of the air conditioner 1000 in the second mode exceeds the first refrigeration time period; if so, execute step 32, otherwise, return to step 29;

[0187] in step 32, control the air conditioner 1000 to switch from the second mode to the self-cleaning mode, and control the air conditioner 1000 to execute the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned;

[0188] in step 33, determine whether the indoor ambient temperature E is within the fourth sub-preset range; if so, execute step 34, otherwise, execute other control logic, for example, above step 13 to step 25;

[0189] in step 34, control the air conditioner 1000 to switch from the self-cleaning mode to the second mode;

[0190] in step 35, determine whether the indoor ambient temperature E is lower than the fourth sub-preset temperature; if so, execute step 36, otherwise, execute step 37;

[0191] in step 36, control the air conditioner 1000 to switch from the second mode to the self-cleaning mode;

[0192] in step 37, determine whether the operation duration of the air conditioner 1000 in the second mode exceeds the second refrigeration time period; if so, execute step 38, otherwise, return to step 35; and

[0193] in step 38, control the air conditioner 1000 to switch from the second mode to the self-cleaning mode, and control the air conditioner 1000 to execute the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[0194] It should be noted that reference can be made to the related description above for the second sub-preset temperature, which will not be repeated here. The third sub-preset temperature can be preset according to the actual requirements. The first refrigeration time period may be the same as or different from the second refrigeration time period, and the two refrigeration time periods can be set according to the actual requirements and are not limited in the present disclosure.

[0195] FIG. 12 is yet another flow chart of execution steps of the controller of the air conditioner according to some embodiments.

[0196] In some embodiments, as shown in FIG. 12, the controller 300 is further configured to execute step 401 to step 403.

[0197] In step 401, it is determined that the heat exchanger to be cleaned is the indoor heat exchanger 101.

[0198] In step 402, if the indoor fan 102 is in an on state at the moment immediately before the air conditioner 1000 switches to the self-cleaning mode, the indoor fan 102 is controlled to continue to operate for a first target duration. Here, after the indoor fan 102 operates for the first target duration, the controller 300 can control the indoor fan 102 to be turned off.

[0199] In step 403, if the air conditioner 1000 is in an off state at the moment immediately before the air conditioner 1000 switches to the self-cleaning mode, the air conditioner 1000 is controlled to continue to be turned off for a second target duration, after the duration of the air conditioner 1000 in the off state reaches the second target duration, the indoor heat exchanger 101 is controlled to operate as the evaporator

for a third target duration, and the indoor fan 102 is controlled to be turned on and operate for the third target duration.

[0200] During the frosting process of the heat exchanger to be cleaned (for example, the indoor heat exchanger 101), the air circulation effect can be improved by means of transient turning-on of the indoor fan 102, and moisture in the air can pass through the heat exchanger to be cleaned, so that more ice can form on the surface of the heat exchanger to be cleaned, and the cleaning of the heat exchanger to be cleaned is facilitated.

[0201] An electric motor of the indoor fan 102 may be a PG electric motor with a Hall element, and is provided with a rotational speed feedback circuit to feed back the rotational speed of the indoor fan 102. The controller 300 can obtain the rotational speed of the indoor fan 102 by means of the rotational speed feedback circuit, so as to determine whether the indoor fan 102 operates. Of course, the controller 300 can determine whether the indoor fan 102 operates in other ways.

[0202] FIG. 13 is yet another flow chart of execution steps of the controller of the air conditioner according to some embodiments. The steps are illustrated below in conjunction with FIG. 13. As shown in FIG. 13, the controller 300 is further configured to execute step 39 to step 45.

[0203] In step 39, the operation mode of the air conditioner 1000 is obtained.

[0204] In step 40, whether the operation mode of the air conditioner 1000 is the first self-cleaning mode is determined. If so, step 41 is executed, otherwise, the operation returns to step 39.

[0205] The controller 300 can determine the operation mode of the air conditioner 1000 according to the received self-cleaning instruction, thereby determining whether the heat exchanger to be cleaned is the indoor heat exchanger 101 or the outdoor heat exchanger 202.

[0206] In step 41, the operation state of the indoor fan 102 at the moment immediately before the air conditioner 1000 switches to the self-cleaning mode is obtained.

[0207] In step 42, whether the indoor fan 102 is in the on state at the moment immediately before the air conditioner 1000 switches to the self-cleaning mode is determined. If so, step 43 is executed, otherwise, step 44 is executed.

[0208] In step 43, the indoor fan 102 is controlled to continue to operate for the first target duration.

[0209] In step 44, whether the air conditioner 1000 is in the off state at the moment immediately before the air conditioner 1000 switches to the self-cleaning mode is determined. If so, step 45 is executed, otherwise, the remaining control logic is executed, for example, step 13 to step 25.

[0210] In step 45, the air conditioner 1000 is controlled to continue to be turned off for the second target duration, after the duration of the air conditioner 1000 in the off state reaches the second target duration, the indoor heat exchanger 101 is controlled to operate as the evaporator for the third target duration, and the indoor fan 102 is controlled to be turned on and operate for the third target duration.

[0211] In some embodiments, after the temperature is within the first preset range and the air conditioner switches from the self-cleaning mode to the first mode, or after the temperature is within the second preset range and the air conditioner switches from the self-cleaning mode to the second mode, the controller 300 is further configured to:

[0212] control the working parameters of the compressor and the expansion valve according to the indoor ambient temperature. Specifically, in the frosting stage, the controller 300 can periodically obtain the indoor ambient temperature, and when it is determined that the indoor ambient temperature reaches the critical temperature, it controls the operating parameters of the compressor and/or the expansion valve to be subjected to first adjustment; and when it is determined that the indoor ambient temperature is lower than the critical temperature, it controls the operating parameters of the compressor and/or the expansion valve to be subjected to second adjustment. It should be noted that reference is made to aforementioned related descriptions for different control solutions under different conditions, which are not elaborated here.

[0213] In some embodiments, step 12 includes controlling the air conditioner 1000 to use different refrigeration freezing modes according to the indoor ambient temperature, including:

[0214] when the indoor ambient temperature is within a fourth preset range, controlling the air conditioner 1000 to use the first refrigeration freezing mode (also referred to as the refrigeration freezing mode 1); and

[0215] when the indoor ambient temperature is within a fifth preset range, controlling the air conditioner 1000 to use the second refrigeration freezing mode (also referred to as the refrigeration freezing mode 2).

[0216] In some embodiments, the fourth preset range includes: the first sub-preset range or the third sub-preset range.

[0217] In some embodiments, the fifth preset range includes: the second sub-preset range or the fourth sub-preset range.

[0218] In some embodiments, the third preset range is divided into a fifth sub-preset range and a sixth sub-preset range, the upper limit value of the fifth sub-preset range is less than the lower limit value of the sixth sub-preset range, then the fourth preset range includes the fifth sub-preset range, and the fifth preset range includes the sixth sub-preset range.

[0219] In some embodiments, in the first refrigeration freezing mode, the controller 300 is configured to:

[0220] control the frequency of the compressor to be a first frequency, or control the wind speed of the blower to be a first wind speed, or control the opening degree of the expansion valve to be a first opening degree, and operate for a preset time;

[0221] obtain the indoor ambient temperature, and if the indoor ambient temperature is higher than or equal to the critical temperature, control the operating parameters of the compressor and the expansion valve of the air conditioner to be subjected to third adjustment; if the indoor ambient temperature is lower than the critical temperature, control the operating parameters of the compressor and the expansion valve of the air conditioner to be subjected to fourth adjustment; and

[0222] when the frosting quitting condition is met, control the rotational speed wind speed of the blower to be a second wind speed (also referred to as a refrigeration low wind rotational speed).

[0223] In some embodiments, in the second refrigeration freezing mode, the controller 300 is configured to:

[0224] control the frequency of the compressor to be a first frequency, or control the wind speed of the blower

to be a first wind speed, or control the opening degree of the expansion valve to be a first opening degree, and operate for a preset time;

[0225] obtain the indoor ambient temperature, and if the indoor ambient temperature is higher than or equal to the critical temperature, control the operating parameters of the compressor and the expansion valve of the air conditioner to be subjected to third adjustment; if the indoor ambient temperature is lower than the critical temperature, control the operating parameters of the compressor and the expansion valve of the air conditioner to be subjected to fourth adjustment; and

[0226] when the frosting quitting condition is met, control the rotational speed wind speed of the blower to be a third wind speed (also referred to as a refrigeration silent rotational speed).

[0227] In some embodiments, the frosting quitting condition includes:

[0228] the indoor coil pipe temperature is lower than a first set temperature and lasts for a first time; or

[0229] the indoor ambient temperature is lower than a second set temperature and lasts for a second time; or

[0230] the duration of the self-cleaning mode reaches a third time.

[0231] FIG. 14 is yet another flow chart of execution steps of the controller of the air conditioner according to some embodiments. In some embodiments, as shown in FIG. 14, the controller 300 is further configured to execute step 46.

[0232] In step 46, after the heat exchanger to be cleaned completes frosting, the air conditioner 1000 is controlled to cause the heat exchanger to be cleaned to operate as the condenser to enable the heat exchanger to be cleaned to be defrosted. Here, the opening degree of the expansion valve 204 during defrosting of the heat exchanger to be cleaned is less than or equal to the opening degree of the expansion valve 204 during frosting of the heat exchanger to be cleaned. Thus, during the defrosting process of the heat exchanger to be cleaned, the refrigerant may have a relatively high temperature, which facilitates defrosting of the surface of the heat exchanger to be cleaned.

[0233] FIG. 15 is yet another flow chart of execution steps of the controller of the air conditioner according to some embodiments. In some embodiments, as shown in FIG. 15, the controller 300 is further configured to execute step 1210.

[0234] In step 1210, before the air conditioner 1000 switches between the first mode and the first self-cleaning mode, or before the air conditioner 1000 switches between the second mode and the second self-cleaning mode, the air conditioner 1000 is controlled to be shut down for a fourth target duration.

[0235] When the air conditioner 1000 is in a frosting stage of the first self-cleaning mode, the flow direction of the refrigerant is the same as the flow direction of the refrigerant in the refrigeration mode. When the air conditioner 1000 is in a frosting stage of the second self-cleaning mode, the flow direction of the refrigerant is the same as the flow direction of the refrigerant in the heating mode. When the air conditioner 1000 switches between the first mode and the first self-cleaning mode, or between the second mode and the second self-cleaning mode, the flow direction of the refrigerant may be changed, so that in order to protect the air conditioner 1000, the controller 300 needs to control the air conditioner 1000 to be shut down when switching the mode.

[0236] In the air conditioner 1000 in some embodiments of the present disclosure, the indoor temperature detection apparatus 400 is arranged indoors to detect the indoor ambient temperature, and the indoor ambient temperature can be monitored in real time when the air conditioner 1000 operates in the self-cleaning mode, so that the operating parameters of the air conditioner 1000 can be adjusted according to the indoor ambient temperature, and the influence of the excessively large change in the indoor ambient temperature during the self-cleaning process on the refrigeration or heating effect of the air conditioner 1000 can be avoided.

[0237] FIG. 16 is a workflow chart of execution steps of the controller of the air conditioner according to some embodiments.

[0238] In some embodiments, as shown in FIG. 16, the controller is configured to execute steps S11-S12:

[0239] S11. controlling the air conditioner to enter the self-cleaning mode in response to the self-cleaning instruction, causing the heat exchanger to be cleaned to implement a function as the evaporator to perform frosting treatment, where the heat exchanger to be cleaned is the outdoor heat exchanger or the indoor heat exchanger;

[0240] specifically, the self-cleaning instruction includes an indoor heat exchanger self-cleaning instruction and an outdoor heat exchanger self-cleaning instruction. When the self-cleaning instruction is the indoor heat exchanger self-cleaning instruction, self-cleaning is performed on the indoor heat exchanger, and the indoor heat exchanger implements the function as the evaporator first to perform frosting treatment, causing ice to form on the surface of the indoor heat exchanger. When the self-cleaning instruction is the outdoor heat exchanger self-cleaning instruction, self-cleaning is performed on the outdoor heat exchanger, and the outdoor heat exchanger implements the function as the evaporator first to perform frosting treatment, causing ice to form on the surface of the outdoor heat exchanger.

[0241] S12. adjusting the operating parameters of the air conditioner according to the indoor ambient temperature.

[0242] Specifically, the self-cleaning mode includes the frosting stage and the defrosting stage, when the air conditioner enters the self-cleaning mode and operates for a period of time, a system operates stably and is in the frosting stage, and the operating parameters of the air conditioner are adjusted according to the monitored indoor ambient temperature, so that the discomfort of the user caused by the excessively large change in the indoor ambient temperature is avoided, and the user experience is improved.

[0243] In some embodiments, the controller is further configured to: control the air conditioner to switch from the self-cleaning mode to the heating mode when the indoor ambient temperature is within a preset low-temperature interval (also referred to as the first preset range). In a preset heating time period after the air conditioner enters the heating mode, if the indoor ambient temperature is detected to reach a preset target temperature, control the air conditioner to switch from the heating mode to the self-cleaning mode, where the preset target temperature is higher than the maximum value of the preset low-temperature interval. In the heating mode, when the indoor ambient temperature does not reach the preset target temperature and the current

time exceeds the preset heating time period, control the air conditioner to switch from the heating mode to the self-cleaning mode, causing the air conditioner to execute self-cleaning until completing same.

[0244] In some embodiments, adjusting the operating parameters of the air conditioner according to the indoor ambient temperature specifically includes controlling the air conditioner to switch between the heating mode and the self-cleaning mode according to the indoor ambient temperature. Specifically, the condition that the indoor ambient temperature is relatively low is considered, when the indoor ambient temperature is within the preset low-temperature interval, it is indicated that the temperature is relatively low, and the comfort of the user is relatively poor, so that the air conditioner switches from the self-cleaning mode to the heating mode to increase the indoor ambient temperature, causing the indoor ambient temperature to rise to a comfort temperature interval of the user.

[0245] In some embodiments, to balance the user comfort and the self-cleaning efficiency, the preset heating time period is set, that is, when the air conditioner switches to the heating mode, it enters the preset heating time period, if the indoor ambient temperature is detected to rise to the preset target temperature within this time period, it is indicated that the heating effect is good, the temperature at which the user is relatively comfortable is reached, then the air conditioner can switch from the heating mode to the self-cleaning mode to continue self-cleaning, and after it switches to the self-cleaning mode, the indoor ambient temperature is still monitored, so as to ensure that the air conditioner switches to the heating mode when the temperature is relatively low. If the indoor ambient temperature still cannot reach the preset target temperature after the preset heating time period ends, it is indicated that the heating effect is not obvious, if the heating mode continues, the rising effect of the indoor ambient temperature is also poor, and the self-cleaning period is prolonged, so that under such condition, in order to ensure the self-cleaning efficiency, after the preset heating time period ends, the air conditioner switches from the heating mode to the self-cleaning mode, in this round of self-cleaning, mode switching according to the indoor ambient temperature is avoided, but the self-cleaning task is directly executed until self-cleaning is completed.

[0246] In some embodiments, the preset low-temperature interval includes a first preset low-temperature interval (also referred to as the first sub-preset range) and a second preset low-temperature interval (also referred to as the second sub-preset range), and the maximum value of the first preset low-temperature interval is less than the minimum value of the second preset low-temperature interval. The preset target temperature at least includes a low suitable temperature and an ideal temperature, the low suitable temperature is lower than the ideal temperature, and the low suitable temperature is higher than the maximum value of the second preset low-temperature interval. The preset heating time period includes a first heating time period and a second heating time period. The preset target temperature corresponding to the first preset low-temperature interval is the low suitable temperature, the preset target temperature corresponding to the second preset low-temperature interval is the ideal temperature, the preset heating time period corresponding to the first preset low-temperature interval is the first heating

time period, and the preset heating time period corresponding to the second preset low-temperature interval is the second heating time period.

[0247] Specifically, to more accurately balance the self-cleaning efficiency and the user experience, the preset low-temperature interval is divided into the first preset low-temperature interval and the second preset low-temperature interval, and the maximum value of the first preset low-temperature interval is less than the minimum value of the second preset low-temperature interval.

[0248] When the air conditioner stably operates in the self-cleaning mode, if the indoor ambient temperature is detected to be within the first preset low-temperature interval, it is indicated that the temperature is relatively low now and is low relative to the second preset low-temperature interval. In order to shorten the self-cleaning period as much as possible while improving the user experience, the preset target temperature is set as the low suitable temperature, the low suitable temperature is lower than the ideal temperature, and the air conditioner can switch to the heating mode to increase the temperature. On this basis, in some embodiments, to balance the user comfort and the self-cleaning efficiency, the first heating time period is set, that is, when the air conditioner switches to the heating mode, it enters the first heating time period, if the indoor ambient temperature is detected to rise to the low suitable temperature within this time period, it is indicated that the heating effect is good, the temperature at which the user is relatively comfortable is reached, then the air conditioner can switch from the heating mode to the self-cleaning mode to continue self-cleaning, and after it switches to the self-cleaning mode, the indoor ambient temperature is still monitored, so as to ensure that the air conditioner switches to the heating mode when the temperature is relatively low. If the indoor ambient temperature still cannot reach the low suitable temperature after the first heating time period ends, it is indicated that the heating effect is not obvious, if the heating mode continues, the rising effect of the indoor ambient temperature is also poor, and the self-cleaning period is prolonged, so that under such condition, in order to ensure the self-cleaning efficiency, after the first heating time period ends, the air conditioner switches from the heating mode to the self-cleaning mode, in this round of self-cleaning, mode switching according to the indoor ambient temperature is avoided, but the self-cleaning task is executed until self-cleaning is completed.

[0249] When the air conditioner stably operates in the self-cleaning mode, if the indoor ambient temperature is detected to be within the second preset low-temperature interval, it is indicated that the temperature is relatively low now but is high relative to the first preset low-temperature interval, and the difficulty in increasing the temperature to the ideal temperature is relatively low, where the ideal temperature is higher than the low suitable temperature. Thus, in order to better improve the user experience, the preset target temperature is set as the ideal temperature, and the air conditioner switches to the heating mode to increase the temperature. On this basis, to better balance the user comfort and the self-cleaning efficiency, the second heating time period is set, that is, when the air conditioner switches to the heating mode, it enters the second heating time period, if the indoor ambient temperature is detected to rise to the ideal temperature within this time period, it is indicated that the heating effect is good, the temperature at which the user is most comfortable is reached, then the air conditioner can

switch from the heating mode to the self-cleaning mode to continue self-cleaning, and after it switches to the self-cleaning mode, the indoor ambient temperature is still monitored, so as to ensure that the air conditioner switches to the heating mode when the temperature is relatively low. If the indoor ambient temperature still cannot reach the ideal temperature after the second heating time period ends, it is indicated that the heating effect is not obvious, if the heating mode continues, the rising effect of the indoor ambient temperature is also poor, and the self-cleaning period is prolonged, so that under such condition, in order to ensure the self-cleaning efficiency, after the second heating time period ends, the air conditioner switches from the heating mode to the self-cleaning mode, in this round of self-cleaning, mode switching according to the indoor ambient temperature is avoided, but the self-cleaning task is directly executed until self-cleaning is completed.

[0250] It is worth noting that the most comfortable temperatures for different users are different, and the ideal temperature may be the relatively comfortable temperature selected on the basis of the requirements of most people, and is not defined here. The low suitable temperature is also the relatively comfortable temperature for human bodies, and can be preset by manufacturers specifically. The specific durations of the first heating time period and the second heating time period may be the same or different, are set by the manufacturers according to the actual requirements, and are not defined here.

[0251] FIG. 17 is another workflow chart of execution steps of the controller of the air conditioner according to some embodiments.

[0252] In some embodiments, as shown in FIG. 17, the controller is configured to execute steps S13-S25:

[0253] S13. when the air conditioner stably operates in the self-cleaning mode, obtaining the indoor ambient temperature E, and then entering step S14;

[0254] S14. determining whether the indoor ambient temperature E is within the first preset low-temperature interval, that is, $E < L$, if so, entering step S15, otherwise, entering S20;

[0255] S15. controlling the air conditioner to switch from the self-cleaning mode to the heating mode, and then entering step S16;

[0256] S16. determining whether the indoor ambient temperature E reaches the low suitable temperature U-M and the current time is within the first heating time period, if so, entering step S17, otherwise, entering step S18;

[0257] S17. controlling the air conditioner to switch from the heating mode to the self-cleaning mode;

[0258] S18. determining whether the current time exceeds the first heating time period, if so, entering step S19, otherwise, returning to step S16;

[0259] S19. controlling the air conditioner to switch from the heating mode to the self-cleaning mode, causing the air conditioner to execute the self-cleaning task until self-cleaning is completed;

[0260] S20. determining whether the indoor ambient temperature E is within the second preset low-temperature interval, that is, $L \leq E < U-M$, if so, entering step S21, otherwise, entering the remaining control logic, for example, subsequent steps S26-S38;

[0261] S21. controlling the air conditioner to switch from the self-cleaning mode to the heating mode, and then entering step S22;

[0262] S22. determining whether the indoor ambient temperature E reaches the ideal temperature U and the current time is within the second heating time period, if so, entering step S23, otherwise, entering step S24;

[0263] S23. controlling the air conditioner to switch from the heating mode to the self-cleaning mode;

[0264] S24. determining whether the current time exceeds the second heating time period, if so, entering step S25, otherwise, returning to step S22; and

[0265] S25. controlling the air conditioner to switch from the heating mode to the self-cleaning mode, causing the air conditioner to execute the self-cleaning task until self-cleaning is completed.

[0266] In some embodiments, the controller is further configured to: control the air conditioner to switch from the self-cleaning mode to the refrigeration mode when the indoor ambient temperature is within a preset high-temperature interval (also referred to as the second preset range); in a preset refrigeration time period after the air conditioner enters the refrigeration mode, if the indoor ambient temperature is detected to reach a preset target temperature, control the air conditioner to switch from the refrigeration mode to the self-cleaning mode, where the preset target temperature is lower than the minimum value of the preset high-temperature interval; and in the refrigeration mode, when the indoor ambient temperature does not reach the preset target temperature and the current time exceeds the preset refrigeration time period, control the air conditioner to switch from the refrigeration mode to the self-cleaning mode, causing the air conditioner to execute self-cleaning until completing same.

[0267] Specifically, adjusting the operating parameters of the air conditioner according to the indoor ambient temperature includes: controlling the air conditioner to switch between the refrigeration mode and the self-cleaning mode according to the indoor ambient temperature. Specifically, the condition that the indoor ambient temperature is relatively high is considered, when the indoor ambient temperature is within the preset high-temperature interval, it is indicated that the temperature is relatively high, and the comfort level of the user is relatively poor, so that the air conditioner switches from the self-cleaning mode to the refrigeration mode to reduce the indoor ambient temperature, causing the indoor ambient temperature to drop to a comfort temperature interval of the user. In some embodiments, to balance the user comfort and the self-cleaning efficiency, the preset refrigeration time period is set, that is, when the air conditioner switches to the refrigeration mode, it enters the preset refrigeration time period, if the indoor ambient temperature is detected to drop to the preset target temperature within this time period, it is indicated that the refrigeration effect is good, the temperature at which the user is relatively comfortable is reached, then the air conditioner can switch from the refrigeration mode to the self-cleaning mode to continue self-cleaning, and after it switches to the self-cleaning mode, the indoor ambient temperature is still monitored, so as to ensure that the air conditioner switches to the refrigeration mode when the temperature is relatively high. If the indoor ambient temperature still cannot reach the preset target temperature after the preset refrigeration time period ends, it is indicated that the refrigeration effect is not

obvious, if the refrigeration mode continues, the dropping effect of the indoor ambient temperature is also poor, and the self-cleaning period is prolonged, so that under such condition, in order to ensure the self-cleaning efficiency, after the preset refrigeration time period ends, the air conditioner switches from the refrigeration mode to the self-cleaning mode, in this round of self-cleaning, mode switching according to the indoor ambient temperature is avoided, but the self-cleaning task is directly executed until self-cleaning is completed.

[0268] In some embodiments, the preset high-temperature interval includes a first preset high-temperature interval (also referred to as the third sub-preset range) and a second preset high-temperature interval (also referred to as the fourth sub-preset range), and the maximum value of the first preset high-temperature interval is less than the minimum value of the second preset high-temperature interval. The preset target temperature at least includes a high suitable temperature and an ideal temperature, the high suitable temperature is higher than the ideal temperature, and the high suitable temperature is lower than the minimum value of the first preset high-temperature interval. The preset refrigeration time period includes a first refrigeration time period and a second refrigeration time period. The preset target temperature corresponding to the first preset high-temperature interval is the ideal temperature, the preset target temperature corresponding to the second preset high-temperature interval is the high suitable temperature, the preset refrigeration time period corresponding to the first preset high-temperature interval is the first refrigeration time period, and the preset refrigeration time period corresponding to the second preset high-temperature interval is the second refrigeration time period.

[0269] Specifically, to more accurately balance the self-cleaning efficiency and the user experience, the preset high-temperature interval is divided into the first preset high-temperature interval and the second preset high-temperature interval, and the maximum value of the first preset high-temperature interval is less than the minimum value of the second preset high-temperature interval.

[0270] When the air conditioner stably operates in the self-cleaning mode, if the indoor ambient temperature is detected to be within the first preset low-temperature interval, it is indicated that the temperature is relatively high now but is low relative to the second preset low-temperature interval, and the difficulty in reducing the temperature to the ideal temperature is relatively low. Thus, in order to better improve the user experience, the preset target temperature is set as the ideal temperature, and the air conditioner switches to the refrigeration mode to reduce the indoor ambient temperature. On this basis, to better balance the user comfort and the self-cleaning efficiency, the first refrigeration time period is set, that is, when the air conditioner switches to the refrigeration mode, it enters the first refrigeration time period. If the indoor ambient temperature is detected to drop to the ideal temperature within this time period, it is indicated that the refrigeration effect is good, the temperature at which the user is most comfortable is reached, then the air conditioner can switch from the refrigeration mode to the self-cleaning mode to continue self-cleaning, and after it switches to the self-cleaning mode, the indoor ambient temperature is still monitored, so as to ensure that the air conditioner switches to the refrigeration mode when the temperature is relatively high. If the indoor ambient tem-

perature still cannot drop to the ideal temperature after the first refrigeration time period ends, it is indicated that the refrigeration effect is not obvious, if the refrigeration mode continues, the dropping effect of the indoor ambient temperature is also poor, and the self-cleaning period is prolonged, so that under such condition, in order to ensure the self-cleaning efficiency, after the first refrigeration time period ends, the air conditioner switches from the refrigeration mode to the self-cleaning mode, in this round of self-cleaning, mode switching according to the indoor ambient temperature is avoided, but the self-cleaning task is directly executed until self-cleaning is completed.

[0271] When the air conditioner stably operates in the self-cleaning mode, if the indoor ambient temperature is detected to be within the second preset low-temperature interval, it is indicated that the temperature is relatively high now and is high relative to the first preset low-temperature interval. In order to shorten the self-cleaning period as much as possible while improving the user experience, the preset target temperature is set as the high suitable temperature, the high suitable temperature is higher than the ideal temperature, and the air conditioner can switch to the refrigeration mode to reduce the temperature. On this basis, in some embodiments, to balance the user comfort and the self-cleaning efficiency, the second refrigeration time period is set, that is, when the air conditioner switches to the refrigeration mode, it enters the second refrigeration time period. If the indoor ambient temperature is detected to drop to the high suitable temperature within this time period, it is indicated that the refrigeration effect is good, the temperature at which the user is relatively comfortable is reached, then the air conditioner can switch from the refrigeration mode to the self-cleaning mode to continue self-cleaning, and after it switches to the self-cleaning mode, the indoor ambient temperature is still monitored, so as to ensure that the air conditioner switches to the refrigeration mode when the temperature is relatively high. If the indoor ambient temperature still cannot reach the high suitable temperature after the first refrigeration time period ends, it is indicated that the refrigeration effect is not obvious, if the refrigeration mode continues, the dropping effect of the indoor ambient temperature is also poor, and the self-cleaning period is prolonged, so that under such condition, in order to ensure the self-cleaning efficiency, after the second refrigeration time period ends, the air conditioner switches from the refrigeration mode to the self-cleaning mode, in this round of self-cleaning, mode switching according to the indoor ambient temperature is avoided, but the self-cleaning task is directly executed until self-cleaning is completed.

[0272] It is worth noting that the most comfortable temperatures for different users are different, and the ideal temperature may be the relatively comfortable temperature selected on the basis of the requirements of most people, and is not defined here. The high suitable temperature is also the relatively comfortable temperature for human bodies, and can be preset by manufacturers specifically. The specific durations of the first refrigeration time period and the second refrigeration time period may be the same or different, are set by the manufacturers according to the actual requirements, and are not defined here.

[0273] FIG. 18 is another workflow chart of execution steps of the controller of the air conditioner according to some embodiments.

[0274] In some embodiments, as shown in FIG. 18, the controller is configured to execute steps S26-S38:

[0275] S26. when the air conditioner stably operates in the self-cleaning mode, obtaining the indoor ambient temperature E, and then entering step S27;

[0276] S27. determining whether the indoor ambient temperature E is within the first preset high-temperature interval, that is, $U+N \leq E < H$, where N is a preset positive number; if so, entering step S28, otherwise, entering S33;

[0277] S28. controlling the air conditioner to switch from the self-cleaning mode to the refrigeration mode, and then entering step S29;

[0278] S29. determining whether the indoor ambient temperature E reaches the ideal temperature U and the current time is within the first refrigeration time period, if so, entering step S30, otherwise, entering step S31;

[0279] S30. controlling the air conditioner to switch from the refrigeration mode to the self-cleaning mode;

[0280] S31. determining whether the current time exceeds the first refrigeration time period, if so, entering step S32, otherwise, returning to step S29;

[0281] S32. controlling the air conditioner to switch from the refrigeration mode to the self-cleaning mode, causing the air conditioner to execute the self-cleaning task until self-cleaning is completed;

[0282] S33. determining whether the indoor ambient temperature E is within the second preset high-temperature interval, that is, $E > H$, if so, entering step S34, otherwise, entering the remaining control logic, for example, steps S13-S25;

[0283] S34. controlling the air conditioner to switch from the self-cleaning mode to the refrigeration mode, and then entering step S35;

[0284] S35. determining whether the indoor ambient temperature E reaches the high suitable temperature $U+M$ and the current time is within the second refrigeration time period, where M is a preset positive number; if so, entering step S36, otherwise, entering step S37;

[0285] S36. controlling the air conditioner to switch from the refrigeration mode to the self-cleaning mode;

[0286] S37. determining whether the current time exceeds the second refrigeration time period, if so, entering step S38, otherwise, returning to step S35; and

[0287] S38. controlling the air conditioner to switch from the refrigeration mode to the self-cleaning mode, causing the air conditioner to execute the self-cleaning task until self-cleaning is completed.

[0288] In some embodiments, the controller is further configured to:

[0289] when the heat exchanger to be cleaned is the indoor heat exchanger, if the indoor fan is in an on state at the moment immediately before the air conditioner switches to the self-cleaning mode, control the indoor fan to continue to operate for a preset operation duration;

[0290] when the heat exchanger to be cleaned is the indoor heat exchanger, if the air conditioner is in an off state at the moment immediately before it switches to the self-cleaning mode, control the air conditioner to maintain the off state for a preset shutdown duration, and after the preset shutdown duration, control the

indoor heat exchanger to perform frosting treatment for a preset frosting duration; and

[0291] during the process of frosting treatment for the preset frosting duration, control the indoor fan to operate.

[0292] In an embodiment, a self-cleaning method for an air conditioner includes the following steps:

[0293] S0: receiving a self-cleaning instruction;

[0294] S1: performing refrigeration frosting for a first preset duration;

[0295] S2: detecting the indoor ambient temperature Env_T , and a controller of the air conditioner performing corresponding control according to the indoor ambient temperature Env_T ;

[0296] S3: continuing frosting;

[0297] S4: determining whether frosting meets end conditions; and

[0298] S5: performing defrosting.

[0299] The air conditioner may be an air conditioner having at least two functions including refrigeration and heating, an air conditioner only having a refrigeration function, a fixed-frequency air conditioner, or a variable-frequency air conditioner.

[0300] In step S0, the air conditioner is in a standby state, or the air conditioner is in an on state.

[0301] In some embodiments, the on state includes a refrigeration mode, a dehumidification mode, a heating mode, an air supply mode, or other operation modes.

[0302] In step S0, the following solution can be used: the air conditioner caches the current operation state after receiving an indoor self-cleaning instruction, begins refrigeration and displays an indoor self-cleaning marking.

[0303] In step S0, the indoor self-cleaning instruction or an outdoor self-cleaning instruction is received.

[0304] In step S1, the following solution can be used: the first preset duration is 1 min, and it enters S2 after 1 min.

[0305] In step S1, the following solution can be used: the first preset duration is started when the operating frequency of the compressor is higher than 0.

[0306] In step S1, the following solution can be used: the blower stops when step S1 begins.

[0307] In step S1, the following solution can be used: the indoor ambient temperature Env_T is detected at the 20th second and the 40th second of the first preset duration respectively: when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment.

[0308] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0309] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0310] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0311] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0312] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0313] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0314] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0315] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0316] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0317] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0318] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0319] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0320] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0321] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0322] In step S1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0323] In step S2, the following solution can be used: the indoor ambient temperature Env_T sent to the outdoor unit during step S2 is continuously updated.

[0324] In step S2, the following solution can be used: according to preset temperature-related parameters L, U, M, N and H, the first detection of the indoor ambient temperature Env_T in step S2 is executed:

[0325] when $Env_T < L$ or $L \leq Env_T < (U-M)$, the air conditioner can be controlled to enter the first sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the first mode;

[0326] when $(U+N) \leq Env_T < H$ or $Env_T \geq H$, the air conditioner can be controlled to enter the second sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the second mode; and

[0327] when $(U-M) \leq Env_T < U$ or $U \leq Env_T < (U+N)$, the air conditioner can be controlled to enter the third

sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the third mode.

[0328] In some embodiments, the first sub-mode, the second sub-mode and the third sub-mode are the heating sub-mode, the refrigeration sub-mode, the state maintaining sub-mode and the air supply sub-mode under the self-cleaning mode, or other sub-modes in which the air conditioner can operation under the self-cleaning mode (including, but are not limited to the “refrigeration freezing mode 1” and the “refrigeration freezing mode 2”).

[0329] In some embodiments, the first mode, the second mode and the third mode may be the heating mode, the refrigeration mode, the state maintaining mode, the air supply mode, or other modes in which the air conditioner can operation.

[0330] In some embodiments, $L=10^{\circ}\text{C.}$, $U-M=15^{\circ}\text{C.}$, $U=24^{\circ}\text{C.}$, $U+N=28^{\circ}\text{C.}$, and $H=30^{\circ}\text{C.}$

[0331] In some embodiments, before entering the first sub-mode, the second sub-mode, the third sub-mode, the first mode, the second mode and the third mode, mode switching protection time can be set.

[0332] In some embodiments, the mode switching protection time is intended for 3-minute shutdown protection. Optionally, in this period, the indoor blower executes a normal heating or refrigeration rule.

[0333] In some embodiments, when the first sub-mode is the heating sub-mode, the indoor blower operates at the automatic wind speed.

[0334] In some embodiments, when the first sub-mode is the heating sub-mode, the air conditioner quits this mode when $\text{Env_T}>U$ or $T_{01}>6\text{ min}$, where T_{01} is the heating sub-mode operation time.

[0335] In some embodiments, when the first sub-mode is the heating sub-mode, the air conditioner quits this mode when $\text{Env_T}>U$ or $T_{01}>3\text{ min}$, where T_{01} is the heating sub-mode operation time.

[0336] In some embodiments, the heating sub-mode operation time is started when the frequency of the compressor is higher than 0.

[0337] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the indoor blower operates at the automatic wind speed.

[0338] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the air conditioner quits this mode when $\text{Env_T}<U$ or $T_{01}>6\text{ min}$, where T_{01} is the refrigeration sub-mode operation time.

[0339] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the air conditioner quits this mode when $\text{Env_T}<U$ or $T_{01}>3\text{ min}$, where T_{01} is the refrigeration sub-mode operation time.

[0340] In some embodiments, the refrigeration sub-mode operation time is started when the frequency of the compressor is higher than 0.

[0341] In some embodiments, when the first mode is the heating mode, the indoor blower operates at the automatic wind speed.

[0342] In some embodiments, when the first mode is the heating mode, the air conditioner quits this mode when $\text{Env_T}>U$ or $T_{01}>6\text{ min}$, where T_{01} is the heating mode operation time.

[0343] In some embodiments, when the first mode is the heating mode, the air conditioner quits this mode when $\text{Env_T}>U$ or $T_{01}>3\text{ min}$, where T_{01} is the heating mode operation time.

[0344] In some embodiments, the heating mode operation time is started when the frequency of the compressor is higher than 0.

[0345] In some embodiments, when the second mode is the refrigeration mode, the indoor blower operates at the automatic wind speed.

[0346] In some embodiments, when the second mode is the refrigeration mode, the air conditioner quits this mode when $\text{Env_T}<U$ or $T_{01}>6\text{ min}$, where T_{01} is the refrigeration mode operation time.

[0347] In some embodiments, when the second mode is the refrigeration mode, the air conditioner quits this mode when $\text{Env_T}<U$ or $T_{01}>3\text{ min}$, where T_{01} is the refrigeration mode operation time.

[0348] In some embodiments, the refrigeration mode operation time is started when the frequency of the compressor is higher than 0.

[0349] In some embodiments, after quitting the first sub-mode, the second sub-mode, the third sub-mode, the first mode, the second mode and the third mode, mode switching protection time can be set.

[0350] In some embodiments, the mode switching protection time is intended for shutdown protection. Optionally, in this period, the blower stops.

[0351] In step S3, the following solution can be used: in step S3, the operation time is started when the frequency of the compressor is higher than 0.

[0352] In step S3, the following solution can be used: in the whole step S3, the indoor blower is kept stopped (under an indoor heat exchanger cleaning condition), or in the whole step S3, the outdoor blower is kept stopped (under an outdoor heat exchanger cleaning condition).

[0353] In the first sub-step S3-1 of step S3, the following step is executed:

[0354] 1) operating refrigeration for 1 min according to the preset operating frequency of the indoor self-cleaning compressor and the preset opening degree of the expansion valve, where the indoor ambient temperature Env_T is detected at the 20th second and the 40th second respectively;

[0355] 2) when $\text{Env_T}\geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment;

[0356] 3) when $\text{Env_T}<t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment;

[0357] 4) after the time of 1 min in step (1), the indoor ambient temperature Env_T is collected again, and the temperature interval of this temperature is determined:

[0358] 5) when $\text{Env_T}<L$, $(U-M)\leq\text{Env_T}<U$, or $(U+N)\leq\text{Env_T}<H$, the “refrigeration freezing mode 1” is executed; and

[0359] 6) when $L\leq\text{Env_T}<(U-M)$, $U\leq\text{Env_T}<(U+N)$, or $\text{Env_T}\geq H$, the “refrigeration freezing mode 2” is executed.

[0360] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0361] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0362] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0363] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0364] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0365] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0366] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0367] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0368] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0369] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0370] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0371] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0372] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0373] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0374] In step S3-1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0375] In step S3, the following solution can be used: when the first indoor ambient temperature Env_T detected value in step S2 is within a specific temperature interval, the sub-step S3-1 is not executed.

[0376] In the second sub-step S3-2 of step S3, the following step is executed: sending the indoor ambient temperature Env_T detected value to the outdoor unit, and on this basis, the outdoor unit determining to execute the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2”.

[0377] In the second sub-step S3-2 of step S3, the indoor self-cleaning marking can be displayed, and/or the indoor coil pipe temperature is fixed at 10°C .

[0378] In the third sub-step S3-3 of step S3, the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2” is executed until frosting ends.

[0379] The “refrigeration freezing mode 1” includes the following steps:

[0380] Step1_1: performing refrigeration operation for 20 s;

[0381] Step1_2: detecting the indoor ambient temperature Env_T : when $\text{Env_T} \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $\text{Env_T} < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment; and

[0382] Step1_3: repeating Step1_2 until the end.

[0383] In some embodiments, in Step1_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0384] In some embodiments, in Step1_2, the rotational speed of the indoor fan is 0.

[0385] In some embodiments, in Step1_2, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0386] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0387] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0388] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0389] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0390] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0391] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0392] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0393] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0394] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0395] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0396] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0397] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0398] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0399] In some embodiments, the t_0 is specifically a value between 0° C. and 28° C.

[0400] The “refrigeration freezing mode 2” includes the following steps:

[0401] Step2_1: performing refrigeration operation for 20 s;

[0402] Step2_2: detecting the indoor ambient temperature Env_T : when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to third adjustment; and when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to fourth adjustment; and

[0403] Step2_3: repeating Step2_2 until the end.

[0404] In some embodiments, in Step2_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0405] In some embodiments, in Step2_2, the rotational speed of the indoor fan is 0.

[0406] In some embodiments, in Step2_2, for the third adjustment and/or the fourth adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0407] In some embodiments, the third adjustment is different from the first adjustment, and/or the fourth adjustment is different from the second adjustment.

[0408] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0409] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 9 Hz.

[0410] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 6 Hz.

[0411] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0412] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0413] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0414] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0415] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 10 Hz.

[0416] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 7 Hz.

[0417] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0418] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 10 steps.

[0419] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0420] In some embodiments, the air conditioner maintains operation for 20 s after the third adjustment and the fourth adjustment.

[0421] In some embodiments, the t_0 is specifically a value between 0° C. and 28° C.

[0422] In step S4, the following solution can be used: when the indoor coil pipe temperature is lower than -19° C. for consecutive 6 min, it is determined that frosting meets the end conditions.

[0423] In step S4, the following solution can be used: when the indoor ambient temperature is lower than 5° C. for consecutive 6 min, it is determined that frosting meets the end conditions.

[0424] In step S4, the following solution can be used: when the frosting time operation duration is longer than 12 min, it is determined that frosting meets the end conditions.

[0425] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s.

[0426] In some embodiments, during the 20 s operation of the indoor blower, the compressor, the expansion valve and/or the other related devices maintain the operating parameters during frosting.

[0427] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 1” meets “frosting end conditions”, the rotational speed of the fan is the “refrigeration low wind rotational speed”.

[0428] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 2” meets the “frosting end conditions”, the rotational speed of the fan is the “refrigeration silent rotational speed”.

[0429] In some embodiments, the “refrigeration low wind rotational speed” is different from the “refrigeration silent rotational speed”.

[0430] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s, then quits frosting, and enters step S5.

[0431] In step S5, the following solution can be used: first, the compressor is shut down for a fixed duration.

[0432] In some embodiments, the fixed duration is 3 min.

[0433] In some embodiments, when the compressor is shut down for the fixed duration, the blower operates for 3 min at the rotational speed corresponding to step S4.

[0434] In some embodiments, the rotational speed corresponding to step S4 refers to the “refrigeration low wind rotational speed” or the “refrigeration silent rotational speed”.

[0435] In some embodiments, when the compressor is shut down for the fixed duration, the opening degree of the expansion valve maintains unchanged.

[0436] In step S5, the following solution can be used: after the compressor is shut down for the fixed duration, the air conditioner performs heating defrosting (the refrigerating and heating air conditioner), or air supply defrosting (the refrigerating and heating air conditioner or the single-refrigerating air conditioner), or natural defrosting (the single-refrigerating air conditioner).

[0437] In some embodiments, during heating defrosting, the heating mode and the indoor self-cleaning marking are sent.

[0438] In some embodiments, a gentle breeze mode is performed indoors (without cold air prevention control).

[0439] In some embodiments, the opening degree of the expansion valve maintains unchanged or is reduced.

[0440] Optionally, the opening degree of the expansion valve is increased.

[0441] In step S5, the following solution can be used: when the indoor coil pipe temperature is kept higher than 40° C. for 1 min, it is determined that the self-cleaning process is completed.

[0442] In step S5, the following solution can be used: when the heating operation time is longer than 7 min, it is determined that the self-cleaning process is completed.

[0443] In step S5, the following solution can be used: first, the compressor is shut down for a fixed duration, or under the suitable condition, first, the compressor is not shut down.

[0444] In some embodiments, the fan is then started for natural wind defrosting, or under the suitable condition, the fan is not started for natural defrosting.

[0445] The above are self-cleaning operation embodiments of the indoor unit, the self-cleaning operation principle and method for the outdoor unit are the same, and a person skilled in the art can reasonably obtain self-cleaning operation embodiments of the outdoor unit according to the above description, which are not repetitively described and recited here.

[0446] In an embodiment, a self-cleaning method for the air conditioner having at least two functions including refrigeration and heating includes the following steps:

[0447] S0: when the air conditioner is in a standby state, receiving a self-cleaning instruction;

[0448] S1: performing refrigeration frosting for a first preset duration;

[0449] S2: detecting the indoor ambient temperature Env_T, and a controller of the air conditioner performing corresponding control according to the indoor ambient temperature Env_T;

[0450] S3: continuing frosting;

[0451] S4: determining whether frosting meets end conditions; and

[0452] S5: performing defrosting.

[0453] In step S0, the following solution can be used: the air conditioner caches the current operation state after receiving an indoor self-cleaning instruction, begins refrigeration and displays a self-cleaning marking.

[0454] In step S0, the indoor self-cleaning instruction or an outdoor self-cleaning instruction is received.

[0455] In step S1, the following solution can be used: the first preset duration is 1 min, and it enters S2 after 1 min.

[0456] In step S1, the following solution can be used: the first preset duration is started when the operating frequency of the compressor is higher than 0.

[0457] In step S1, the following solution can be used: the blower stops when step S1 begins.

[0458] In step S1, the following solution can be used: the indoor ambient temperature Env_T is detected at the 20th second and the 40th second of the first preset duration respectively: when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment.

[0459] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0460] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0461] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0462] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0463] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0464] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0465] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0466] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0467] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0468] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0469] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0470] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0471] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0472] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0473] In step S1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0474] In step S2, the following solution can be used: the indoor ambient temperature Env_T sent to the outdoor unit during step S2 is continuously updated.

[0475] In step S2, the following solution can be used: according to preset temperature-related parameters L, U, M, N and H, the first detection of the indoor ambient temperature Env_T in step S2 is executed:

[0476] when $\text{Env_T} < L$ or $L \leq \text{Env_T} < (U-M)$, the air conditioner can be controlled to enter the first sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the first mode;

[0477] when $(U+N) \leq \text{Env_T} < H$ or $\text{Env_T} \geq H$, the air conditioner can be controlled to enter the second sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the second mode; and

[0478] when $(U-M) \leq \text{Env_T} < U$ or $U \leq \text{Env_T} < (U+N)$, the air conditioner can be controlled to enter the third sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the third mode.

[0479] In some embodiments, the first sub-mode, the second sub-mode and the third sub-mode are the heating sub-mode, the refrigeration sub-mode, the state maintaining sub-mode and the air supply sub-mode under the self-cleaning mode, or other sub-modes in which the air conditioner can operation under the self-cleaning mode (including, but are not limited to the “refrigeration freezing mode 1” and the “refrigeration freezing mode 2”).

[0480] In some embodiments, the first mode, the second mode and the third mode may be the heating mode, the refrigeration mode, the state maintaining mode, the air supply mode, or other modes in which the air conditioner can operation.

[0481] In some embodiments, $L=10^{\circ}\text{C}$., $U-M=15^{\circ}\text{C}$., $U=24^{\circ}\text{C}$., $U+N=28^{\circ}\text{C}$., and $H=30^{\circ}\text{C}$.

[0482] In some embodiments, before the first sub-mode, the second sub-mode, the third sub-mode, the first mode, the second mode and the third mode, mode switching protection time can be set.

[0483] In some embodiments, the mode switching protection time is intended for 3-minute shutdown protection. Optionally, in this period, the indoor blower executes a normal heating or refrigeration rule.

[0484] In some embodiments, when the first sub-mode is the heating sub-mode, the indoor blower operates at the automatic wind speed.

[0485] In some embodiments, when the first sub-mode is the heating sub-mode, the air conditioner quits this mode when $\text{Env_T} > U$ or $T_{01} > 6\text{ min}$, where T_{01} is the heating sub-mode operation time.

[0486] In some embodiments, when the first sub-mode is the heating sub-mode, the air conditioner quits this mode when $\text{Env_T} > U$ or $T_{01} > 3\text{ min}$, where T_{01} is the heating sub-mode operation time.

[0487] In some embodiments, the heating sub-mode operation time is started when the frequency of the compressor is higher than 0.

[0488] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the indoor blower operates at the automatic wind speed.

[0489] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 6\text{ min}$, where T_{01} is the refrigeration sub-mode operation time.

[0490] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 3\text{ min}$, where T_{01} is the refrigeration sub-mode operation time.

[0491] In some embodiments, the refrigeration sub-mode operation time is started when the frequency of the compressor is higher than 0.

[0492] In some embodiments, when the first mode is the heating mode, the indoor blower operates at the automatic wind speed.

[0493] In some embodiments, when the first mode is the heating mode, the air conditioner quits this mode when $\text{Env_T} > U$ or $T_{01} > 6\text{ min}$, where T_{01} is the heating mode operation time.

[0494] In some embodiments, when the first mode is the heating mode, the air conditioner quits this mode when $\text{Env_T} > U$ or $T_{01} > 3\text{ min}$, where T_{01} is the heating mode operation time.

[0495] In some embodiments, the heating mode operation time is started when the frequency of the compressor is higher than 0.

[0496] In some embodiments, when the second mode is the refrigeration mode, the indoor blower operates at the automatic wind speed.

[0497] In some embodiments, when the second mode is the refrigeration mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 6\text{ min}$, where T_{01} is the refrigeration mode operation time.

[0498] In some embodiments, when the second mode is the refrigeration mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 3\text{ min}$, where T_{01} is the refrigeration mode operation time.

[0499] In some embodiments, the refrigeration mode operation time is started when the frequency of the compressor is higher than 0.

[0500] In some embodiments, after the first sub-mode, the second sub-mode, the third sub-mode, the first mode, the second mode and the third mode, mode switching protection time can be set.

[0501] In some embodiments, the mode switching protection time is intended for shutdown protection. Optionally, in this period, the blower stops.

[0502] In step S3, the following solution can be used: in step S3, the operation time is started when the frequency of the compressor is higher than 0.

[0503] In step S3, the following solution can be used: in the whole step S3, the indoor blower is kept stopped (an indoor heat exchanger cleaning condition), or in the whole step S3, the outdoor blower is kept stopped (an outdoor heat exchanger cleaning condition).

[0504] In the first sub-step S3-1 of step S3, the following step is executed:

[0505] 1) operating refrigeration for 1 min according to the preset operating frequency of the indoor self-cleaning compressor and the preset opening degree of the expansion valve, where the indoor ambient temperature Env_T is detected at the 20th second and the 40th second respectively;

[0506] 2) when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment;

[0507] 3) when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment;

[0508] 4) after the time of 1 min in step (1), the indoor ambient temperature Env_T is collected again, and the temperature interval of this temperature is determined:

[0509] 5) when $Env_T < L$, $(U-M) \leq Env_T < U$, or $(U+N) \leq Env_T < H$, the “refrigeration freezing mode 1” is executed; and

[0510] 6) when $L \leq Env_T < (U-M)$, $U \leq Env_T < (U+N)$, or $Env_T \geq H$, the “refrigeration freezing mode 2” is executed.

[0511] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0512] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0513] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0514] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0515] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0516] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0517] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0518] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0519] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0520] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0521] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0522] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0523] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0524] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0525] In step S3-1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0526] In step S3, the following solution can be used: when the first indoor ambient temperature Env_T detected value in step S2 is within a specific temperature interval, the sub-step S3-1 is not executed.

[0527] In the second sub-step S3-2 of step S3, the following step is executed: sending the indoor ambient temperature Env_T detected value to the outdoor unit, and on this basis, the outdoor unit determining to execute the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2”.

[0528] In the second sub-step S3-2 of step S3, the indoor self-cleaning marking can be displayed, and/or the indoor coil pipe temperature is fixed at 10°C .

[0529] In the third sub-step S3-3 of step S3, the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2” is executed until frosting ends.

[0530] The “refrigeration freezing mode 1” includes the following steps:

[0531] Step1_1: performing refrigeration operation for 20 s;

[0532] Step1_2: detecting the indoor ambient temperature Env_T : when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment; and

[0533] Step1_3: repeating Step1_2 until the end.

[0534] In some embodiments, in Step1_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0535] In some embodiments, in Step1_2, the rotational speed of the indoor fan is 0.

[0536] In some embodiments, in Step1_2, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0537] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0538] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0539] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0540] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0541] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0542] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0543] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0544] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0545] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0546] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0547] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0548] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0549] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0550] In some embodiments, the t_0 is specifically a value between 0° C. and 28° C.

[0551] The “refrigeration freezing mode 2” includes the following steps:

[0552] Step2_1: performing refrigeration operation for 20 s;

[0553] Step2_2: detecting the indoor ambient temperature Env_T : when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to third adjustment; and when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to fourth adjustment; and

[0554] Step2_3: repeating Step2_2 until the end.

[0555] In some embodiments, in Step2_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0556] In some embodiments, in Step2_2, the rotational speed of the indoor fan is 0.

[0557] In some embodiments, in Step2_2, for the third adjustment and/or the fourth adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0558] In some embodiments, the third adjustment is different from the first adjustment, and/or the fourth adjustment is different from the second adjustment.

[0559] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0560] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 9 Hz.

[0561] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 6 Hz.

[0562] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0563] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0564] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0565] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0566] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 10 Hz.

[0567] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 7 Hz.

[0568] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0569] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 10 steps.

[0570] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0571] In some embodiments, the air conditioner maintains operation for 20 s after the third adjustment and the fourth adjustment.

[0572] In some embodiments, the t_0 is specifically a value between 0°C . and 28°C .

[0573] In step S4, the following solution can be used: when the indoor coil pipe temperature is lower than -19°C . for consecutive 6 min, it is determined that frosting meets the end conditions.

[0574] In step S4, the following solution can be used: when the indoor ambient temperature is lower than 5°C . for consecutive 6 min, it is determined that frosting meets the end conditions.

[0575] In step S4, the following solution can be used: when the frosting time operation duration is longer than 12 min, it is determined that frosting meets the end conditions.

[0576] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s.

[0577] In some embodiments, during the 20 s operation of the indoor blower, the compressor, the expansion valve and/or the other related devices maintain the operating parameters during frosting.

[0578] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 1” meets “frosting end conditions”, the rotational speed of the fan is the “refrigeration low wind rotational speed”.

[0579] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 2” meets the “frosting end conditions”, the rotational speed of the fan is the “refrigeration silent rotational speed”.

[0580] In some embodiments, the “refrigeration low wind rotational speed” is different from the “refrigeration silent rotational speed”.

[0581] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s, then quits frosting, and enters step S5.

[0582] In step S5, the following solution can be used: first, the compressor is shut down for a fixed duration.

[0583] In some embodiments, the fixed duration is 3 min.

[0584] In some embodiments, when the compressor is shut down for the fixed duration, the blower operates for 3 min at the rotational speed corresponding to step S4.

[0585] In some embodiments, the rotational speed corresponding to step S4 refers to the “refrigeration low wind rotational speed” or the “refrigeration silent rotational speed”.

[0586] In some embodiments, when the compressor is shut down for the fixed duration, the opening degree of the expansion valve maintains unchanged.

[0587] In step S5, the following solution can be used: after the compressor is shut down for the fixed duration, the air conditioner performs heating defrosting or air supply defrosting.

[0588] In some embodiments, during heating defrosting, the heating mode and the indoor self-cleaning marking are sent.

[0589] In some embodiments, a gentle breeze is blown indoors (without cold air prevention control).

[0590] In some embodiments, the opening degree of the expansion valve maintains unchanged or is reduced.

[0591] Optionally, the opening degree of the expansion valve is increased.

[0592] In step S5, the following solution can be used: when the indoor coil pipe temperature is kept higher than 40°C . for 1 min, it is determined that the self-cleaning process is completed.

[0593] In step S5, the following solution can be used: when the heating operation time is longer than 7 min, it is determined that the self-cleaning process is completed.

[0594] In step S5, the following solution can be used: first, the compressor is shut down for a fixed duration, or under the suitable condition, first, the compressor is not shut down.

[0595] In some embodiments, the fan is then started for natural wind defrosting, or under the suitable condition, the fan is not started for natural defrosting.

[0596] The above are self-cleaning operation embodiments of the indoor unit, the self-cleaning operation principle and method for the outdoor unit are the same, and a person skilled in the art can reasonably obtain self-cleaning operation embodiments of the outdoor unit according to the above description, which are not repetitively described and recited here.

[0597] In an embodiment, a self-cleaning method for the air conditioner having at least two functions including refrigeration and heating includes the following steps:

[0598] S0: when the air conditioner is in an on state, receiving a self-cleaning instruction;

[0599] S1: performing refrigeration frosting for a first preset duration;

[0600] S2: detecting the indoor ambient temperature Env_T , and a controller of the air conditioner performing corresponding control according to the indoor ambient temperature Env_T ;

[0601] S3: continuing frosting;

[0602] S4: determining whether frosting meets end conditions; and

[0603] S5: performing defrosting.

[0604] In step S0, the following solution can be used: the air conditioner caches the current operation state after receiving an indoor self-cleaning instruction, begins refrigeration and displays a self-cleaning marking.

[0605] In step S0, the indoor self-cleaning instruction or an outdoor self-cleaning instruction is received.

[0606] In step S0, when the air conditioner is in the on state, the operation mode of the air conditioner may be the heating mode, the refrigeration mode, the state maintaining mode, the air supply mode, or other modes in which the air conditioner can operation.

[0607] In step S1, the following solution can be used: the first preset duration is 1 min, and it enters S2 after 1 min.

[0608] In step S1, the following solution can be used: the first preset duration is started when the operating frequency of the compressor is higher than 0.

[0609] In step S1, the following solution can be used: the blower stops when step S1 begins.

[0610] In step S1, the following solution can be used: the indoor ambient temperature Env_T is detected at the 20th

second and the 40th second of the first preset duration respectively: when $\text{Env_T} \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $\text{Env_T} < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment.

[0611] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0612] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0613] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0614] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0615] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0616] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0617] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0618] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0619] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0620] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0621] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0622] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0623] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0624] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0625] In step S1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0626] In step S2, the following solution can be used: the indoor ambient temperature Env_T sent to the outdoor unit during step S2 is continuously updated.

[0627] In step S2, the following solution can be used: according to preset temperature-related parameters L, U, M, N and H, the first detection of the indoor ambient temperature Env_T in step S2 is executed:

[0628] when $\text{Env_T} < L$ or $L \leq \text{Env_T} < (U-M)$, the air conditioner can be controlled to enter the first sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the first mode;

[0629] when $(U+N) \leq \text{Env_T} < H$ or $\text{Env_T} \geq H$, the air conditioner can be controlled to enter the second sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the second mode; and

[0630] when $(U-M) \leq \text{Env_T} < U$ or $U \leq \text{Env_T} < (U+N)$, the air conditioner can be controlled to enter the third sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the third mode.

[0631] In some embodiments, the first sub-mode, the second sub-mode and the third sub-mode are the heating sub-mode, the refrigeration sub-mode, the state maintaining sub-mode and the air supply sub-mode under the self-cleaning mode, or other sub-modes in which the air conditioner can operation under the self-cleaning mode (including, but are not limited to the “refrigeration freezing mode 1” and the “refrigeration freezing mode 2”).

[0632] In some embodiments, the first mode, the second mode and the third mode may be the heating mode, the refrigeration mode, the state maintaining mode, the air supply mode, or other modes in which the air conditioner can operation.

[0633] In some embodiments, $L=10^\circ\text{C}$., $U-M=15^\circ\text{C}$., $U=24^\circ\text{C}$., $U+N=28^\circ\text{C}$., and $H=30^\circ\text{C}$.

[0634] In some embodiments, before switching of the first sub-mode, the second sub-mode, the third sub-mode, the first mode, the second mode and the third mode, mode switching protection time can be set.

[0635] In some embodiments, the mode switching protection time is intended for 3-minute shutdown protection. Optionally, in this period, the indoor blower executes a normal heating or refrigeration rule.

[0636] In some embodiments, when the first sub-mode is the heating sub-mode, the indoor blower operates at the automatic wind speed.

[0637] In some embodiments, when the first sub-mode is the heating sub-mode, the air conditioner quits this mode when $\text{Env_T} > U$ or $T_{01} > 6\text{ min}$, where T_{01} is the heating sub-mode operation time.

[0638] In some embodiments, when the first sub-mode is the heating sub-mode, the air conditioner quits this mode when $\text{Env_T} > U$ or $T_{01} > 3\text{ min}$, where T_{01} is the heating sub-mode operation time.

[0639] In some embodiments, the heating sub-mode operation time is started when the frequency of the compressor is higher than 0.

[0640] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the indoor blower operates at the automatic wind speed.

[0641] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the air conditioner quits this

mode when $\text{Env_T} < U$ or $T_{01} > 6$ min, where T_{01} is the refrigeration sub-mode operation time.

[0642] In some embodiments, when the second sub-mode is the refrigeration sub-mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 3$ min, where T_{01} is the refrigeration sub-mode operation time.

[0643] In some embodiments, the refrigeration sub-mode operation time is started when the frequency of the compressor is higher than 0.

[0644] In some embodiments, when the first mode is the heating mode, the indoor blower operates at the automatic wind speed.

[0645] In some embodiments, when the first mode is the heating mode, the air conditioner quits this mode when $\text{Env_T} > U$ or $T_{01} > 6$ min, where T_{01} is the heating mode operation time.

[0646] In some embodiments, when the first mode is the heating mode, the air conditioner quits this mode when $\text{Env_T} > U$ or $T_{01} > 3$ min, where T_{01} is the heating mode operation time.

[0647] In some embodiments, the heating mode operation time is started when the frequency of the compressor is higher than 0.

[0648] In some embodiments, when the second mode is the refrigeration mode, the indoor blower operates at the automatic wind speed.

[0649] In some embodiments, when the second mode is the refrigeration mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 6$ min, where T_{01} is the refrigeration mode operation time.

[0650] In some embodiments, when the second mode is the refrigeration mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 3$ min, where T_{01} is the refrigeration mode operation time.

[0651] In some embodiments, the refrigeration mode operation time is started when the frequency of the compressor is higher than 0.

[0652] In some embodiments, after switching of the first sub-mode, the second sub-mode, the third sub-mode, the first mode, the second mode and the third mode, mode switching protection time can be set.

[0653] In some embodiments, the mode switching protection time is intended for shutdown protection. Optionally, in this period, the blower stops.

[0654] In step S3, the following solution can be used: in step S3, the operation time is started when the frequency of the compressor is higher than 0.

[0655] In step S3, the following solution can be used: in the whole step S3, the indoor blower is kept stopped (an indoor heat exchanger cleaning condition), or in the whole step S3, the outdoor blower is kept stopped (an outdoor heat exchanger cleaning condition).

[0656] In the first sub-step S3-1 of step S3, the following step is executed:

[0657] 1) operating refrigeration for 1 min according to the preset operating frequency of the indoor self-cleaning compressor and the preset opening degree of the expansion valve, where the indoor ambient temperature Env_T is detected at the 20th second and the 40th second respectively;

[0658] 2) when $\text{Env_T} \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment;

[0659] 3) when $\text{Env_T} < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment;

[0660] 4) after the time of 1 min in step (1), the indoor ambient temperature Env_T is collected again, and the temperature interval of this temperature is determined:

[0661] 5) when $\text{Env_T} < L$, $(U-M) \leq \text{Env_T} < U$, or $(U+N) \leq \text{Env_T} < H$, the “refrigeration freezing mode 1” is executed; and

[0662] 6) when $L \leq \text{Env_T} < (U-M)$, $U \leq \text{Env_T} < (U+N)$, or $\text{Env_T} \geq H$, the “refrigeration freezing mode 2” is executed.

[0663] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0664] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0665] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0666] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0667] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0668] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0669] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0670] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0671] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0672] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0673] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0674] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0675] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0676] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0677] In step S3-1, the following solution can be used: the t_0 is specifically a value between 0° C. and 28° C.

[0678] In step S3, the following solution can be used: when the first indoor ambient temperature Env_T detected value in step S2 is within a specific temperature interval, the sub-step S3-1 is not executed.

[0679] In the second sub-step S3-2 of step S3, the following step is executed: sending the indoor ambient temperature Env_T detected value to the outdoor unit, and on this basis, the outdoor unit determining to execute the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2”.

[0680] In the second sub-step S3-2 of step S3, the indoor self-cleaning marking can be displayed, and/or the indoor coil pipe temperature is fixed at 10° C.

[0681] In the third sub-step S3-3 of step S3, the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2” is executed until frosting ends.

[0682] The “refrigeration freezing mode 1” includes the following steps:

[0683] Step1_1: performing refrigeration operation for 20 s;

[0684] Step1_2: detecting the indoor ambient temperature Env_T: when $\text{Env_T} \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $\text{Env_T} < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment; and

[0685] Step1_3: repeating Step1_2 until the end.

[0686] In some embodiments, in Step1_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0687] In some embodiments, in Step1_2, the rotational speed of the indoor fan is 0.

[0688] In some embodiments, in Step1_2, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0689] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0690] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0691] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0692] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0693] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0694] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0695] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0696] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0697] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0698] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0699] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0700] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0701] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0702] In some embodiments, the t_0 is specifically a value between 0° C. and 28° C.

[0703] The “refrigeration freezing mode 2” includes the following steps:

[0704] Step2_1: performing refrigeration operation for 20 s;

[0705] Step2_2: detecting the indoor ambient temperature Env_T: when $\text{Env_T} \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to third adjustment; and when $\text{Env_T} < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to fourth adjustment; and

[0706] Step2_3: repeating Step2_2 until the end.

[0707] In some embodiments, in Step2_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0708] In some embodiments, in Step2_2, the rotational speed of the indoor fan is 0.

[0709] In some embodiments, in Step2_2, for the third adjustment and/or the fourth adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0710] In some embodiments, the third adjustment is different from the first adjustment, and/or the fourth adjustment is different from the second adjustment.

[0711] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0712] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 9 Hz.

[0713] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 6 Hz.

[0714] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0715] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0716] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0717] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0718] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 10 Hz.

[0719] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 7 Hz.

[0720] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0721] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 10 steps.

[0722] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0723] In some embodiments, the air conditioner maintains operation for 20 s after the third adjustment and the fourth adjustment.

[0724] In some embodiments, the t_0 is specifically a value between 0° C. and 28° C.

[0725] In step S4, the following solution can be used: when the indoor coil pipe temperature is lower than -19° C. for consecutive 6 min, it is determined that frosting meets the end conditions.

[0726] In step S4, the following solution can be used: when the indoor ambient temperature is lower than 5° C. for consecutive 6 min, it is determined that frosting meets the end conditions.

[0727] In step S4, the following solution can be used: when the frosting time operation duration is longer than 12 min, it is determined that frosting meets the end conditions.

[0728] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s.

[0729] In some embodiments, during the 20 s operation of the indoor blower, the compressor, the expansion valve and/or the other related devices maintain the operating parameters during frosting.

[0730] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 1” meets “frosting end conditions”, the rotational speed of the fan is the “refrigeration low wind rotational speed”.

[0731] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 2” meets the “frosting end conditions”, the rotational speed of the fan is the “refrigeration silent rotational speed”.

[0732] In some embodiments, the “refrigeration low wind rotational speed” is different from the “refrigeration silent rotational speed”.

[0733] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s, then quits frosting, and enters step S5.

[0734] In step S5, the following solution can be used: first, the compressor is shut down for a fixed duration.

[0735] In some embodiments, the fixed duration is 3 min.

[0736] In some embodiments, when the compressor is shut down for the fixed duration, the blower operates for 3 min at the rotational speed corresponding to step S4.

[0737] In some embodiments, the rotational speed corresponding to step S4 refers to the “refrigeration low wind rotational speed” or the “refrigeration silent rotational speed”.

[0738] In some embodiments, when the compressor is shut down for the fixed duration, the opening degree of the expansion valve maintains unchanged.

[0739] In some embodiments, the fan is then started for natural wind defrosting, or under the suitable condition, the fan is not started for natural defrosting.

[0740] The above are self-cleaning operation embodiments of the indoor unit, the self-cleaning operation principle and method for the outdoor unit are the same, and a person skilled in the art can reasonably obtain self-cleaning operation embodiments of the outdoor unit according to the above description, which are not repetitively described and recited here.

[0741] In an embodiment, a self-cleaning method for the air conditioner only having the refrigeration function includes the following steps:

[0742] S0: when the air conditioner is in a standby state, receiving a self-cleaning instruction;

[0743] S1: performing refrigeration frosting for a first preset duration;

[0744] S2: detecting the indoor ambient temperature Env_T , and a controller of the air conditioner performing corresponding control according to the indoor ambient temperature Env_T ;

[0745] S3: continuing frosting;

[0746] S4: determining whether frosting meets end conditions; and

[0747] S5: performing defrosting.

[0748] In step S0, the following solution can be used: the air conditioner caches the current operation state after receiving an indoor self-cleaning instruction, begins refrigeration and displays a self-cleaning marking.

[0749] In step S0, the indoor self-cleaning instruction or an outdoor self-cleaning instruction is received.

[0750] In step S1, the following solution can be used: the first preset duration is 1 min, and it enters S2 after 1 min.

[0751] In step S1, the following solution can be used: the first preset duration is started when the operating frequency of the compressor is higher than 0.

[0752] In step S1, the following solution can be used: the blower stops when step S1 begins.

[0753] In step S1, the following solution can be used: the indoor ambient temperature Env_T is detected at the 20th second and the 40th second of the first preset duration respectively: when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment.

[0754] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0755] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0756] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0757] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0758] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0759] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0760] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0761] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0762] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0763] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0764] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0765] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0766] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0767] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0768] In step S1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0769] In step S2, the following solution can be used: the indoor ambient temperature Env_T sent to the outdoor unit during step S2 is continuously updated.

[0770] In step S2, the following solution can be used: according to preset temperature-related parameters L, U, M, N and H, the first detection of the indoor ambient temperature Env_T in step S2 is executed:

[0771] when $Env_T < L$ or $L \leq Env_T < (U-M)$, the air conditioner can be controlled to enter the first sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the first mode;

[0772] when $(U+N) \leq Env_T < H$ or $Env_T \geq H$, the air conditioner can be controlled to enter the second sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the second mode; and

[0773] when $(U-M) \leq Env_T < U$ or $U \leq Env_T < (U+N)$, the air conditioner can be controlled to enter the third sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the third mode.

[0774] In some embodiments, the first sub-mode, the second sub-mode and the third sub-mode are the heating sub-mode, the refrigeration sub-mode, the state maintaining sub-mode, the air supply sub-mode and the refrigeration free operation sub-mode under the self-cleaning mode, or other sub-modes in which the air conditioner can operation under the self-cleaning mode (including, but are not limited to the "refrigeration freezing mode 1" and the "refrigeration freezing mode 2").

[0775] In some embodiments, the first mode, the second mode and the third mode may be the heating mode, the refrigeration mode, the state maintaining mode, the air supply mode, or other modes in which the air conditioner can operation (including, but are not limited to a "refrigeration free operation mode").

[0776] In some embodiments, $L=10^\circ\text{C}$., $U-M=15^\circ\text{C}$., $U=24^\circ\text{C}$., $U+N=28^\circ\text{C}$., and $H=30^\circ\text{C}$.

[0777] In some embodiments, when the second sub-mode is the refrigeration free operation sub-mode, the indoor blower operates at the automatic wind speed.

[0778] In some embodiments, when the second sub-mode is the refrigeration free operation sub-mode, the air conditioner quits this mode when $Env_T < U$ or $T_{01} > 6\text{ min}$, where T_{01} is the refrigeration free operation sub-mode operation time.

[0779] In some embodiments, when the second sub-mode is the refrigeration free operation sub-mode, the air condi-

tioner quits this mode when $\text{Env_T} < \text{U}$ or $\text{T_01} > 3$ min, where T_01 is the refrigeration free operation sub-mode operation time.

[0780] In some embodiments, the refrigeration free operation sub-mode operation time is started when the frequency of the compressor is higher than 0.

[0781] In some embodiments, when the second mode is the refrigeration free operation mode, the indoor blower operates at the automatic wind speed.

[0782] In some embodiments, when the second mode is the refrigeration free operation mode, the air conditioner quits this mode when $\text{Env_T} < \text{U}$ or $\text{T_01} > 6$ min, where T_01 is the refrigeration free operation mode operation time.

[0783] In some embodiments, when the second mode is the refrigeration free operation mode, the air conditioner quits this mode when $\text{Env_T} < \text{U}$ or $\text{T_01} > 3$ min, where T_01 is the refrigeration free operation mode operation time.

[0784] In some embodiments, the refrigeration free operation mode operation time is started when the frequency of the compressor is higher than 0.

[0785] In some embodiments, during the operation time of the first sub-mode, the second sub-mode, the third sub-mode, the first mode, the second mode and the third mode, the blower stops.

[0786] In step S3, the following solution can be used: in step S3, the operation time is started when the frequency of the compressor is higher than 0.

[0787] In step S3, the following solution can be used: in the whole step S3, the indoor blower is kept stopped (under an indoor heat exchanger cleaning condition), or in the whole step S3, the outdoor blower is kept stopped (under an outdoor heat exchanger cleaning condition).

[0788] In the first sub-step S3-1 of step S3, the following step is executed:

[0789] 1) operating refrigeration for 1 min according to the preset operating frequency of the indoor self-cleaning compressor and the preset opening degree of the expansion valve, where the indoor ambient temperature Env_T is detected at the 20th second and the 40th second respectively;

[0790] 2) when $\text{Env_T} \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment;

[0791] 3) when $\text{Env_T} < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment;

[0792] 4) after the time of 1 min in step (1), the indoor ambient temperature Env_T is collected again, and the temperature interval of this temperature is determined:

[0793] 5) when $\text{Env_T} < \text{L}$, $(\text{U}-\text{M}) \leq \text{Env_T} < \text{U}$, or $(\text{U}+\text{N}) \leq \text{Env_T} < \text{H}$, the “refrigeration freezing mode 1” is executed; and

[0794] 6) when $\text{L} \leq \text{Env_T} < (\text{U}-\text{M})$, $\text{U} \leq \text{Env_T} < (\text{U}+\text{N})$, or $\text{Env_T} \geq \text{H}$, the “refrigeration freezing mode 2” is executed.

[0795] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0796] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0797] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0798] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0799] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0800] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0801] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0802] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0803] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0804] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0805] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0806] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0807] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0808] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0809] In step S3-1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0810] In step S3, the following solution can be used: when the first indoor ambient temperature Env_T detected value in step S2 is within a specific temperature interval, the sub-step S3-1 is not executed.

[0811] In the second sub-step S3-2 of step S3, the following step is executed: sending the indoor ambient temperature Env_T detected value to the outdoor unit, and on this basis, the outdoor unit determining to execute the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2”.

[0812] In the second sub-step S3-2 of step S3, the indoor self-cleaning marking can be displayed, and/or the indoor coil pipe temperature is fixed at 10°C .

[0813] In the third sub-step S3-3 of step S3, the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2” is executed until frosting ends.

[0814] The “refrigeration freezing mode 1” includes the following steps:

[0815] Step1_1: performing refrigeration operation for 20 s;

[0816] Step1_2: detecting the indoor ambient temperature Env_T: when $\text{Env_T} \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $\text{Env_T} < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment; and

[0817] Step1_3: repeating Step1_2 until the end.

[0818] In some embodiments, in Step1_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0819] In some embodiments, in Step1_2, the rotational speed of the indoor fan is 0.

[0820] In some embodiments, in Step1_2, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0821] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0822] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0823] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0824] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0825] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0826] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0827] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0828] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0829] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0830] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0831] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0832] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0833] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0834] In some embodiments, the t_0 is specifically a value between 0°C . and 28°C .

[0835] The “refrigeration freezing mode 2” includes the following steps:

[0836] Step2_1: performing refrigeration operation for 20 s;

[0837] Step2_2: detecting the indoor ambient temperature Env_T: when $\text{Env_T} \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to third adjustment; and when $\text{Env_T} < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to fourth adjustment; and

[0838] Step2_3: repeating Step2_2 until the end.

[0839] In some embodiments, in Step2_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0840] In some embodiments, in Step2_2, the rotational speed of the indoor fan is 0.

[0841] In some embodiments, in Step2_2, for the third adjustment and/or the fourth adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0842] In some embodiments, the third adjustment is different from the first adjustment, and/or the fourth adjustment is different from the second adjustment.

[0843] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0844] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 9 Hz.

[0845] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 6 Hz.

[0846] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0847] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0848] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0849] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0850] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 10 Hz.

[0851] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 7 Hz.

[0852] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0853] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 10 steps.

[0854] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0855] In some embodiments, the air conditioner maintains operation for 20 s after the third adjustment and the fourth adjustment.

[0856] In some embodiments, the t_0 is specifically a value between 0° C. and 28° C.

[0857] In step S4, the following solution can be used: when the indoor coil pipe temperature is lower than -19° C. for consecutive 6 min, it is determined that frosting meets the end conditions.

[0858] In step S4, the following solution can be used: when the indoor ambient temperature is lower than 5° C. for consecutive 6 min, it is determined that frosting meets the end conditions.

[0859] In step S4, the following solution can be used: when the frosting time operation duration is longer than 12 min, it is determined that frosting meets the end conditions.

[0860] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s.

[0861] In some embodiments, during the 20 s operation of the indoor blower, the compressor, the expansion valve and/or the other related devices maintain the operating parameters during frosting.

[0862] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 1” meets “frosting end conditions”, the rotational speed of the fan is the “refrigeration low wind rotational speed”.

[0863] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 2” meets the “frosting end conditions”, the rotational speed of the fan is the “refrigeration silent rotational speed”.

[0864] In some embodiments, the “refrigeration low wind rotational speed” is different from the “refrigeration silent rotational speed”.

[0865] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s, then quits frosting, and enters step S5.

[0866] In step S5, the following solution can be used: first, the compressor is shut down for a fixed duration.

[0867] In some embodiments, the fixed duration is 3 min.

[0868] In some embodiments, when the compressor is shut down for the fixed duration, the blower operates for 3 min at the rotational speed corresponding to step S4.

[0869] In some embodiments, the rotational speed corresponding to step S4 refers to the “refrigeration low wind rotational speed” or the “refrigeration silent rotational speed”.

[0870] In some embodiments, when the compressor is shut down for the fixed duration, the opening degree of the expansion valve maintains unchanged.

[0871] In some embodiments, the fan is then started for natural wind defrosting, or under the suitable condition, the fan is not started for natural defrosting.

[0872] The above are self-cleaning operation embodiments of the indoor unit, the self-cleaning operation principle and method for the outdoor unit are the same, and a person skilled in the art can reasonably obtain self-cleaning operation embodiments of the outdoor unit according to the above description, which are not repetitively described and recited here.

[0873] In an embodiment, a self-cleaning method for the air conditioner only having a refrigeration function includes the following steps:

[0874] S0: when the air conditioner is in an on state, receiving a self-cleaning instruction;

[0875] S1: performing refrigeration frosting for a first preset duration;

[0876] S2: detecting the indoor ambient temperature Env_T , and a controller of the air conditioner performing corresponding control according to the indoor ambient temperature Env_T ;

[0877] S3: continuing frosting;

[0878] S4: determining whether frosting meets end conditions; and

[0879] S5: performing defrosting.

[0880] In step S0, the following solution can be used: the air conditioner caches the current operation state after receiving an indoor self-cleaning instruction, begins refrigeration and displays a self-cleaning marking.

[0881] In step S0, the indoor self-cleaning instruction or an outdoor self-cleaning instruction is received.

[0882] In step S0, when the air conditioner is in the on state, the air conditioner is in one of the heating mode, the refrigeration mode, the state maintaining mode, the air supply mode, or other modes in which the air conditioner can operation.

[0883] In step S1, the following solution can be used: the first preset duration is 1 min, and it enters S2 after 1 min.

[0884] In step S1, the following solution can be used: the first preset duration is started when the operating frequency of the compressor is higher than 0.

[0885] In step S1, the following solution can be used: the blower stops when step S1 begins.

[0886] In step S1, the following solution can be used: the indoor ambient temperature Env_T is detected at the 20th second and the 40th second of the first preset duration respectively: when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment.

[0887] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air

conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0888] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0889] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0890] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0891] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0892] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0893] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0894] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0895] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0896] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0897] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0898] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0899] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0900] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0901] In step S1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0902] In step S2, the following solution can be used: the indoor ambient temperature Env_T sent to the outdoor unit during step S2 is continuously updated.

[0903] In step S2, the following solution can be used: according to preset temperature-related parameters L , U , M , N and H , the first detection of the indoor ambient temperature Env_T in step S2 is executed:

[0904] when $\text{Env_T} < L$ or $L \leq \text{Env_T} < (U-M)$, the air conditioner can be controlled to enter the first sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the first mode;

[0905] when $(U+N) \leq \text{Env_T} < H$ or $\text{Env_T} \geq H$, the air conditioner can be controlled to enter the second sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the second mode; and

[0906] when $(U-M) \leq \text{Env_T} < U$ or $U \leq \text{Env_T} < (U+N)$, the air conditioner can be controlled to enter the third sub-mode from the self-cleaning mode, or the air conditioner can be controlled to switch from the self-cleaning mode to the third mode.

[0907] In some embodiments, the first sub-mode, the second sub-mode and the third sub-mode are the heating sub-mode, the refrigeration sub-mode, the state maintaining sub-mode, the air supply sub-mode and the refrigeration free operation sub-mode under the self-cleaning mode, or other sub-modes in which the air conditioner can operation under the self-cleaning mode (including, but are not limited to the “refrigeration freezing mode 1” and the “refrigeration freezing mode 2”).

[0908] In some embodiments, the first mode, the second mode and the third mode may be the heating mode, the refrigeration mode, the state maintaining mode, the air supply mode, or other modes in which the air conditioner can operation (including, but are not limited to a “refrigeration free operation mode”).

[0909] In some embodiments, $L=10^{\circ}\text{C}$., $U-M=15^{\circ}\text{C}$., $U=24^{\circ}\text{C}$., $U+N=28^{\circ}\text{C}$., and $H=30^{\circ}\text{C}$.

[0910] In some embodiments, when the second sub-mode is the refrigeration free operation sub-mode, the indoor blower operates at the automatic wind speed.

[0911] In some embodiments, when the second sub-mode is the refrigeration free operation sub-mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 6$ min, where T_{01} is the refrigeration free operation sub-mode operation time.

[0912] In some embodiments, when the second sub-mode is the refrigeration free operation sub-mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 3$ min, where T_{01} is the refrigeration free operation sub-mode operation time.

[0913] In some embodiments, the refrigeration free operation sub-mode operation time is started when the frequency of the compressor is higher than 0.

[0914] In some embodiments, when the second mode is the refrigeration free operation mode, the indoor blower operates at the automatic wind speed.

[0915] In some embodiments, when the second mode is the refrigeration free operation mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 6$ min, where T_{01} is the refrigeration free operation mode operation time.

[0916] In some embodiments, when the second mode is the refrigeration free operation mode, the air conditioner quits this mode when $\text{Env_T} < U$ or $T_{01} > 3$ min, where T_{01} is the refrigeration free operation mode operation time.

[0917] In some embodiments, the refrigeration free operation mode operation time is started when the frequency of the compressor is higher than 0.

[0918] In some embodiments, during the first sub-mode, the second sub-mode, the third sub-mode, the first mode, the second mode and the third mode, the blower stops.

[0919] In step S3, the following solution can be used: in step S3, the operation time is started when the frequency of the compressor is higher than 0.

[0920] In step S3, the following solution can be used: in the whole step S3, the indoor blower is kept stopped (an indoor heat exchanger cleaning condition), or in the whole step S3, the outdoor blower is kept stopped (an outdoor heat exchanger cleaning condition).

[0921] In the first sub-step S3-1 of step S3, the following step is executed:

[0922] 1) operating refrigeration for 1 min according to the preset operating frequency of the indoor self-cleaning compressor and the preset opening degree of the expansion valve, where the indoor ambient temperature Env_T is detected at the 20th second and the 40th second respectively;

[0923] 2) when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment;

[0924] 3) when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment;

[0925] 4) after the time of 1 min in step (1), the indoor ambient temperature Env_T is collected again, and the temperature interval of this temperature is determined:

[0926] 5) when $Env_T < L$, $(U-M) \leq Env_T < U$, or $(U+N) \leq Env_T < H$, the “refrigeration freezing mode 1” is executed; and

[0927] 6) when $L \leq Env_T < (U-M)$, $U \leq Env_T < (U+N)$, or $Env_T \geq H$, the “refrigeration freezing mode 2” is executed.

[0928] In some embodiments, for the first adjustment and/or the second adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0929] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0930] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0931] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0932] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0933] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0934] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0935] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0936] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0937] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0938] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0939] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0940] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0941] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0942] In step S3-1, the following solution can be used: the t_0 is specifically a value between 0°C . and 28°C .

[0943] In step S3, the following solution can be used: when the first indoor ambient temperature Env_T detected value in step S2 is within a specific temperature interval, the sub-step S3-1 is not executed.

[0944] In the second sub-step S3-2 of step S3, the following step is executed: sending the indoor ambient temperature Env_T detected value to the outdoor unit, and on this basis, the outdoor unit determining to execute the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2”.

[0945] In the second sub-step S3-2 of step S3, the indoor self-cleaning marking can be displayed, and/or the indoor coil pipe temperature is fixed at 10°C .

[0946] In the third sub-step S3-3 of step S3, the “refrigeration freezing mode 1” or the “refrigeration freezing mode 2” is executed until frosting ends.

[0947] The “refrigeration freezing mode 1” includes the following steps:

[0948] Step1_1: performing refrigeration operation for 20 s;

[0949] Step1_2: detecting the indoor ambient temperature Env_T : when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to first adjustment; and when $Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to second adjustment; and

[0950] Step1_3: repeating Step1_2 until the end.

[0951] In some embodiments, in Step1_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0952] In some embodiments, in Step1_2, the rotational speed of the indoor fan is 0.

[0953] In some embodiments, in Step1_2, for the first adjustment and/or the second adjustment, the controller of

the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0954] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0955] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0956] In some embodiments, the first adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0957] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0958] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0959] In some embodiments, the first adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0960] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-10 Hz.

[0961] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 8 Hz.

[0962] In some embodiments, the second adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 5 Hz.

[0963] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-10 steps.

[0964] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0965] In some embodiments, the second adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0966] In some embodiments, the air conditioner maintains operation for 20 s after the first adjustment and the second adjustment.

[0967] In some embodiments, the t_0 is specifically a value between 0° C. and 28° C.

[0968] The “refrigeration freezing mode 2” includes the following steps:

[0969] Step2_1: performing refrigeration operation for 20 s;

[0970] Step2_2: detecting the indoor ambient temperature Env_T : when $Env_T \geq t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to third adjustment; and when

$Env_T < t_0$, the frequency of the compressor and the opening degree of the expansion valve are subjected to fourth adjustment; and

[0971] Step2_3: repeating Step2_2 until the end.

[0972] In some embodiments, in Step2_1, the compressor executes the preset self-cleaning frequency, the expansion valve executes the preset self-cleaning opening degree, and the rotational speed of the indoor fan is 0.

[0973] In some embodiments, in Step2_2, the rotational speed of the indoor fan is 0.

[0974] In some embodiments, in Step2_2, for the third adjustment and/or the fourth adjustment, the controller of the air conditioner sends out clear control signals and controls the compressor and the expansion valve to maintain the related states and/or adjust the related states.

[0975] In some embodiments, the third adjustment is different from the first adjustment, and/or the fourth adjustment is different from the second adjustment.

[0976] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0977] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 9 Hz.

[0978] In some embodiments, the third adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 6 Hz.

[0979] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0980] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0981] In some embodiments, the third adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 6 steps.

[0982] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be in a range of 0-12 Hz.

[0983] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 10 Hz.

[0984] In some embodiments, the fourth adjustment includes increasing or reducing the operating frequency of the compressor by a corresponding value, where the value may be 7 Hz.

[0985] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be in a range of 0-15 steps.

[0986] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 10 steps.

[0987] In some embodiments, the fourth adjustment includes increasing or reducing the opening degree of the expansion valve by a corresponding value, where the value may be 8 steps.

[0988] In some embodiments, the air conditioner maintains operation for 20 s after the third adjustment and the fourth adjustment.

[0989] In some embodiments, the t_0 is specifically a value between 0° C. and 28° C.

[0990] In step S4, the following solution can be used: when the indoor coil pipe temperature is lower than -19° C. for consecutive 6 min, it is determined that frosting meets the end conditions.

[0991] In step S4, the following solution can be used: when the indoor ambient temperature is lower than 5° C. for consecutive 6 min, it is determined that frosting meets the end conditions.

[0992] In step S4, the following solution can be used: when the frosting time operation duration is longer than 12 min, it is determined that frosting meets the end conditions.

[0993] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s.

[0994] In some embodiments, during the 20 s operation of the indoor blower, the compressor, the expansion valve and/or the other related devices maintain the operating parameters during frosting.

[0995] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 1” meets “frosting end conditions”, the rotational speed of the fan is the “refrigeration low wind rotational speed”.

[0996] In some embodiments, during the 20 s operation of the indoor blower, when the “refrigeration freezing mode 2” meets the “frosting end conditions”, the rotational speed of the fan is the “refrigeration silent rotational speed”.

[0997] In some embodiments, the “refrigeration low wind rotational speed” is different from the “refrigeration silent rotational speed”.

[0998] In step S4, the following solution can be used: after it is determined that frosting meets the end conditions, the indoor blower starts operation for 20 s, then quits frosting, and enters step S5.

[0999] In step S5, the following solution can be used: first, the compressor is shut down for a fixed duration.

[1000] In some embodiments, the fixed duration is 3 min.

[1001] In some embodiments, when the compressor is shut down for the fixed duration, the blower operates for 3 min at the rotational speed corresponding to step S4.

[1002] In some embodiments, the rotational speed corresponding to step S4 refers to the “refrigeration low wind rotational speed” or the “refrigeration silent rotational speed”.

[1003] In some embodiments, when the compressor is shut down for the fixed duration, the opening degree of the expansion valve maintains unchanged.

[1004] In some embodiments, the fan is then started for natural wind defrosting, or under the suitable condition, the fan is not started for natural defrosting.

[1005] The above are self-cleaning operation embodiments of the indoor unit, the self-cleaning operation principle and method for the outdoor unit are the same, and a person skilled in the art can reasonably obtain self-cleaning

operation embodiments of the outdoor unit according to the above description, which are not repetitively described and recited here.

[1006] The self-cleaning control method for the air conditioner in some embodiments of the present application is implemented by the controller of the air conditioner, and includes steps S61-S62:

[1007] S61. controlling the air conditioner to enter the self-cleaning mode, causing the heat exchanger to be cleaned to implement a function as the evaporator to enable the heat exchanger to be cleaned to perform frosting treatment, where the heat exchanger to be cleaned is the outdoor heat exchanger or the indoor heat exchanger; and

[1008] S62. adjusting the operating parameters of the air conditioner according to the obtained indoor ambient temperature.

[1009] In some embodiments, the adjusting the operating parameters of the air conditioner according to the obtained indoor ambient temperature includes:

[1010] when the indoor ambient temperature is in a preset low-temperature interval, controlling the air conditioner to switch from the self-cleaning mode to the heating mode;

[1011] in a preset heating time period after the air conditioner enters the heating mode, if the indoor ambient temperature is detected to reach a preset target temperature, controlling the air conditioner to switch from the heating mode to the self-cleaning mode, where the preset target temperature is higher than the maximum value of the preset low-temperature interval; and

[1012] in the heating mode, when the indoor ambient temperature does not reach the preset target temperature and the current time exceeds the preset heating time period, controlling the air conditioner to switch from the heating mode to the self-cleaning mode, causing the air conditioner to execute self-cleaning until completion.

[1013] In some embodiments, the preset low-temperature interval includes a first preset low-temperature interval and a second preset low-temperature interval, and the maximum value of the first preset low-temperature interval is less than the minimum value of the second preset low-temperature interval. The preset target temperature at least includes a low suitable temperature and an ideal temperature, the low suitable temperature is lower than the ideal temperature, and the low suitable temperature is higher than the maximum value of the second preset low-temperature interval. The preset heating time period includes a first heating time period and a second heating time period; the preset target temperature corresponding to the first preset low-temperature interval is the low suitable temperature, the preset target temperature corresponding to the second preset low-temperature interval is the ideal temperature, the preset heating time period corresponding to the first preset low-temperature interval is the first heating time period, and the preset heating time period corresponding to the second preset low-temperature interval is the second heating time period.

[1014] In some embodiments, the adjusting the operating parameters of the air conditioner according to the obtained indoor ambient temperature includes:

- [1015] when the indoor ambient temperature is in a preset high-temperature interval, controlling the air conditioner to switch from the self-cleaning mode to the refrigeration mode;
- [1016] in a preset refrigeration time period after the air conditioner enters the refrigeration mode, if the indoor ambient temperature is detected to reach a preset target temperature, controlling the air conditioner to switch from the refrigeration mode to the self-cleaning mode, where the preset target temperature is lower than the minimum value of the preset high-temperature interval; and
- [1017] in the refrigeration mode, when the indoor ambient temperature does not reach the preset target temperature and the current time exceeds the preset refrigeration time period, controlling the air conditioner to switch from the refrigeration mode to the self-cleaning mode, causing the air conditioner to execute self-cleaning until completion.
- [1018] In some embodiments, the preset high-temperature interval includes a first preset high-temperature interval and a second preset high-temperature interval, and the maximum value of the first preset high-temperature interval is less than the minimum value of the second preset high-temperature interval. The preset target temperature at least includes a high suitable temperature and an ideal temperature, the high suitable temperature is higher than the ideal temperature, and the high suitable temperature is lower than the minimum value of the first preset high-temperature interval. The preset refrigeration time period includes a first refrigeration time period and a second refrigeration time period. The preset target temperature corresponding to the first preset high-temperature interval is the ideal temperature, the preset target temperature corresponding to the second preset high-temperature interval is the high suitable temperature, the preset refrigeration time period corresponding to the first preset high-temperature interval is the first refrigeration time period, and the preset refrigeration time period corresponding to the second preset high-temperature interval is the second refrigeration time period.
- [1019] In some embodiments, the method further includes:
- [1020] when the heat exchanger to be cleaned is the indoor heat exchanger, if the indoor fan is in an on state at the moment immediately before the air conditioner switches to the self-cleaning mode, controlling the indoor fan to continue to operate for a preset operation duration;
- [1021] when the heat exchanger to be cleaned is the indoor heat exchanger, if the air conditioner is in an off state at the moment immediately before switching to the self-cleaning mode, controlling the air conditioner to maintain the off state for a preset shutdown duration, and after the preset shutdown duration, controlling the indoor heat exchanger to perform frosting treatment for a preset frosting duration; and
- [1022] during the process of frosting treatment for the preset frosting duration, controlling the indoor fan to operate.
- [1023] In some embodiments, the method further includes:
- [1024] after the frosting treatment ends, controlling the air conditioner to enter the defrosting stage of the heat exchanger to be cleaned, causing the heat exchanger to be cleaned to implement a function as the condenser to enable the heat exchanger to be cleaned to perform the defrosting treatment, where the pressure reducer is the expansion valve, and the opening degree of the expansion valve during the defrosting treatment is less than or equal to the opening degree of the expansion valve during the frosting treatment.
- [1025] In some embodiments, the method further includes:
- [1026] before controlling the air conditioner to switch between the heating mode and the indoor heat exchanger self-cleaning mode, controlling the air conditioner to be shut down for a preset shutdown duration; and
- [1027] before controlling the air conditioner to switch between the refrigeration mode and the outdoor heat exchanger self-cleaning mode, controlling the air conditioner to be shut down for a preset shutdown duration, where the self-cleaning mode includes the indoor heat exchanger self-cleaning mode and the outdoor heat exchanger self-cleaning mode.
- [1028] It is worth noting that for the specific steps of the methods in the above embodiments, reference can be made to the working processes of the air conditioner in the above embodiments, and these steps are not repeated here.
- [1029] Some embodiments of the present disclosure further provide a self-cleaning control method for an air conditioner, and the method is applied to a controller. The air conditioner is of a structure similar to that of the air conditioner 1000. For example, the air conditioner includes the indoor unit 100, the outdoor unit 200, and the indoor temperature detection apparatus 400. The indoor unit 100 includes the indoor heat exchanger 101 and the indoor fan 102. The outdoor unit 200 includes the compressor 201, the outdoor heat exchanger 202, and the expansion valve 204. The method includes step 71 to step 72.
- [1030] In step 71, the air conditioner is controlled to enter a self-cleaning mode in response to a received self-cleaning instruction, causing the heat exchanger to be cleaned to operate as an evaporator to cause ice to form on a surface of the heat exchanger to be cleaned. The heat exchanger to be cleaned here is the outdoor heat exchanger or the indoor heat exchanger. The self-cleaning instruction includes a first instruction and a second instruction. The self-cleaning mode includes a first self-cleaning mode and a second self-cleaning mode.
- [1031] The user can input the self-cleaning instruction by means of the button, or the user can also input the self-cleaning instruction by means of the wander lead controller or the remote controller, or the user can also preset the self-cleaning instruction that can be triggered in a timed manner in the air conditioner, thereby implementing the timed self-cleaning function of the air conditioner. The triggering of the self-cleaning instruction is not limited in the present disclosure.
- [1032] When the controller 300 receives the first instruction, the controller 300 controls the air conditioner 1000 to enter the first self-cleaning mode. Under such condition, the indoor heat exchanger 101 is the heat exchanger to be cleaned, and the controller 300 can control the flow direction of the refrigerant by means of the four-way valve 205 to enable the flow direction of the refrigerant to be the same as the flow direction of the refrigerant in the refrigeration mode, causing the indoor heat exchanger 101 to operate as the evaporator to cause ice to form on the surface of the indoor heat exchanger 101. Then the controller 300 changes the flow direction of the refrigerant again by means of the

four-way valve 205, causing the indoor heat exchanger 101 to operate as the condenser, thereby defrosting the indoor heat exchanger 101.

[1033] When the controller 300 receives the second instruction, the controller 300 controls the air conditioner 1000 to enter the second self-cleaning mode. Under such condition, the outdoor heat exchanger 202 is the heat exchanger to be cleaned, and the controller 300 can control the flow direction of the refrigerant by means of the four-way valve 205 to enable the flow direction of the refrigerant to be the same as the flow direction of the refrigerant in the heating mode, causing the outdoor heat exchanger 202 to operate as the evaporator to cause ice to form on the surface of the outdoor heat exchanger 202. Then the controller 300 changes the flow direction of the refrigerant again by means of the four-way valve 205, causing the outdoor heat exchanger 202 to operate as the condenser, thereby defrosting the outdoor heat exchanger 202.

[1034] It should be noted that the controller can also control the air conditioner to enter the other modes to enable the heat exchanger to be cleaned to serve as the evaporator, and this is not limited in the present disclosure.

[1035] In step 72, the operating parameters of the air conditioner are adjusted according to the indoor ambient temperature.

[1036] The operation process of the air conditioner in the self-cleaning mode includes a frosting stage and a defrosting stage. When the air conditioner enters the self-cleaning mode, the air conditioner first enters the frosting stage.

[1037] The frosting stage refers to that the heat exchanger to be cleaned operates as the evaporator, causing ice to form on the surface of the heat exchanger to be cleaned. When an ice layer on the surface of the heat exchanger to be cleaned can meet the cleaning requirement, the air conditioner enters the defrosting stage. For example, if the duration of the heat exchanger to be cleaned operating as the evaporator reaches a preset time, or the temperature of the coil pipe in the heat exchanger to be cleaned reaches a preset temperature, it indicates that the ice on the surface of the heat exchanger to be cleaned is enough to meet the cleaning requirements, the controller determines that the heat exchanger to be cleaned completes frosting, and the air conditioner enters the defrosting stage.

[1038] The defrosting stage refers to that the heat exchanger to be cleaned operates as the condenser, causing the ice on the surface of the heat exchanger to be cleaned to melt, thereby cleaning the heat exchanger to be cleaned. When the ice layer on the surface of the heat exchanger to be cleaned completely melts, the air conditioner exits the defrosting stage. For example, when the heat exchanger to be cleaned is defrosted for a preset time, or the temperature of the coil pipe in the heat exchanger to be cleaned reaches a preset temperature, the controller determines that the heat exchanger to be cleaned completes defrosting, the air conditioner exits the defrosting stage, and thus the air conditioner completes cleaning of the heat exchanger to be cleaned.

[1039] In the frosting stage, the controller can adjust the operating parameters of the air conditioner according to the indoor ambient temperature, thereby avoiding the influence of the excessively large change of the indoor ambient temperature on the refrigeration or heating effect of the air conditioner.

[1040] For the different heat exchangers to be cleaned, the flow directions of the refrigerant in the frosting stage are different, causing the indoor ambient temperature to drop or rise in the frosting stage, and then the indoor ambient temperature is prone to being too low or too high. Thus, to solve the problem, for the indoor ambient temperature, a first preset range and a second preset range are preset. The first preset range corresponds to the excessively low indoor ambient temperature. The second preset range corresponds to the excessively high indoor ambient temperature.

[1041] In some embodiments, step 72 includes step 721 to step 723.

[1042] In step 721, it is determined that the indoor ambient temperature is in the first preset range, and the air conditioner is controlled to switch from the self-cleaning mode to the first mode.

[1043] In step 722, it is determined that the operation duration of the air conditioner in the first mode is within a first preset duration and the indoor ambient temperature reaches a first preset temperature, the air conditioner is controlled to switch from the first mode to the self-cleaning mode. The first preset temperature here is higher than the upper limit value of the first preset range. It should be noted that if the operation duration of the air conditioner in the first mode is within the first preset duration and the indoor ambient temperature does not reach the first preset temperature, the controller controls the air conditioner to maintain the first mode.

[1044] In step 723, it is determined that the operation duration of the air conditioner in the first mode exceeds the first preset duration and the indoor ambient temperature is lower than the first preset temperature, and the air conditioner is controlled to switch from the first mode to the self-cleaning mode, thereby continuing to clean the heat exchanger to be cleaned until the air conditioner completes cleaning of the heat exchanger to be cleaned.

[1045] In some embodiments, it is determined that the operation duration of the air conditioner in the first mode exceeds the first preset duration and the indoor ambient temperature reaches the first preset temperature, and the air conditioner is controlled to switch from the first mode to the self-cleaning mode, thereby continuing to clean the heat exchanger to be cleaned until the air conditioner completes cleaning of the heat exchanger to be cleaned.

[1046] Under the condition that the indoor ambient temperature is in the first preset range, the indoor ambient temperature is relatively low, and the controller needs to control the air conditioner to switch from the self-cleaning mode to the first mode, so as to increase the indoor ambient temperature and enable the indoor ambient temperature to rise again to a suitable temperature range. Besides, in order to improve the self-cleaning efficiency while meeting the user's requirements, the first preset duration is preset.

[1047] Under the condition that the air conditioner switches from the self-cleaning mode to the first mode, if the indoor ambient temperature rises to the first preset temperature within the first preset duration, the heating effect of the air conditioner is relatively good, and the indoor ambient temperature can meet user's requirements. Under such condition, the controller can control the air conditioner to switch from the first mode to the self-cleaning mode. After the air conditioner switches from the first mode to the self-cleaning mode, the controller still monitors the indoor ambient temperature by means of the corresponding indoor temperature

detection apparatus, thereby timely controlling the air conditioner to switch to the first mode when the indoor ambient temperature is relatively low.

[1048] Alternatively, under the condition that the air conditioner switches from the self-cleaning mode to the first mode, if the operation duration of the air conditioner in the first mode exceeds the first preset duration and the indoor ambient temperature is still lower than the first preset temperature, the heating effect of the air conditioner is not obvious. Under such condition, if the air conditioner continues the first mode, the indoor ambient temperature rises slowly, resulting in prolonging the self-cleaning time. Thus, under such condition, in order to improve the self-cleaning efficiency, after the operation duration of the air conditioner in the first mode exceeds the first preset duration, the controller can control the air conditioner to switch from the first mode to the self-cleaning mode. Besides, after the air conditioner switches from the first mode to the self-cleaning mode, the controller no longer switches the operation modes according to the indoor ambient temperature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[1049] In some embodiments, the first preset range includes a first sub-preset range and a second sub-preset range, and the upper limit value of the first sub-preset range is less than the lower limit value of the second sub-preset range. The first preset temperature includes a first sub-preset temperature and a second sub-preset temperature, the first sub-preset temperature is lower than the second sub-preset temperature, and the first sub-preset temperature is higher than the upper limit value of the second sub-preset range. The first preset duration includes a first heating time period and a second heating time period. The first sub-preset temperature and the first heating time period correspond to the first sub-preset range, and the second sub-preset temperature and the second heating time period correspond to the second sub-preset range. Thus, by dividing the first preset range into the first sub-preset range and the second sub-preset range, the self-cleaning efficiency can be further improved while the user's requirements are met.

[1050] In some embodiments, step 72 further includes step 724 to step 726.

[1051] In step 724, it is determined that the indoor ambient temperature is in the second preset range, and the air conditioner is controlled to switch from the self-cleaning mode to the second mode.

[1052] In step 725, it is determined that the operation duration of the air conditioner in the second mode is within a second preset duration and the indoor ambient temperature is lower than or equal to a second preset temperature, and the air conditioner is controlled to switch from the second mode to the self-cleaning mode. The second preset temperature here is lower than the lower limit value of the second preset range. It should be noted that if the operation duration of the air conditioner in the second mode is within the second preset duration and the indoor ambient temperature does not reach the second preset temperature, the controller controls the air conditioner to maintain the second mode.

[1053] In step 726, it is determined that the operation duration of the air conditioner in the second mode exceeds the second preset duration and the indoor ambient temperature is higher than the second preset temperature, and the air conditioner is controlled to switch from the second mode to the self-cleaning mode, thereby continuing to clean the heat

exchanger to be cleaned until the air conditioner completes cleaning of the heat exchanger to be cleaned.

[1054] In some embodiments, it is determined that the operation duration of the air conditioner in the second mode exceeds the second preset duration and the indoor ambient temperature reaches the second preset temperature, and the air conditioner is controlled to switch from the second mode to the self-cleaning mode, thereby continuing to clean the heat exchanger to be cleaned until the air conditioner completes cleaning of the heat exchanger to be cleaned.

[1055] Under the condition that the indoor ambient temperature is in the second preset range, the indoor ambient temperature is relatively high, and the controller needs to control the air conditioner to switch from the self-cleaning mode to the second mode, so as to reduce the indoor ambient temperature and enable the indoor ambient temperature to drop to a suitable temperature range. Besides, in order to improve the self-cleaning efficiency while meeting the user's requirements, the second preset duration is preset.

[1056] Under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the indoor ambient temperature drops to (for example, be lower than or equal to) the second preset temperature within the second preset duration, the refrigeration effect of the air conditioner 1000 is relatively good, and the indoor ambient temperature can meet user's requirements. Under such condition, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode. After the air conditioner 1000 switches from the second mode to the self-cleaning mode, the controller 300 still monitors the indoor ambient temperature by means of the indoor temperature detection apparatus 400, thereby timely controlling the air conditioner 1000 to switch to the second mode when the indoor ambient temperature is relatively high.

[1057] Alternatively, under the condition that the air conditioner 1000 switches from the self-cleaning mode to the second mode, if the operation duration of the air conditioner 1000 in the second mode exceeds the second preset duration and the indoor ambient temperature is still higher than the second preset temperature, the refrigeration effect of the air conditioner 1000 is not obvious. Under such condition, if the air conditioner 1000 continues the second mode, the indoor ambient temperature drops slowly, resulting in prolonging the self-cleaning time. Thus, under such condition, in order to improve the self-cleaning efficiency, after the operation duration of the air conditioner 1000 in the second mode exceeds the second preset duration, the controller 300 can control the air conditioner 1000 to switch from the second mode to the self-cleaning mode. Besides, after the air conditioner 1000 switches from the second mode to the self-cleaning mode, the controller 300 no longer switches the operation modes according to the indoor ambient temperature, but directly executes the self-cleaning task until completing the cleaning of the heat exchanger to be cleaned.

[1058] In some embodiments, the second preset range includes a third sub-preset range and a fourth sub-preset range, and the upper limit value of the third sub-preset range is less than the lower limit value of the fourth sub-preset range. The second preset temperature includes a third sub-preset temperature and a fourth sub-preset temperature, the third sub-preset temperature is higher than the fourth sub-preset temperature, and the third sub-preset temperature is lower than the lower limit value of the third sub-preset

range. The second preset duration includes a first refrigeration time period and a second refrigeration time period. The third sub-preset temperature and the first refrigeration time period correspond to the third sub-preset range, and the fourth sub-preset temperature and the second refrigeration time period correspond to the fourth sub-preset range. Thus, by dividing the second preset range into the third sub-preset range and the fourth sub-preset range, the self-cleaning efficiency can be further improved while the user's requirements are met.

[1059] In some embodiments, the method further includes step 701 to step 703.

[1060] In step 701, it is determined that the heat exchanger to be cleaned is the indoor heat exchanger.

[1061] In step 702, if the indoor fan is in an on state at the moment immediately before the air conditioner switches to the self-cleaning mode, the indoor fan is controlled to continue to operate for a first target duration. After the indoor fan operates for the first target duration, the controller can control the indoor fan to be turned off.

[1062] In step 703, if the air conditioner is in an off state at the moment immediately before the air conditioner switches to the self-cleaning mode, the air conditioner is controlled to continue to be turned off for a second target duration, after the duration of the air conditioner in the off state reaches the second target duration, the indoor heat exchanger is controlled to operate as the evaporator for a third target duration, and the indoor fan is controlled to be turned on and operate for the third target duration.

[1063] During the frosting process of the heat exchanger to be cleaned (for example, the indoor heat exchanger 101), the air circulation effect can be improved by means of transient turning-on of the indoor fan 102, and moisture in the air can pass through the heat exchanger to be cleaned, so that more ice can form on the surface of the heat exchanger to be cleaned, and the cleaning of the heat exchanger to be cleaned is facilitated.

[1064] In some embodiments, the method further includes step 81.

[1065] In step 81, after the heat exchanger to be cleaned completes frosting, the air conditioner is controlled to cause the heat exchanger to be cleaned to operate as the condenser to enable the heat exchanger to be cleaned to be defrosted. Here, the opening degree of the expansion valve during defrosting of the heat exchanger to be cleaned is less than or equal to the opening degree of the expansion valve during frosting of the heat exchanger to be cleaned. Thus, during the defrosting process of the heat exchanger to be cleaned, the refrigerant may have a relatively high temperature, which facilitates defrosting of the surface of the heat exchanger to be cleaned.

[1066] In some embodiments, the method further includes step 7210.

[1067] In step 7210, before the air conditioner switches between the first mode and the first self-cleaning mode, or before the air conditioner switches between the second mode and the second self-cleaning mode, the air conditioner is controlled to be shut down for a fourth target duration.

[1068] When the air conditioner is in a frosting stage of the first self-cleaning mode, the flow direction of the refrigerant is the same as the flow direction of the refrigerant in the refrigeration mode. When the air conditioner is in a frosting stage of the second self-cleaning mode, the flow direction of the refrigerant is the same as the flow direction of the

refrigerant in the heating mode. When the air conditioner switches between the first mode and the first self-cleaning mode, or between the second mode and the second self-cleaning mode, the flow direction of the refrigerant may be changed, so that in order to protect the air conditioner, it needs to be shut down when switching the mode.

[1069] In the self-cleaning method for the air conditioner in some embodiments of the present disclosure, the indoor temperature detection apparatus is arranged indoors to detect the indoor ambient temperature, and the indoor ambient temperature can be monitored in real time when the air conditioner operates in the self-cleaning mode, so that the operating parameters of the air conditioner can be adjusted according to the indoor ambient temperature, and the influence of the excessively large change in the indoor ambient temperature during the self-cleaning process on the refrigeration or heating effect of the air conditioner can be avoided.

[1070] A person skilled in the art should understand that the disclosure scope of the present disclosure is not limited to the above specific embodiments, and some elements of the embodiments may be amended and replaced without departing from the spirit of the present disclosure. The scope of the present disclosure is limited by the appended claims.

What is claimed is:

1. An air conditioner, comprising:

an indoor unit, which comprises an indoor heat exchanger;

an outdoor unit, which comprises a compressor, an outdoor heat exchanger, and an expansion valve;

an indoor temperature detection apparatus, which is configured to detect an indoor ambient temperature; and

a controller, which is configured to:

control the air conditioner to enter a self-cleaning mode in response to a received self-cleaning instruction, causing a heat exchanger to be cleaned to operate as an evaporator to cause the heat exchanger to be cleaned to perform a frosting treatment;

adjust an operating parameter of the air conditioner according to the indoor ambient temperature in a frosting treatment stage; and

control the air conditioner to enter a defrosting stage of the heat exchanger to be cleaned after the frosting treatment stage ends;

wherein the heat exchanger to be cleaned is the outdoor heat exchanger or the indoor heat exchanger, and wherein the self-cleaning instruction comprises a first instruction and a second instruction, the first instruction is configured to instruct cleaning of the indoor heat exchanger, and the second instruction is configured to instruct cleaning of the outdoor heat exchanger.

2. The air conditioner of claim 1, wherein the self-cleaning instruction is the first instruction, wherein a step of adjusting the operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, comprises:

determining that the indoor ambient temperature is in a first preset range, and controlling the air conditioner to switch from the self-cleaning mode to a first mode or a first sub-mode;

determining that an operation duration of the air conditioner in the first mode or the first sub-mode is within a first preset duration, and controlling the air conditioner to switch from the first mode or the first sub-

mode back to the self-cleaning mode if the indoor ambient temperature is detected to reach a first preset temperature, wherein the first preset temperature is greater than an upper limit value of the first preset range; or

controlling the air conditioner to switch from the first mode or the first sub-mode back to the self-cleaning mode to cause the air conditioner to execute a self-cleaning task until completion, if the operation duration of the air conditioner in the first mode or the first sub-mode exceeds the first preset duration.

3. The air conditioner of claim 2, wherein the first preset range comprises a first sub-preset range and a second sub-preset range, and an upper limit value of the first sub-preset range is less than a lower limit value of the second sub-preset range;

wherein the first preset temperature comprises a first sub-preset temperature and a second sub-preset temperature;

wherein the first preset duration comprises a first heating time period and a second heating time period;

wherein the first sub-preset temperature and the first heating time period correspond to the first sub-preset range, and the second sub-preset temperature and the second heating time period correspond to the second sub-preset range.

4. The air conditioner of claim 3, wherein determining that the operation duration of the air conditioner in the first mode or the first sub-mode is within the first preset duration, and controlling the air conditioner to switch from the first mode or the first sub-mode back to the self-cleaning mode if the indoor ambient temperature is detected to reach the first preset temperature, comprises:

determining whether the indoor ambient temperature is within the first sub-preset range; if so, controlling the air conditioner to switch from the self-cleaning mode to the first mode or the first sub-mode; and controlling the air conditioner to switch from the first mode or the first sub-mode back to the self-cleaning mode when the indoor ambient temperature reaches the first sub-preset temperature; and

if the indoor ambient temperature is not within the first sub-preset range, determining whether the indoor ambient temperature is within the second sub-preset range; if so, controlling the air conditioner to switch from the self-cleaning mode to the first mode or the first sub-mode; and controlling the air conditioner to switch from the first mode or the first sub-mode back to the self-cleaning mode when the indoor ambient temperature reaches the second sub-preset temperature.

5. The air conditioner of claim 3, wherein controlling the air conditioner to switch from the first mode or the first sub-mode back to the self-cleaning mode to cause the air conditioner to execute the self-cleaning task until completion, if the operation duration of the air conditioner in the first mode or the first sub-mode exceeds the first preset duration, comprises:

determining whether the indoor ambient temperature is within the first sub-preset range; if so, controlling the air conditioner to switch from the self-cleaning mode to the first mode or the first sub-mode; and controlling the air conditioner to switch from the first mode or the first sub-mode back to the self-cleaning mode to cause the air conditioner to execute the self-cleaning task until

completion, if the operation duration of the air conditioner in the first mode or the first sub-mode exceeds the first heating time period; and

if the indoor ambient temperature is not within the first sub-preset range, determining whether the indoor ambient temperature is within the second sub-preset range; if so, controlling the air conditioner to switch from the self-cleaning mode to the first mode or the first sub-mode; and controlling the air conditioner to switch from the first mode or the first sub-mode back to the self-cleaning mode to cause the air conditioner to execute the self-cleaning task until completion, if the operation duration of the air conditioner in the first mode or the first sub-mode exceeds the second heating time period.

6. The air conditioner of claim 1, wherein the self-cleaning instruction is the second instruction, wherein a step of adjusting the operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, comprises:

determining that the indoor ambient temperature is in a second preset range, and controlling the air conditioner to switch from the self-cleaning mode to a second mode or a second sub-mode;

determining that an operation duration of the air conditioner in the second mode or the second sub-mode is within a second preset duration, and controlling the air conditioner to switch from the second mode or the second sub-mode back to the self-cleaning mode if the indoor ambient temperature is detected to be lower than a second preset temperature, wherein the second preset temperature is lower than a lower limit value of the second preset range; or

controlling the air conditioner to switch from the second mode or the second sub-mode back to the self-cleaning mode to cause the air conditioner to execute a self-cleaning task until completion, if the operation duration of the air conditioner in the second mode or the second sub-mode exceeds the second preset duration.

7. The air conditioner of claim 6, wherein the second preset range comprises a third sub-preset range and a fourth sub-preset range, and an upper limit value of the third sub-preset range is less than a lower limit value of the fourth sub-preset range;

wherein the second preset temperature comprises a third sub-preset temperature and a fourth sub-preset temperature;

wherein the second preset duration comprises a first refrigeration time period and a second refrigeration time period;

wherein the third sub-preset temperature and the first refrigeration time period correspond to the third sub-preset range, and the fourth sub-preset temperature and the second refrigeration time period correspond to the fourth sub-preset range.

8. The air conditioner of claim 7, wherein determining that the operation duration of the air conditioner in the second mode or the second sub-mode is within the second preset duration, and controlling the air conditioner to switch from the second mode or the second sub-mode back to the self-cleaning mode if the indoor ambient temperature is detected to be lower than the second preset temperature, comprises:

determining whether the indoor ambient temperature is within the third sub-preset range; if so, controlling the air conditioner to switch from the self-cleaning mode to the second mode or the second sub-mode; and controlling the air conditioner to switch from the second mode or the second sub-mode back to the self-cleaning mode when the indoor ambient temperature is lower than the third sub-preset temperature; and

if the indoor ambient temperature is not within the third sub-preset range, determining whether the indoor ambient temperature is within the fourth sub-preset range; if so, controlling the air conditioner to switch from the self-cleaning mode to the second mode or the second sub-mode; and controlling the air conditioner to switch from the second mode or the second sub-mode back to the self-cleaning mode when the indoor ambient temperature is lower than the fourth sub-preset temperature.

9. The air conditioner of claim 7, wherein controlling the air conditioner to switch from the second mode or the second sub-mode back to the self-cleaning mode to cause the air conditioner to execute the self-cleaning task until completion, if the operation duration of the air conditioner in the second mode or the second sub-mode exceeds the second preset duration, comprises:

determining whether the indoor ambient temperature is within the third sub-preset range; if so, controlling the air conditioner to switch from the self-cleaning mode to the second mode or the second sub-mode; and controlling the air conditioner to switch from the second mode or the second sub-mode back to the self-cleaning mode to cause the air conditioner to execute the self-cleaning task until completion, if the operation duration of the air conditioner in the second mode or the second sub-mode exceeds the first refrigeration time period; and

if the indoor ambient temperature is not within the third sub-preset range, determining whether the indoor ambient temperature is within the fourth sub-preset range; if so, controlling the air conditioner to switch from the self-cleaning mode to the second mode or the second sub-mode; and controlling the air conditioner to switch from the second mode or the second sub-mode back to the self-cleaning mode to cause the air conditioner to execute the self-cleaning task until completion, if the operation duration of the air conditioner in the second mode or the second sub-mode exceeds the second refrigeration time period.

10. The air conditioner of claim 1, wherein a step of adjusting the operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, comprises:

periodically obtaining the indoor ambient temperature in the frosting treatment stage;

controlling an operating parameter of at least one of the compressor and the expansion valve to be subjected to first adjustment when determining that the indoor ambient temperature reaches a critical temperature; and

controlling the operating parameter of at least one of the compressor and the expansion valve to be subjected to second adjustment when determining that the indoor ambient temperature is lower than the critical temperature;

wherein the first adjustment and the second adjustment each comprise increasing or decreasing one of frequency of the compressor and an opening degree of the expansion valve; or simultaneously increasing or decreasing the frequency of the compressor and the opening degree of the expansion valve.

11. The air conditioner of claim 1, wherein a step of adjusting the operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, comprises:

performing a refrigeration frosting process for a first preset duration, performing more than one indoor ambient temperature detections within the first preset duration, controlling an operating parameter of at least one of the compressor and the expansion valve to be subjected to first adjustment when the indoor ambient temperature is greater than or equal to a critical temperature; and controlling the operating parameter of at least one of the compressor and the expansion valve to be subjected to second adjustment when the indoor ambient temperature is lower than the critical temperature;

detecting the indoor ambient temperature, and controlling the air conditioner to switch modes based on the detected indoor ambient temperature;

continuing the refrigeration frosting process, and

determining whether the refrigeration frosting process reaches a termination condition, wherein the termination condition comprises:

a duration of the heat exchanger to be cleaned operating as the evaporator reaches a preset time, or a temperature of a coil pipe in the heat exchanger to be cleaned reaches a preset temperature, or the temperature of the coil pipe in the heat exchanger to be cleaned reaches the preset temperature and lasts for a preset time, or the indoor ambient temperature reaches a preset temperature and lasts for a preset time.

12. The air conditioner of claim 11, wherein detecting the indoor ambient temperature, and controlling the air conditioner to switch modes based on the detected indoor ambient temperature comprises:

determining that the indoor ambient temperature is within a first preset range, and controlling the air conditioner to switch from the self-cleaning mode to a first mode or a first sub-mode; or

determining that the indoor ambient temperature is within a second preset range, and controlling the air conditioner to switch from the self-cleaning mode to a second mode or a second sub-mode; or

determining that the indoor ambient temperature is within a third preset range, and controlling the air conditioner to switch from the self-cleaning mode to a third mode;

wherein continuing the refrigeration frosting process comprises:

performing the frosting treatment for the first preset duration, and performing more than one indoor ambient temperature detections within the first preset duration, controlling the operating parameter of at least one of the compressor and the expansion valve to be subjected to the first adjustment when the indoor ambient temperature is greater than or equal to the critical temperature; and controlling the operating parameter of at least one of the compressor and the expansion valve to be

subjected to the second adjustment when the indoor ambient temperature is lower than the critical temperature; and

continuing to detect the indoor ambient temperature, and determining to execute a first refrigeration freezing mode or a second refrigeration freezing mode based on the indoor ambient temperature.

13. The air conditioner of claim **1**, wherein after a step of adjusting the operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, the controller is further configured to:

control a fan to be started when the frosting treatment is completed, wherein when the self-cleaning instruction is the first instruction, the fan is an indoor fan; and when the self-cleaning instruction is the second instruction, the fan is an outdoor fan.

14. The air conditioner of claim **1**, wherein a step of controlling the air conditioner to enter the defrosting stage of the heat exchanger to be cleaned after the frosting treatment stage ends, comprises:

controlling the air conditioner to enter the defrosting stage of the heat exchanger to be cleaned after the frosting treatment stage ends, causing the heat exchanger to be cleaned to implement a function as a condenser to enable the heat exchanger to be cleaned to perform defrosting treatment, wherein an opening degree of the expansion valve during the defrosting treatment is less than or equal to an opening degree of the expansion valve during the frosting treatment.

15. A self-cleaning control method for an air conditioner, wherein the method is applied to a controller of the air conditioner, wherein the air conditioner comprises an indoor unit, an outdoor unit, and an indoor temperature detection apparatus; the indoor unit comprises an indoor heat exchanger, wherein the outdoor unit comprises a compressor, an outdoor heat exchanger, and an expansion valve, wherein the indoor temperature detection apparatus is configured to detect an indoor ambient temperature, and wherein the method comprises:

controlling the air conditioner to enter a self-cleaning mode in response to a received self-cleaning instruction, causing a heat exchanger to be cleaned to operate as an evaporator to cause the heat exchanger to be cleaned to perform a frosting treatment;

adjusting an operating parameter of the air conditioner according to the indoor ambient temperature in a frosting treatment stage; and

controlling the air conditioner to enter a defrosting stage of the heat exchanger to be cleaned after the frosting treatment stage ends;

wherein the heat exchanger to be cleaned is the outdoor heat exchanger or the indoor heat exchanger, and wherein the self-cleaning instruction comprises a first instruction and a second instruction, the first instruction is configured to instruct cleaning of the indoor heat exchanger, and the second instruction is configured to instruct cleaning of the outdoor heat exchanger.

16. The method of claim **15**, wherein adjusting the operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, comprises:

periodically obtaining the indoor ambient temperature in the frosting treatment stage;

controlling an operating parameter of at least one of the compressor and the expansion valve to be subjected to first adjustment when determining that the indoor ambient temperature reaches a critical temperature; and

controlling the operating parameter of at least one of the compressor and the expansion valve to be subjected to second adjustment when determining that the indoor ambient temperature is lower than the critical temperature;

wherein the first adjustment and the second adjustment each comprise increasing or decreasing one of frequency of the compressor and an opening degree of the expansion valve; or simultaneously increasing or decreasing the frequency of the compressor and the opening degree of the expansion valve.

17. The method of claim **15**, wherein adjusting the operating parameter of the air conditioner according to the indoor ambient temperature in the frosting treatment stage, comprises:

performing a refrigeration frosting process for a first preset duration, performing more than one indoor ambient temperature detections within the first preset duration, controlling an operating parameter of at least one of the compressor and the expansion valve to be subjected to first adjustment when the indoor ambient temperature is greater than or equal to a critical temperature; and controlling the operating parameter of at least one of the compressor and the expansion valve to be subjected to second adjustment when the indoor ambient temperature is lower than the critical temperature;

detecting the indoor ambient temperature, and controlling the air conditioner to switch modes based on the detected indoor ambient temperature;

continuing the refrigeration frosting process, and

determining whether the refrigeration frosting process reaches a termination condition, wherein the termination condition comprises:

a duration of the heat exchanger to be cleaned operating as the evaporator reaches a preset time, or a temperature of a coil pipe in the heat exchanger to be cleaned reaches a preset temperature, or the temperature of the coil pipe in the heat exchanger to be cleaned reaches the preset temperature and lasts for a preset time, or the indoor ambient temperature reaches a preset temperature and lasts for a preset time.

18. The method of claim **17**, wherein detecting the indoor ambient temperature, and controlling the air conditioner to switch modes based on the detected indoor ambient temperature, comprises:

determining that the indoor ambient temperature is within a first preset range, and controlling the air conditioner to switch from the self-cleaning mode to a first mode or a first sub-mode; or

determining that the indoor ambient temperature is within a second preset range, and controlling the air conditioner to switch from the self-cleaning mode to a second mode or a second sub-mode; or

determining that the indoor ambient temperature is within a third preset range, and controlling the air conditioner to switch from the self-cleaning mode to a third mode;

wherein continuing the refrigeration frosting process comprises:

performing frosting treatment for the first preset duration, and performing more than one indoor ambient temperature detections within the first preset duration, controlling the operating parameter of at least one of the compressor and the expansion valve to be subjected to the first adjustment when the indoor ambient temperature is greater than or equal to the critical temperature; and controlling the operating parameter of at least one of the compressor and the expansion valve to be subjected to the second adjustment when the indoor ambient temperature is lower than the critical temperature; and

continuing to detect the indoor ambient temperature, and determining to execute a first refrigeration freezing mode or a second refrigeration freezing mode based on the indoor ambient temperature.

19. The method of claim **15**, wherein after adjusting the operating parameter of the air conditioner according to the

indoor ambient temperature in the frosting treatment stage, the method further comprises:

controlling a fan to be started when the frosting treatment is completed, wherein when the self-cleaning instruction is the first instruction, the fan is an indoor fan; and when the self-cleaning instruction is the second instruction, the fan is an outdoor fan.

20. The method of claim **15**, wherein controlling the air conditioner to enter the defrosting stage of the heat exchanger to be cleaned after the frosting treatment stage ends, comprises:

controlling the air conditioner to enter the defrosting stage of the heat exchanger to be cleaned after the frosting treatment stage ends, causing the heat exchanger to be cleaned to implement a function as a condenser to enable the heat exchanger to be cleaned to perform defrosting treatment, wherein an opening degree of the expansion valve during the defrosting treatment is less than or equal to an opening degree of the expansion valve during the frosting treatment.

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